OECD Economic Surveys

Denmark

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OECD ECONOMIC SURVEYS

2002-2003

Denmark

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BASIC STATISTICS OF DENMARK

THE LAND

Area (sq. km)	43 094
Agricultural area (sq. km) (1997)	26 470

Population of major cities, 2002, thousands:	
Copenhagen	501
Århus	289
Odense	184
Ålborg	162

THE PEOPLE

Population, 2002, thousands	5 368
Number of inhabitants per sq. km	123
Population, annual net natural increase (average 1997-2001, thousands)	7.7
Natural increase rate per 1 000 inhabitants, 2001	1.3

Total employment, 2002, thousands	2 772
By sector:	
Agriculture	100
Industry	435
Construction	166
Market services	1 084
Community, social and personal services	987

THE PRODUCTION

Gross domestic product, 2002:	
Kr billion	1 358.3
Per capita (US$)	32 092

Gross fixed capital formation, 2002:	
Kr billion	270.6
Per cent of GDP	19.9
Per capita (US$)	6 394

THE GOVERNMENT

Public consumption, 2002, per cent of GDP	26.1
General government current revenue, per cent of GDP	56.5
Public gross fixed capital investment, per cent of GDP	1.8

Composition of Parliament:	Number of seats
Social Democrats	52
Liberals	56
Danish People's Party	22
Conservatives	16
Socialist Peoples's Party	12
Social Liberals	9
Left Alliance	4
Christian Democrats	4
North Atlantic	3
Other	1
Total	179

Last general elections: 20.11.2001

Next general elections: 20.11.2005 (at the latest)

THE FOREIGN TRADE

Exports, 2000	
Exports of goods and services, per cent of GDP	44.2
Main merchandise exports (per cent of total):	
Agricultural products	9.9
of which: Products of animal origin	8.1
Manufactured products	75.8
of which:	
Machinery and instruments	27.8
Other manufactured products	48.0
Other products	14.3

Imports, 2000	
Imports of goods and services, per cent of GDP	38.3
Main merchandise imports (per cent of total):	
Intermediate goods for agriculture	2.4
Intermediate goods for other sectors	40.8
Fuels and lubricants	4.3
Capital goods	14.4
Transport equipment	5.9
Consumer goods	28.3

THE CURRENCY

Monetary unit: Krone

Currency units per US$:	
Year 2002	7.884
April 2003	6.831

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

•

The economic situation and policies of Denmark were reviewed by the Committee on 23 April 2003. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 7 May 2003.

•

The Secretariat's draft report was prepared for the Committee by Deborah Roseveare, Martin Jørgensen and Richard Herd under the supervision of Peter Jarrett.

•

The previous Survey of Denmark was issued in February 2002.

Assessment and recommendations

The underlying macroeconomic position is sound and policies should remain prudent

The Danish economy has a sound underlying macroeconomic position that has enabled it to cope well with the current international slowdown. Assuming that the projected recovery in global demand transpires, the pace of activity is expected to gradually pick up from its current rate of around 1½ per cent to around 2½ per cent growth next year, sufficient to reverse the recent up-tick in unemployment. With a current budget surplus of 1½ per cent of GDP and inflation running just above 2 per cent, the economy is well-balanced. But Denmark's output gap is currently smaller than that of the euro area and is likely to close sooner. Monetary conditions could thus be slightly looser than desirable given available capacity, leading to some risk of future overheating. Against this backdrop, any additional stimulus from fiscal policy would be unhelpful, so that tax cuts planned for 2004 to 2007 should be combined with restrained growth in public expenditure. Prudent macroeconomic policies are in place and need to be maintained. But the principal focus of policy in Denmark is appropriately set on addressing longer term economic challenges, and this is also the main focus of this *Survey*.

Ensuring fiscal sustainability by raising labour supply is the key policy challenge

The foremost challenge is to prepare for the rapid ageing of the population between now and 2040, when more than one in four adults will be over 65 years old. Gains in productivity should still be sufficient to generate continuing growth in national income per capita, but rising retirement income transfers and other age-related public outlays risk undermining fiscal sustainability. Curtailing expenditure growth may prove difficult, given the deep attachment to a comprehensive welfare state, while raising already high taxes would risk being counterproductive by making work

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even less attractive relative to leisure. The government's strategy therefore rightly focuses on ensuring fiscal sustainability by continuing to reduce public debt and expanding future labour supply through further raising participation rates and/or increasing hours worked, thereby both expanding the tax base and lowering the transfer burden. While rising living standards generally lead to an increasing preference for leisure, in Denmark the choice between work and leisure is heavily biased towards the latter by the tax/benefit system. Equity concerns have constrained the use of market-oriented incentives to achieve better labour supply outcomes. But the capacity for further expanding administrative measures instead may be limited. At least in the longer run, some shift towards more market-based instruments would help to safeguard the future of the welfare state.

Several avenues can be pursued to raise labour supply

Changing the relevant work incentives so as to encourage people to spend more time in work over their lifetime requires a multi-pronged approach. The government's "More People in Work" package makes a good start, but action is warranted in several other areas to address: late entry to the labour market; a significant proportion of prime-age workers drawing public income support; lower participation among immigrants from less developed countries; early retirement; and lower average hours worked per person.

Passage through education needs to be accelerated

A significant proportion of young Danes are still studying into their late 20s, though many of them combine study with working. Lowering the age of full entry into the labour market would not only add to the effective labour supply but would also be cost-effective and welfare-enhancing. One contribution would be to refocus the voluntary 10th form at the end of compulsory education, currently attended by a majority of children, so that it instead provides more intensive remedial assistance to the small percentage of pupils that have fallen behind, while other students pass directly into upper-secondary education. In tertiary education, study paths tend to be drawn out and completed at a later age than in many other countries. Virtually all the costs of tuition are borne by the taxpayer, while generous study grants and subsidised loans are provided.

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But as additional qualifications provide a relatively modest boost to after-tax earnings in Denmark, charging significant tuition fees would probably be counter-productive and might discourage people from undertaking such education at all, despite the social return from having a well-qualified workforce. Instead, encouraging students to choose courses carefully, complete them expeditiously and start earning quickly may need to involve measures to tighten the provisions of the grants and loans scheme. These could include shortening the length of time for which study assistance is provided to more closely reflect the minimum time required to complete the study programme, linking continued support to passing courses, and adjusting the mix of income support from grants to loans.

Economic incentives to work need further strengthening

Participation rates could be boosted further, even for people of prime working age. The "More People in Work" scheme streamlines the management of unemployment but the elements addressing the weak economic incentives to work are confined to a sub-group of those receiving social assistance. The government's stated desire to promote self-reliance is undermined if the net gains from taking a job are meagre, and even in some circumstances negative. Unemployment insurance and cash benefits are high relative to other OECD countries, particularly for those whose earnings prospects are relatively modest. The earned income tax credit that will be implemented as from 2004 should strengthen the incentive to move into work somewhat. Such incentives could be strengthened further, without attendant budgetary costs, by tightening eligibility and/or gradually reducing benefit rates, through:

- Shortening the length of unemployment spell covered by unemployment insurance benefits.
- Calculating unemployment insurance benefits on the basis of income earned over the previous 12 months, thereby gradually reducing benefits as the unemployment spell lengthens. This would strengthen the incentive to accept a job offer even if it involves a lower rate of pay and would also to some extent limit the capacity for seasonal workers to misuse the system.
- Instituting longer work requirements before qualifying for full unemployment insurance benefits.

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– Making further adjustments to the formulae used to calculate cash benefits so as to provide a larger margin between income on benefits and potential income from work.

The effective retirement age needs to rise

Keeping the effective dependency ratio as low as possible requires every effort to encourage those who are able to work to do so. Despite this, the average effective age of retirement has fallen somewhat over the last two decades, while life expectancy has risen: thus retirement represents a steadily rising share of adult lifetimes. At the same time, the voluntary early retirement scheme pays participants up to three or four times more in benefits than the net present value of their contributions; this sends the wrong signal by making early retirement more attractive. Improving the incentives to work by actions on several fronts could lead to a significant increase in the effective retirement age:

– Giving consideration to abolishing the voluntary early retirement scheme altogether, noting that other systems already take care of people unable to find work or with reduced work capacity, or, as a second best, making it a strictly defined-contribution system.
– Applying activation measures to all workers right up to the statutory age of retirement to signal clearly that people are expected to continue to work.
– Ensuring that other forms of income support such as early retirement (disability) pensions are not used as an alternative channel for permanent exit from the labour force.

Reversing the earlier reduction in the statutory retirement age from 67 to 65, and then changing the statutory age, along with the age criteria for any early retirement scheme, in line with increases in life expectancy would also help to ease the impact of ageing on public finances and make room for further supply-oriented income tax cuts.

Immigrants need to be better integrated

An inflow of migrants could, in principle, also alleviate the ageing burden. But Denmark's experience with migrants arriving in the country since the 1960s has been very mixed. Highly skilled immigrants from more developed countries have generally integrated well, work to support themselves and make a positive lifetime contribution to public finances.

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Well-qualified immigrants from less developed countries have had more difficulty in getting work commensurate with their apparent educational attainment. Low-skilled immigrants from such countries tend to be poorly integrated and are more often reliant on state assistance, creating a net drain on the public purse over their lifetimes. Furthermore, although descendants seem to be doing better than their parents, they still tend to lag behind native Danes. In response to these challenges, recent revisions to Denmark's migration policy have, in particular, involved:

– Stemming the inflow of low skilled immigrants without demonstrated means of supporting themselves.
– Putting more effort into integrating foreigners, including those already in Denmark, helping them first to become self-reliant and ultimately to attain economic outcomes indistinguishable from those of native Danes.
– Easing entry for highly skilled immigrants.

Attracting more highly skilled migrants will be difficult

Denmark's entry policies are now somewhat closer in spirit to the more selective immigration policies already practised by some other OECD countries. But the government can do little more to attract highly skilled foreign workers than remove obstacles to their entry and make it easier for immigrants with foreign qualifications to use their skills. Denmark will remain at a disadvantage in competing to attract such foreigners because of language barriers, compressed wages and high taxes. In principle, these same factors could encourage well-educated Danes to emigrate, although there is little evidence of this happening on a large scale thus far. In any case, what is important for the sustainability of public finances is the net migration balance of highly skilled workers, irrespective of nationality.

Low skilled immigrants find it hard to integrate

Lack of education and skills is a particular problem for a significant proportion of immigrants because low paid, unskilled jobs that migrants might take up in other countries are relatively scarce, once again because of the compressed wage structure and high tax wedges. This lack of suitable jobs makes it harder for these immigrants to take a major step towards full integration. At the same time, it is difficult to envisage such low-wage jobs being created while unemployment benefits and welfare payments remain high and

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collective agreements embody high negotiated minimum wages. Integrating these groups into the labour market may therefore require a combination of greater efforts to improve skills, stronger activation policies and perhaps also time-limited subsidies, all of which have budgetary costs.

Language skills are crucial for integration

Improving educational attainment and local language skills are both key determinants of integration. But it is difficult to obtain Danish qualifications without a solid language base. Good Danish language skills are also more important if the jobs available require a reasonable amount of communication and interaction with others. At the same time, immigrants tend to cluster together, reducing both the need and the opportunities for learning the host language in daily life. Thus, a countervailing insistence that immigrants make every effort to learn the language, matched with provision of suitable facilities, would help to speed integration. Recent reforms have already moved in this direction: the introduction programme for new arrivals and increased emphasis on activation should both help, although such efforts need to be carefully monitored and evaluated to ensure they are delivering cost-efficient results. More consistent application of activation efforts across municipalities, along with wider diffusion of best practices might also help to improve integration outcomes.

Greater efforts should be made to lift the educational attainment of descendants

Special efforts to help improve the opportunities for second-generation immigrants are also necessary in order to avoid perpetuating the disadvantages faced by their parents. The government's increased emphasis on pre-school in Danish for non-native children is appropriate: earlier remedial action as soon as children start slipping behind in the education system would also help. It is still too early to determine how large a labour market gap will remain between descendants and their native counterparts, as most adult descendants are still young. Nevertheless, active promotion of the value of educational achievement for immigrants, backed up with wages that more fully reward human capital, might also prompt ethnic communities to more actively encourage their children to succeed in the Danish education system.

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Ensuring fiscal sustainability has become harder

Labour supply, the budgetary options available in the short and medium term, and ultimately, the sustainability of government finances are intimately related. Denmark is in the enviable position of currently having a structural budget surplus of close to 2 per cent of GDP, although it will have to contain expenditures as intended in order to maintain this position over the coming years. The task of maintaining sustainability is not made easier by the nature of the tax cuts scheduled for 2004 to 2007, the erosion of revenues due to the "tax freeze", and widespread expectations of increasing public service standards.

Tax cuts should focus more tightly on reducing high marginal tax rates

High marginal tax rates on labour income reduce the attractiveness of choosing work over leisure for many employed. The agreed tax cuts, once fully implemented, will mitigate this problem to some extent. They raise the threshold for the intermediate tax rate and provide an earned income tax credit to all workers. Yet the top tax rate remains 63 per cent and still applies to around 40 per cent of full-time workers. Raising significantly the threshold for the top tax rate, or even abolishing that tax bracket altogether would involve immediate budget costs of around 1 per cent of GDP, but these costs would be attenuated in the longer run by the stronger work incentives created for higher-skilled workers. In contrast, the announced tax measures will increase after-tax incomes for all employed by a modest amount, and significantly lower marginal tax rates for around one-third of workers. These measures have a different distributional impact, but are estimated to generate less extra employment for the revenue foregone than could have been achieved by raising the threshold for the top tax rate.

The "tax freeze" limits expenditure growth, but is also a straightjacket

The government's "tax freeze" is designed to constrain public expenditure growth and reduce the overall tax burden. Its terms are very strict, which makes the government's commitment credible and easy to monitor. But it erodes real revenue from property and excise taxes that are set in nominal terms, which – even though the reduction of excise duties and, to a lesser extent, property tax also have some positive indirect impact on the labour supply decision – puts an increasing proportion of the overall tax

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burden onto labour income. Having a rule adjusting nominal taxes in line with inflation while gradually cutting income taxes would have a greater impact on labour supply, and hence on public finances. Likewise, better economic outcomes could be obtained by more loosely interpreting the freeze to allow revenue-neutral tax packages that would shift more of the tax burden from labour income to immobile factors such as property, although this would reduce the transparency of the tax freeze to some extent.

Yet it should work to constrain spending by lower levels of government

A key objective of the tax freeze is to avoid increases in the average income tax rate of local authorities. This in turn implies a stricter expenditure prioritisation by the local authorities, who are responsible for a large proportion of public outlays. Since local authorities set their own income taxes and can thus undermine national objectives to reduce the tax burden on labour, it makes sense to limit their room for manoeuvre. Until this year, the government lacked an effective instrument, since budget agreements negotiated with associations representing local authorities were not binding on individual municipalities or counties. However, it has now added a sanctions mechanism to encourage co-operative behaviour among local authorities. This should ensure that average local tax rates are kept stable, consistent with the "tax freeze", but it remains to be seen how this will work out over time as authorities gain experience in "playing the game". One possible outcome would be that the required coordination to balance tax cuts and tax increases fails to develop, with the result that current variations in local tax rates become locked in. This would effectively negate the rationale for local taxing powers and might call for a different mechanism, for example, the use of tradable tax permits.

Measures to slow growth in public outlays need to be spelt out

Real public consumption has tended to creep upwards over the years, and it could prove difficult to slow the pace in the face of public expectations of higher public service standards in the future. While the tax freeze and the fiscal surplus target together should put greater downward pressure on spending, measures and/or mechanisms to pinpoint where and how spending will be trimmed remain unspecified. Allowing more extensive application of user

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charges to cover part of the cost of publicly supplied services could help to make provision more efficient and responsive. It would also be prudent to reconsider various benefit schemes or to prepare for additional trimming of outlays to ensure fiscal sustainability if the assumed increase in labour supply between now and 2010 fails to materialise. This requires a shift in approach, since Denmark has so far focussed on preparing to meet its future spending obligations by maintaining budget surpluses and reducing public debt, rather than acting directly to reduce those future obligations themselves.

Climate change policy has been placed on a more efficient basis…

Another aspect of sustainable development concerns the environment, where Denmark has set high ambitions, and a stronger focus on cost-effectiveness is now emerging. A number of policies for reducing greenhouse gases have entailed significant costs, but the government's new strategy to counter climate change emphasises using the EU emission trading system to minimise the cost of lowering emissions. Assistance to the wind-power programme is now limited to the government's estimate of the externalities stemming from carbon emissions. Finally, it rules out additional measures in areas where taxation is already high. These steps are welcome. However, the government will need to assess carefully the quantity of emissions allowances that will be granted in order to ensure that Denmark meets its burden-sharing agreement. In addition, policies could still be improved by ensuring that abatement costs are more uniform across the economy, using the allowance price in the EU trading system as a benchmark. Consideration should also be given to finding a way to capture a significant part of the value of such allowances for society as whole, thereby providing the revenue to reduce a number of distorting taxes.

… but further re-orientations of policies are required in other areas

Further efforts in achieving sustainable development in an efficient fashion are also warranted in other areas including:

– To lessen nitrogen concentrations in waterways and coastal areas, economic instruments, such as taxing commercial fertiliser inputs with rebates based on the nitrogen content of output, should be used. Tax rates should be set on the basis of cost-benefit analysis.

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– To deal with waste more efficiently, a more rational and coherent system is needed. This should be based on rigorous assessment of the costs and benefits of alternative disposal methods. The use of economic instruments, such as the packaging tax, should be realistically calibrated according to the corresponding environmental externalities, with an emphasis on the use of incentives rather than regulations.

Summing up

Denmark is weathering the international slowdown reasonably well, and the pace of activity should pick up over the remainder of this year. The underlying macroeconomic situation is sound, although being slightly out of step with the euro area monetary conditions may lead to a risk of future overheating. The announced tax cuts could provide undue stimulus to the economy at the same time, if not balanced by appropriate expenditure restraint. Denmark's main challenges in coming years involve changing incentive structures so as to expand the labour supply. This would raise productive capacity and keep the effective dependency ratio within manageable proportions over the longer term, thereby ensuring fiscal sustainability despite increasing pressure arising from changes in the composition of the population. Current policy settings involve high levels of income support in one form or another to a proportion of recipients who could rely more extensively on their own resources, given a better set of incentives to do so. Specific measures could help to speed up entry into the labour force, provide greater economic rewards for working, and postpone retirement. Special efforts could also make it easier for immigrants to become more self-reliant and integrate more quickly and effectively in the future. In sum, a variety of measures – some already taken and others recommended in this *Survey* – would help the economy to continue on a sustainable path and improve overall outcomes.

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I. Key challenges

The Danish economy has continued to perform well despite the weak international economic climate. The macroeconomic policy framework provides a sound underpinning for a policy stance that aims to guide the country safely through short-term developments by steadfastly focussing on medium-term priorities rather than allowing itself to be blown off course. This steady-as-she-goes approach enables policymakers to concentrate most of their efforts on addressing the longer-term challenges it faces.

The most difficult task will be preparing to meet the economic pressures arising from the ageing population. Denmark is better placed than most OECD countries, since public pensions for the elderly are designed to provide a basic level of income support, and most workers will have private occupational pensions available to draw on in their retirement. Nevertheless, the working-age population is set to start shrinking in just a few years, and the dependency ratio will climb quite steeply. At the same time, there are some public expectations of, and demands for, improvements in the quality of public services, which are often consumed to a significantly greater extent by older people. These two developments risk putting an inevitable squeeze on public finances.

Fortunately, there is already a high degree of awareness of these likely future pressures, and the importance of maintaining fiscal sustainability already features in budget discussions. Medium term projections and policy orientations have a long tradition and for the past several years, the authorities have published longer-term scenarios that spell out the labour supply requirements necessary to deliver a sustainable fiscal position, given assumptions about demands on public expenditures. This is an important first step in demonstrating the gap between current policies and the further policy efforts required. Policies that help to fill the gap in labour supply will not only boost national income, but are also crucial to keeping public finances in good shape.

The rest of this chapter starts with a brief review of the current macroeconomic situation, policies and prospects. It then examines longer-term challenges, starting with prospective demographic developments, their implications for growth, and how these and other pressures will bear on public finances.

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It then examines fiscal sustainability, which is a critical element of Denmark's long-term outlook. Lastly, the chapter considers the objectives of delivering sustainable development concerning the environment.

Macroeconomic situation, policies and prospects

Denmark is in the enviable position of not facing any immediate macroeconomic problems. It is weathering the current international slowdown relatively well, and indeed the global situation has served to ease pressures on capacity that were starting to emerge. This is in large part due to the sound macroeconomic policy framework, so that the economy is able to ride through such slowdowns relatively easily, without the added burden of having to deal with any underlying macroeconomic imbalances:

- The government has been running a steady budget surplus since 1997 (Figure 1) and has established a framework that presents current budget decisions within a long-term context (see below).
- The steady reduction of public debt relative to GDP since the early 1990s and the continued fixed exchange rate regime have helped to reduce interest spreads and pressure on real long-term interest rates, which are now around half the level they were in the mid-1990s, making business investment more attractive and bringing forth significantly higher capital stock growth in recent years.
- Trend labour efficiency growth for the business sector has also risen to rates of around 2½ per cent, and looking through yearly fluctuations, annual private sector productivity growth seems to be stable at around 2 per cent. As a result, the economy has been able to support strong wage growth without significant loss of competitiveness.
- Standardised unemployment is just below the OECD's estimated structural unemployment rate, which is slightly under 5 per cent. There is a very small negative output gap, which is expected to close once again within the next year or so, and inflation is moderate.
- Household balance sheets are solid, with net assets equal to slightly more than annual disposable income at the end of 2002. The household savings rate has returned to around 6 per cent now that all the measures of the Whitsun tax package instituted in 1998 have worked through (see OECD, 2000).
- The current account has remained comfortably in surplus almost every year of the past decade and now stands at around 3 per cent of GDP. At the same time, higher international trade flows relative to GDP indicate that the country has become more internationally oriented in recent years.

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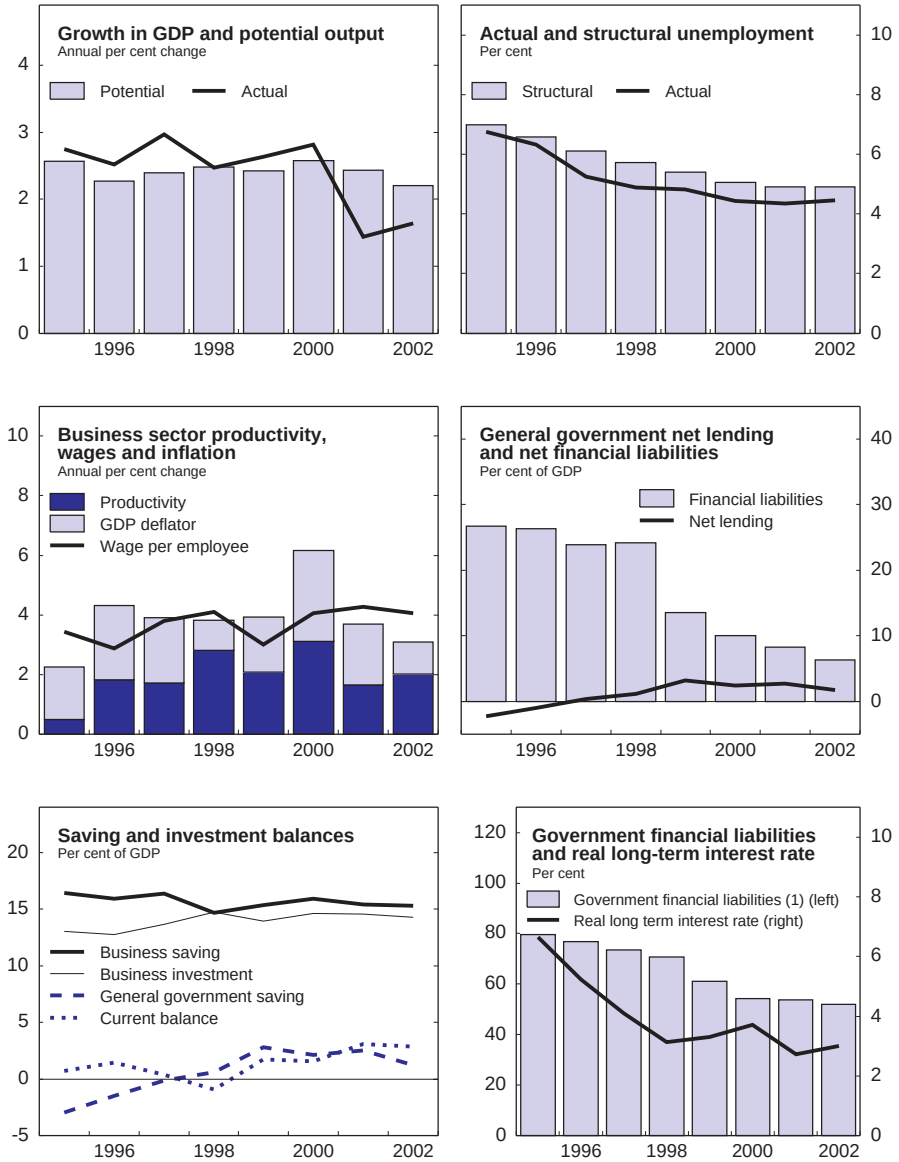
Figure 1. **Indicators of macroeconomic balances**



1. Per cent of GDP.
Source: OECD.

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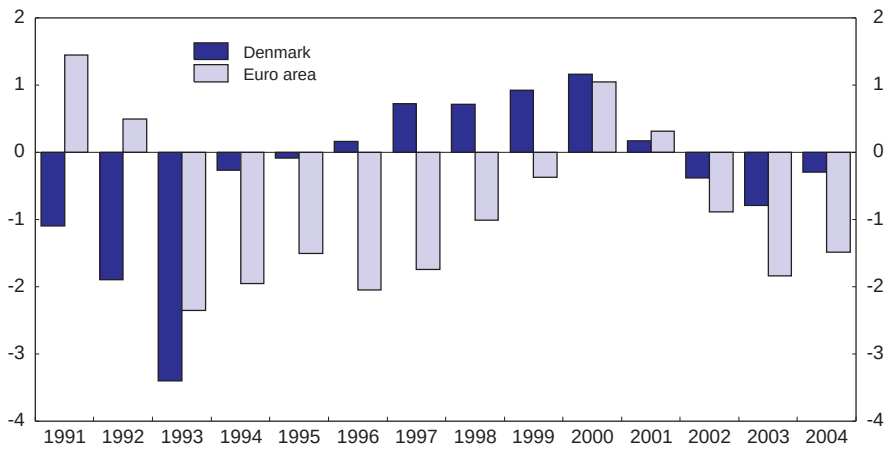
The Danish economy thus remains in a healthy state, despite the recent deceleration in GDP to growth of around 1½ per cent in 2002 and the prospects of a pick-up being postponed until the second half of this year (see Annex I). GDP is nevertheless projected to grow around 2½ per cent in 2004. In this context, the unemployment rate has risen a little, and total hours worked have shrunk. However, some easing of labour market pressures is welcome, since the Danish economy was showing some signs of skilled labour shortages, with associated upward pressures on wages, before the slowdown.

The main macroeconomic focus for policymakers over the next two years or so will be to keep the economy from overheating as the recovery gathers steam and output expands faster than its potential rate of about 2 per cent. Monetary conditions may exacerbate the situation, as Denmark and the euro area may again be at different stages of their economic cycles (Figure 2). Denmark conducts a fixed exchange rate policy *vis-à-vis* the euro area within the EU exchange rate mechanism (ERM2). In normal circumstances, this implies that the Danish National Bank changes interest rates in step with the European Central Bank,[1] and Danish short term rates are currently 0.2 per cent above those of the ECB. The European Central Bank would be expected to set rates appropriate for the euro area, taking into account the negative output gap and developments in inflation in the zone.

Figure 2. **Divergent output gaps**
Per cent of potential GDP



Source: OECD.

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But for Denmark, this could result in slightly looser monetary conditions than desirable and tend to add monetary stimulus to the economy when it is already operating at around its potential.

In this situation, fiscal policy will need to remain fairly neutral, if not biased towards restriction, to avoid exacerbating the risk of excess demand. The Danish authorities have successfully pursued a broadly neutral fiscal policy stance in recent years, following the tightening associated with the Whitsun tax package. Relatively strong automatic stabilisers have played a part in keeping the economy on an even keel, although their interpretation has been made more difficult by volatile tax flows associated with asset prices (see OECD, 2003a). In the present cyclical position it is important to ensure that the announced income tax cuts to be phased in from next year are offset with appropriate restraints on expenditure growth in order to contain the fiscal stimulus to the economy.

The Government's tax freeze (see Chapter IV) means that if the fiscal policy stance does need to be tightened to damp activity, it cannot be done through the typical channel of raising taxes. This places a greater burden on preventing periods of overheating and, if necessary, finding ways of bearing down on public expenditure instead. With decentralised control over outlays, that would be difficult to manage within a short time frame. But given these constraints and the desire of the government to bear down on public expenditure in any case, it makes more sense to concentrate on meeting the medium- and longer-term requirements to raise the labour supply and preserving long-run fiscal sustainability. At this point there are no reasons to expect more than modest variations in Denmark's economic cycle, and the recent easing of labour market pressures does not suggest any need to change this longer-term orientation of policy, or to try to fine-tune the economy.[2]

Longer-term challenges

The fundamental long-term challenge is to ensure fiscal sustainability in the face of demographic and other developments. This section starts by setting out the likely demographic developments and their impact on the size and composition of the working age population and on labour supply. Trends in labour force participation and hours worked, and the possible implications of these for long-term prospects are discussed, since they may exacerbate the pure demographic effects. The likely consequences for growth of GDP per person are then covered, before turning to the elements and mechanisms shaping the future pressures on public finances. Fiscal sustainability and possible risks are then examined, taking into account the current objectives of the government to 2010, set out in its medium-term survey. Alternative strategies that would involve putting more effort into reducing the magnitude of future obligations, in contrast to the Danish strategy of primarily concentrating on ensuring that these obligations can be

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financed are briefly explored. Lastly, the chapter also briefly touches on another ultimate concern, environmental aspects of sustainable development, and ends with some concluding remarks.

Demographic prospects

Almost all OECD countries face significant changes in the size and composition of the population in the years ahead. The common pattern of ageing populations basically stems from the same combination of a baby boom in the early post-war period succeeded by declining fertility rates and increasing life expectancy. In general, the main features of this process are the following (Dang *et al.*, 2001):

- Very modest growth or declines in total population.
- A fall in the working-age population and an increase in the number of elderly, thereby nearly doubling, on average, the old-age dependency ratio (the ratio of elderly to the working-age population).[3]
- A rise in the average age of the working-age population and of elderly.
- A small decline in the ratio of youth to the working-age population.

The impact on growth potential and especially public finances from these developments will be substantial in most countries, given the current institutional settings. Output growth will, other things equal, slow as labour forces shrink. Public expenditure will increase with the number of elderly, while the reduction in labour forces tends to erode tax bases, thus raising concerns about intra- and inter-generational income distribution.[4]

Developments in individual countries will depend not only on the specific demographic and labour force characteristics but also to a varying extent on the size and composition of net immigration, though the levels of immigration that would be required to make the demographic picture outlined above distinctly more favourable are probably not imaginable for most countries (Roseveare *et al.*, 1996). Nevertheless, it will make a significant difference whether immigrants are highly qualified and able to counteract some of the negative effect on labour supply from the ageing of the population or instead arrive with little potential for becoming self-supporting without a major amount of public assistance. Thus, another element in the efforts to address the challenges to public finances is the design of immigration policies and the ability to integrate immigrants quickly and efficiently once they arrive (see Chapter III).

Although Denmark will undergo significant demographic changes this century of the sort mentioned above, the projected shifts are less extreme than in some other OECD countries (Figure 3). However, two distinct changes stand out: a higher proportion of elderly in the population and a higher proportion of immigrants and their descendants, especially from less developed countries (LDCs).

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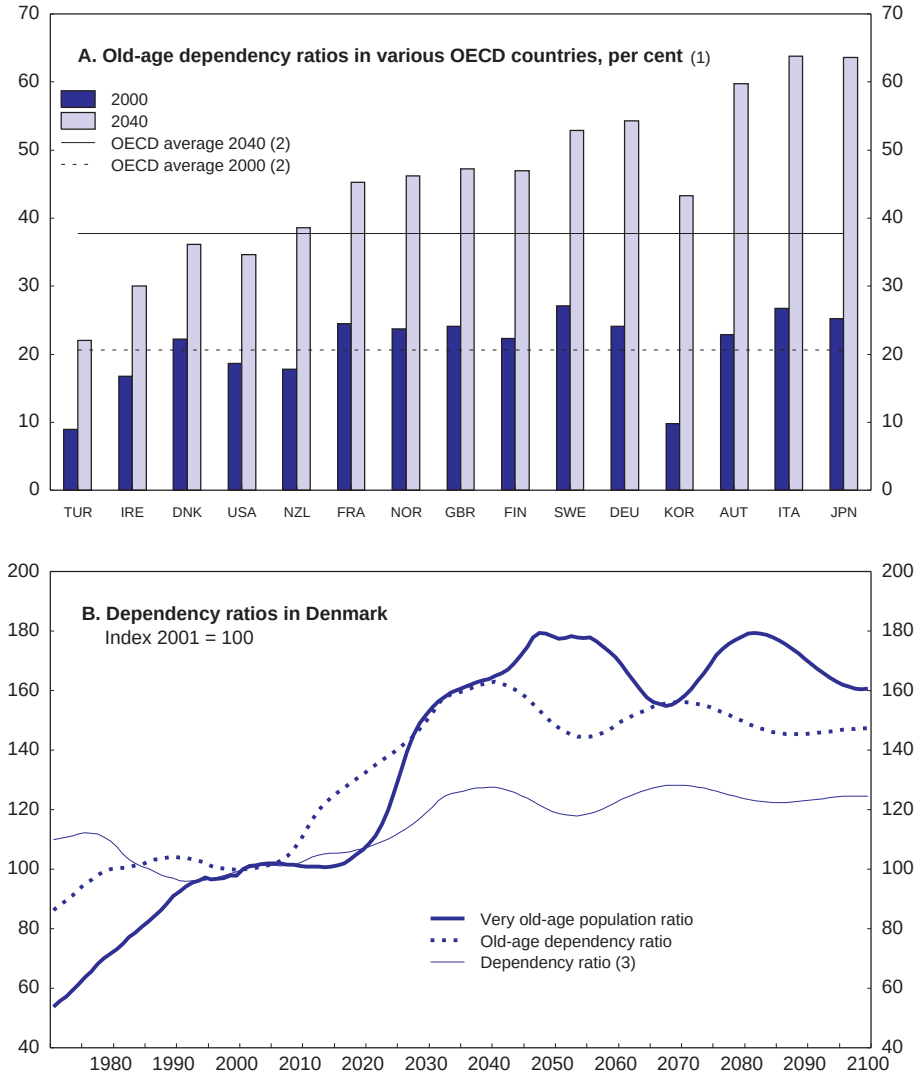
Figure 3. **Dependency ratios**



1. Ranked in ascending order of the difference between 2000 and 2040.
2. Unweighted.
3. Ratio of 0-14 and 65+ year-olds to 15-64 year-olds.

Source: United Nations; DREAM (for Denmark); OECD calculations.

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These prospects are quantified in the DREAM population projection,[5] which forms the basis for most policy planning in Denmark. It assumes a slowly increasing fertility rate, a modest increase in life expectancy during the century and a net immigration of 10 000 persons per year (Annex II). The projection implies the following (Table 1):

- The population aged 0-14 years decreases by 11 per cent from 2000 to 2020, but rises again thereafter.
- The working-age population (15-64 years) falls by 5 per cent from 2000 to 2040, but reverses this decline during the rest of the projection period.
- The number of people aged 65 and above rises more than 50 per cent in the period 2000-40 and remains near that higher level thereafter.

The main "ageing" effect will come after 2010 when the old-age dependency ratio begins a steep increase that will leave it 60 per cent higher in 2040. By the end of the century the old-age dependency ratio would be around 50 per cent higher than today, indicating that the older age structure will become a permanent demographic feature. Furthermore, the share of people above 80 years in the population of people aged 65 or above (the "very old population ratio") almost doubles in the period 2020-50.

Demographic effects on future labour supply

The ageing population and a relative shift towards groups of immigrants and their descendants will both have a stronger negative impact on labour supply than the reduction in the absolute size of the working-age population might suggest. This is because of the impact of age-, gender- and origin-specific labour force participation patterns. Labour force participation in Denmark ranks among the highest in the OECD, especially because with three-quarters of all 15 to 64 year-old women in the labour force, only other Nordic countries have higher female participation rates (OECD, 2002a). But given the current set of policies, there is no reason to expect that age- and gender-specific participation rates would alter spontaneously: shifts in the age structure within the working-age population will thus have almost inevitable consequences for total labour supply, though policy changes could influence their magnitude.

Perhaps the most dramatic impact comes from the sharp drop in participation for both men and women at the age of 60 (Figure 4), once they become entitled to voluntary early retirement benefits (see Chapter II). Thus, as the share of the working-age population above the age of 60 years rises, a substantial drain of the labour force can be expected. Labour market participation is also low for 18-30 year-olds, since a large proportion of Danes start higher education and spend a long time completing it (see Chapter II). However, since the ratio of

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Table 1. **Population projections**[1]

	2000	2020	2040	2100
	1 000 persons			
By age group				
0-14 years	981	872	942	1 074
15-64 years	3 558	3 589	3 384	3 611
65 years and above	790	1 052	1 223	1 182
Total	5 330	5 513	5 548	5 867
	Index 2000 = 100			
0-14 years	100.0	88.9	96.0	109.5
15-64 years	100.0	100.8	95.1	101.5
65 years and above	100.0	133.1	154.7	149.6
Total	100.0	103.4	104.1	110.1
	Proportion of total population, per cent			
0-14 years	18.4	15.8	17.0	18.3
15-64 years	66.8	65.1	61.0	61.5
65 years and above	14.8	19.1	22.0	20.1
	1 000 persons			
By origin				
Immigrants from LDCs	147	323	481	618
Immigrants from MDCs	150	205	240	257
Descendants from LDCs	60	181	327	570
Descendants from MDCs	21	40	62	116
Remaining population	4 952	4 764	4 439	4 305
Total	5 330	5 513	5 548	5 867
	Index 2000 = 100			
Immigrants from LDCs	100.0	219.0	326.6	419.8
Immigrants from MDCs	100.0	137.2	160.7	171.8
Descendants from LDCs	100.0	302.0	546.1	953.6
Descendants from MDCs	100.0	190.0	290.9	547.0
Remaining population	100.0	96.2	89.6	86.9
Total	100.0	103.4	103.4	110.1
	Proportion of total population, per cent			
Immigrants from LDCs	2.8	5.9	8.7	10.5
Immigrants from MDCs	2.8	3.7	4.3	4.4
Descendants from LDCs	1.1	3.3	5.9	9.7
Descendants from MDCs	0.4	0.7	1.1	2.0
Remaining population	92.9	86.4	80.0	73.4

1. Population as of 1 January of the year.
Source: DREAM (2002).

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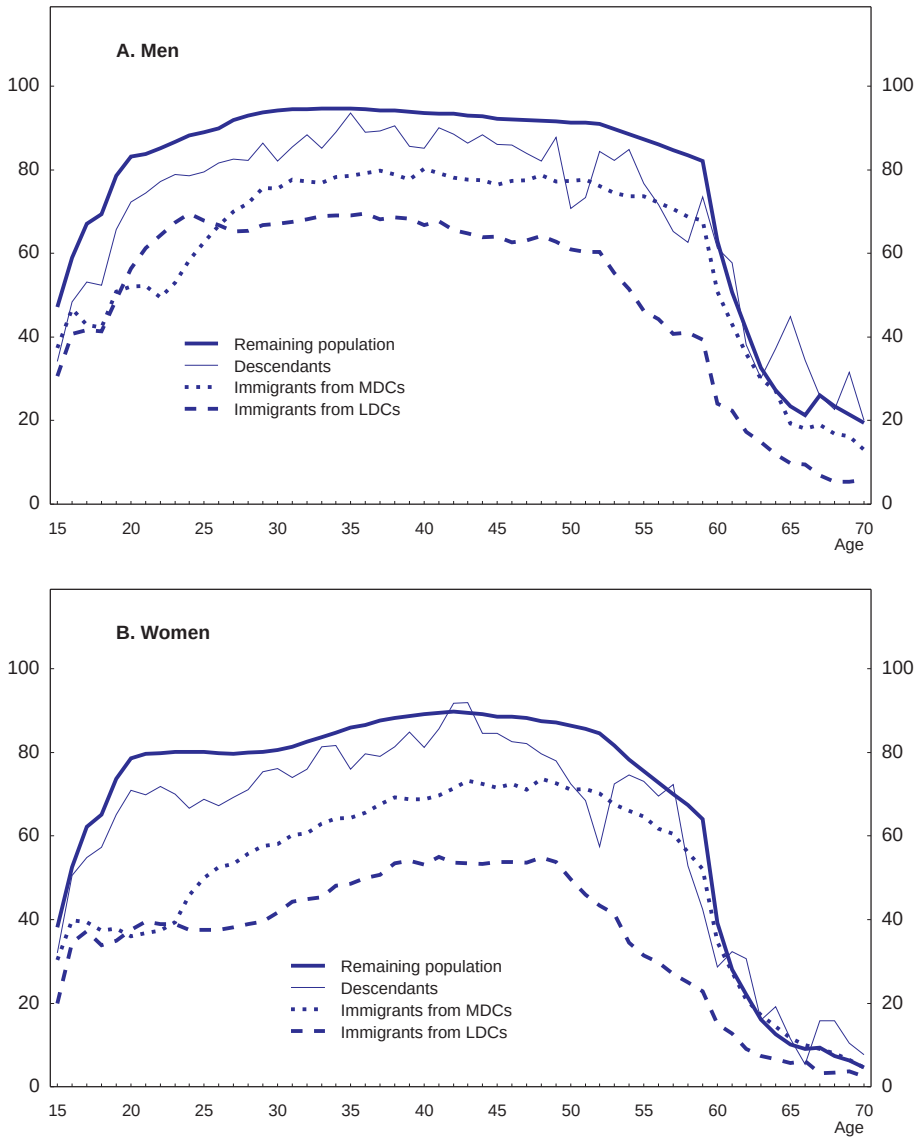
Figure 4. **Participation rates by age, gender and origin**
Per cent, 2001



Source: Ministry of Finance; Statistics Denmark.

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18-30 year-olds to 15-64 year-olds is projected to fluctuate around its current level, it is unlikely to have a significant effect on labour supply assuming present participation patterns continue.

The profiles of age-specific participation rates for immigrants and their descendants are fairly similar to those of the remaining population. For descendants, the levels for both men and women are also close to those of the remaining population, although there are, as yet, relatively few adult descendants from LDCs from which to extrapolate.[6] However, for immigrants the difference in levels is marked. Lower labour market participation among immigrants and descendants seems generally to reflect lower qualifications (see Chapter III). Indeed, controlling for various characteristics, including education, labour market experience and the number of children in the household, immigrants have almost the same participation and employment rates as the remaining population (Det Økonomiske Råd, 2002a).

The ageing effects on labour supply will appear imminently, and the working-age population is expected to start shrinking from around 2010, although the major impact will not emerge until after 2020. By 2040, when the negative impact is largest, ageing effects alone are projected to reduce the labour force cumulatively by around 7 per cent (Figure 5). Of that, 5 percentage points would be due to the fall in the number of 15-64 year-olds, while the rest would come from the effect of assuming cohort-constant participation rates for different age groups and changes in the age structure within the working-age population. However, when changes in the composition of the population stemming from net immigration are included, the reduction would be almost 11 per cent given the present differences in labour participation patterns between groups of various origins. These prospects stand in sharp contrast to the experience of the 1980s and 1990s, when demographic changes made a significant positive contribution to the labour force.

These combined effects from changes in the age and origin composition of the population will act as a severe strain on labour supply in the future. Given present participation and assumed immigration patterns, a steadily smaller part of the population will be producing the country's output and providing for the rest, thus raising concerns about the distribution of burdens between and within generations. The nature of the population changes suggests that policy action addressing the pure population effects will necessarily be limited to the immigration dimension, since fertility and mortality rates are commonly believed to be largely beyond reasonable policy influence. In contrast, policies can make a significant difference to participation within the working-age population and even redefine the boundaries of the working-age population itself.

Trends in labour force participation and hours worked

There is evidence of a slightly increasing aggregate participation rate and longer average hours being worked in the last five to six years. However, long-term

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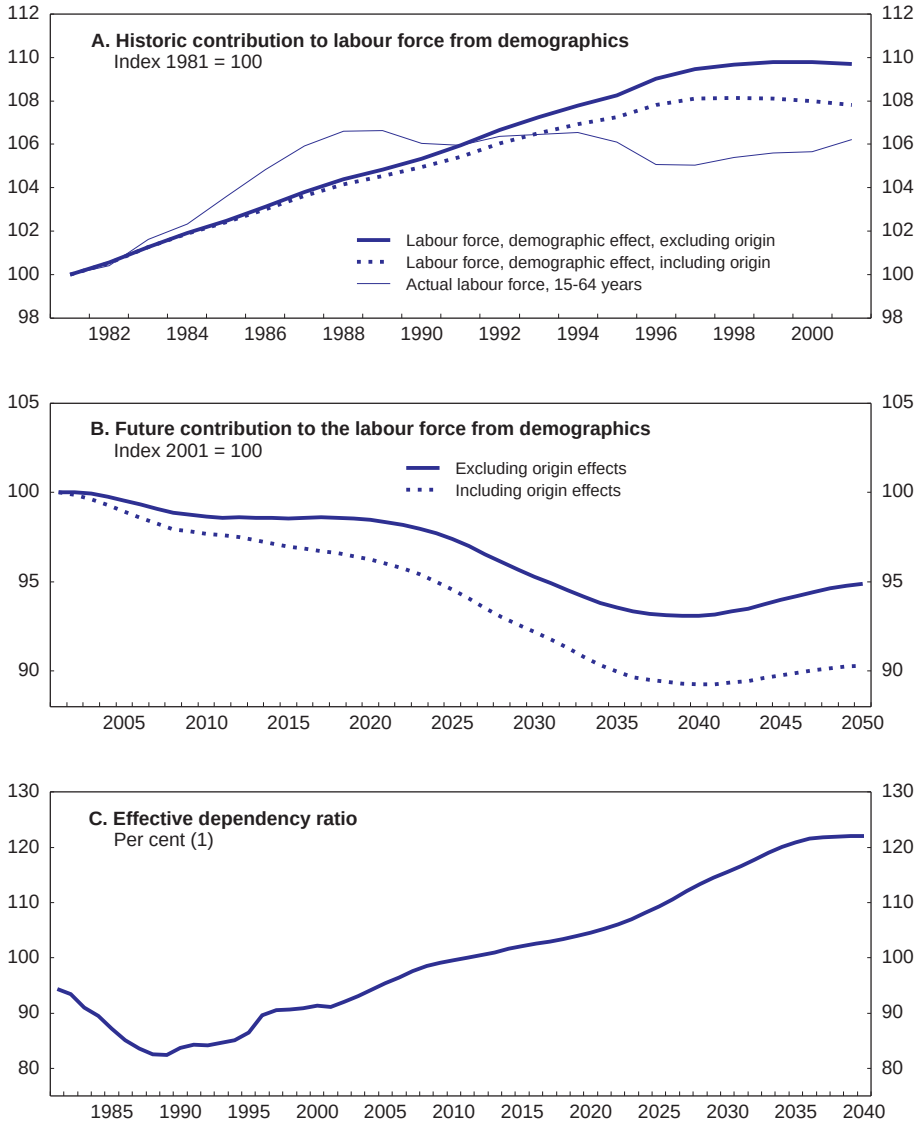
Figure 5. **Demographic impact on the labour force**



1. The ratio of inactive people to those in the labour force.
Source: DREAM; Statistics Denmark; OECD calculations.

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trends in both labour force participation and hours worked suggest that assuming even constant age-specific participation rates in the future may prove to be overly optimistic. While the demographics have been feeding into higher labour supply until now, changes in participation rates have in general worked in the opposite direction. Aggregate labour participation has declined by 3 percentage points since the end of the 1980s, despite the positive demographics and a favourable business cycle position since 1994, reflecting a continued decrease in male participation and a break in the trend towards higher female participation (Figure 6).

The changes in aggregate participation rates reflect not only demographic effects but also substantial shifts in age-specific participation rates in the past two decades. The developments are most striking for those aged 60 years and older, reflecting the tendency to earlier retirement for both men and women, and among the under-30 year-olds, who are spending more time in education. But rates have even fallen for men between 30 and 50 years of age, partly reflecting easier terms for entitlement to early retirement (disability) pension and the introduction of different paid leave schemes but also the inflow of new immigrants with lower participation rates.[7]

If the long-term trends towards lower age-specific participation rates for these age groups were to continue, they could significantly erode future labour supply, for example:

- An extension of the tendency to earlier retirement for 60-64 year-olds, reflected in steadily declining participation rates for this age-group in the period until 2020, would reduce the number of people in the labour force by almost 1½ per cent.
- If the proportion of young people in tertiary education increases further, and participation rates for 18-30 year-olds continue to fall with the same speed, the labour force would shrink by a further 2¾ per cent. (However, this estimate does not take into account a higher probability of labour participation for highly educated people in later stages of their lives.)

While these calculations are only illustrative, they do emphasise the possible impact of further facilitating temporary or permanent withdrawal from the labour force. They also suggest that restricting the existing opportunities for withdrawal on public income support could be necessary to maintain the slight increase in aggregate participation in the last few years. In fact, the recent experience is almost entirely attributable to rising participation rates among 50-59 year-olds following the closure of entry to the transitional early retirement benefit scheme in 1996. Thus, in order to counteract the effect of changes in the size and composition of the population, policy measures would need to have a substantial impact on behavioural patterns and therefore probably need to restrict what might be broadly perceived as fundamental rights in the welfare state.

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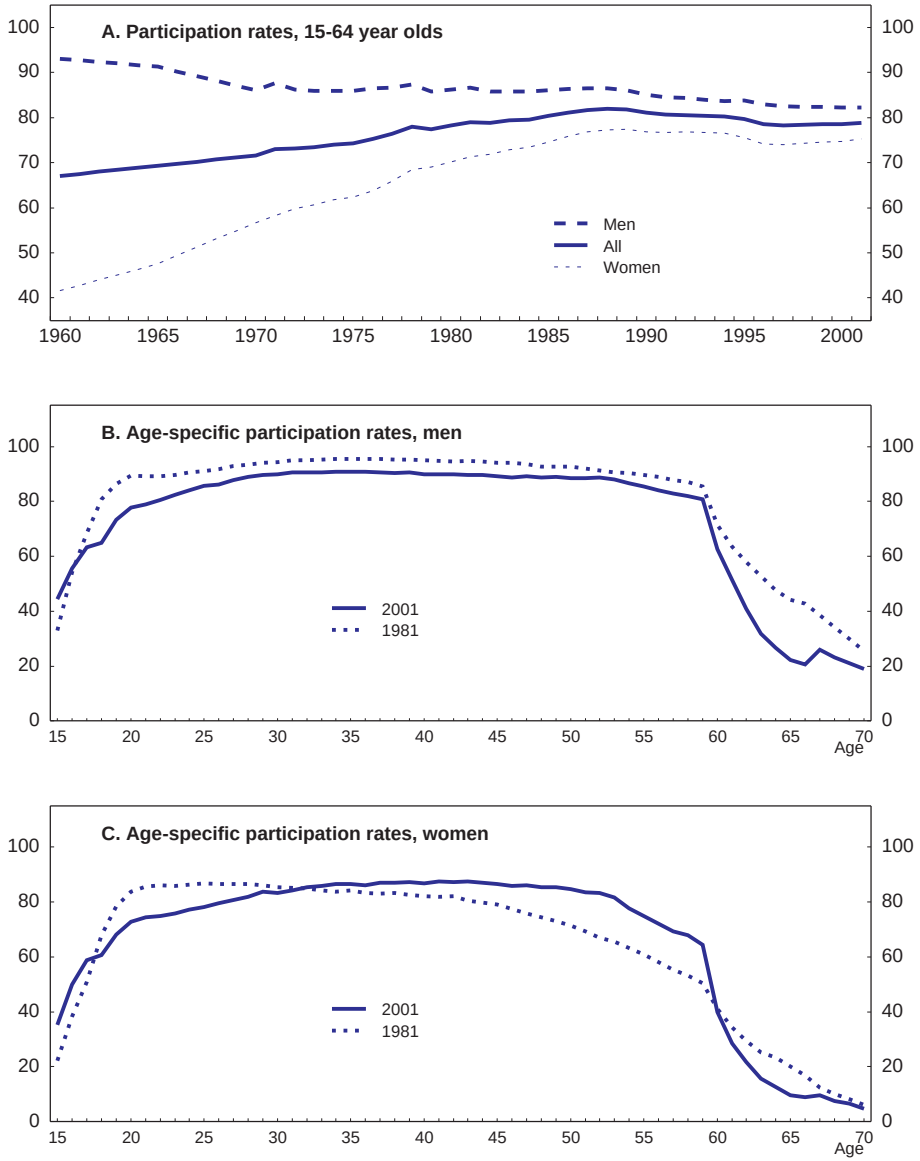
Figure 6. **Participation rates over time**

Per cent



Source: DREAM; Statistics Denmark; OECD calculations.

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Working hours are another element of total labour resources applied to producing output. As well as the fall in labour market participation, average annual working hours per employee have fallen markedly (Figure 7). This is primarily a result of extensions of annual vacation, reductions of working hours agreed in collective bargaining and a higher proportion of part-time jobs in the early phases of female entry into the labour market.[8] Since 1950, the number of vacation weeks has been increased from two to five, and normal working hours per week have been reduced from 48 to 37 hours. In 1998, the government introduced additional "holiday" days, primarily for parents, when it intervened in the widespread dispute affecting the private sector. In subsequent collective agreements these days were extended to other employees, as well as other parts of the labour market, and the number of days per person has been increased, so that a total of five extra "holiday" days will be at most employees' disposal in 2004. That would constitute a decrease in annual working hours of 2 per cent if employees choose to use all the extra days, rather than taking them in the form of extra pay.

The decrease in actual average working hours per year is less than the official reductions.[9] This is partly explained by a falling frequency of part-time jobs among women, as women's integration into the labour market has deepened and the gap between male and female participation has gradually narrowed. Lower marginal tax rates introduced in tax reforms since the mid-1980s may also have contributed to curbing the fall in average working hours. Actual average working hours have in fact increased since 1995, mainly due to a fall in the frequency of part-time employment, although some business-cycle effects might also be at work. Nevertheless, in 2001 women were still 50 per cent more likely to be working part-time than men, suggesting that at least some potential for higher average annual working hours in the future may still exist.

However, that potential might be offset by the changing composition of the labour force. On average, older people work fewer hours per week than others (Figure 8). If this pattern holds steady in the future, a higher proportion of the labour force will work fewer hours and the annual average will decrease. Furthermore, high marginal taxes on earned income might still exert a downward pressure on working hours for persons in full-time jobs (see Chapter IV), and the general increase in prosperity from productivity growth would probably partly be realised through reduced working hours (increased leisure). This is confirmed by a recent Gallup opinion poll, which indicates that almost 50 per cent of wage earners would prefer more leisure to higher wages or employer pension contributions, and that these preferences increase with annual household income and age.

Furthermore, the tendency to more flexible working time could facilitate a substitution towards more leisure, although more flexibility in organising working hours might at the same time make it easier for households to take on additional work. Some recent collective agreements provide the scope to choose between

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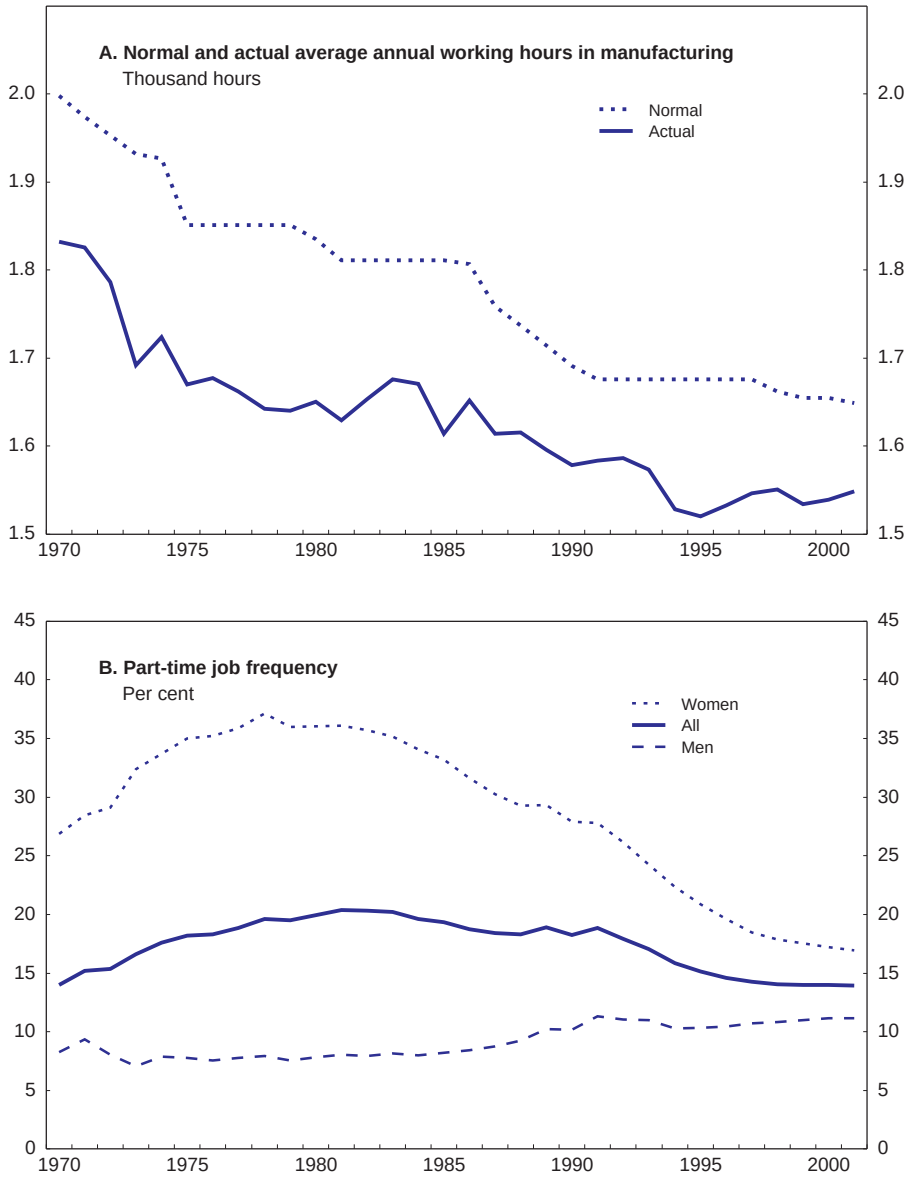
Figure 7. **Evolution of working time**



Source: Ministry of Finance; Statistics Denmark.

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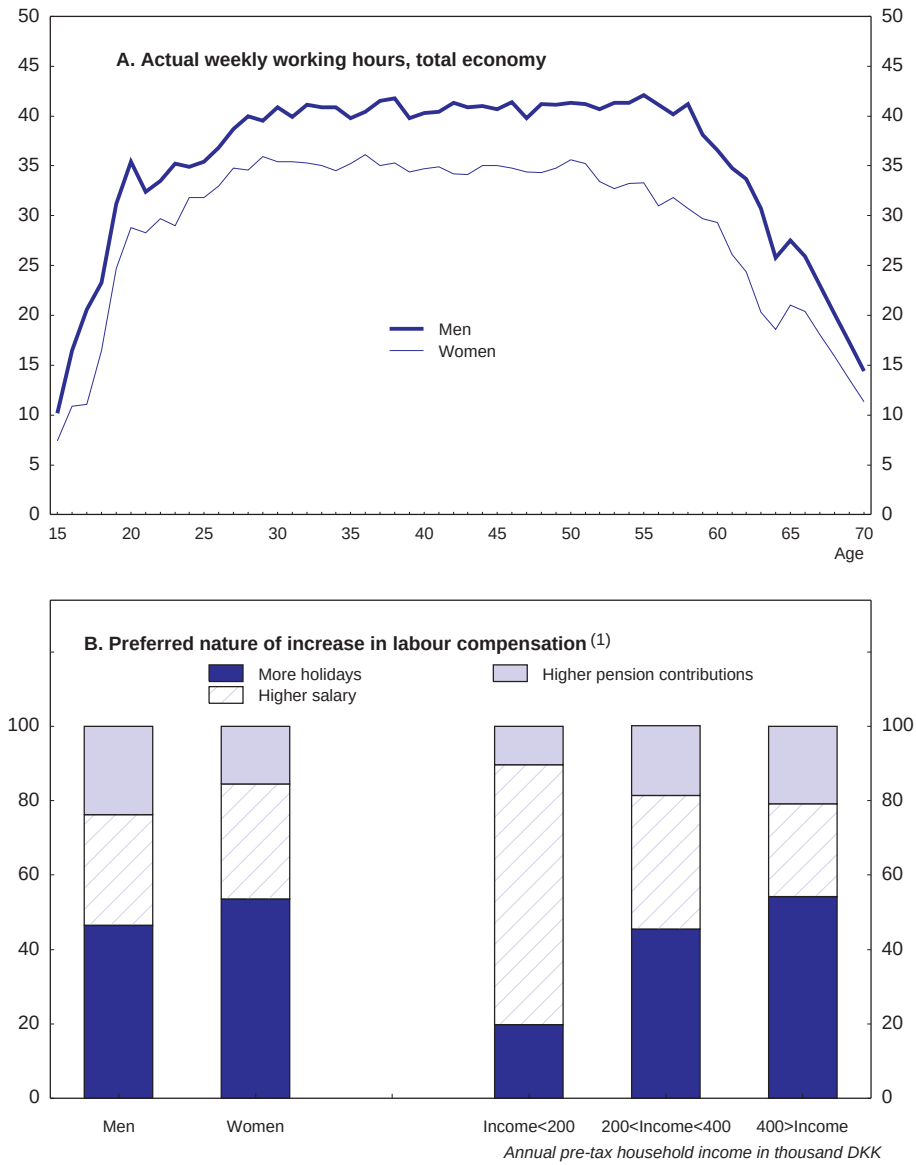
Figure 8. **Weekly working hours and preference for leisure**



1. The share of people naming the particular element (per cent).
Source: Ministry of Finance; Statistics Denmark; Gallup and Ugebrevet A4.

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wages and leisure within a certain envelope, and there is a risk that such arrangements will spread across the labour market, as most Danes seem to prefer this type of free choice (Ugebrevet A4, 2003), thus potentially enlarging the decrease in working hours.

Overall, predictions of future changes in working hours are quite uncertain, although the risk of further reductions seems to dominate. The demographic effect on its own is likely to add to the fall in total working hours coming from lower average participation rates and fewer people of working age. If all these forces were to act together, total working hours might be almost 12 per cent lower in 2040 than today, while continued long-term trends in labour market participation and official working hours (if reflected in actual working hours) could bring a total reduction of 19 per cent (Table 2). In contrast, the fall in total hours worked in the last 40 years can be estimated to around 10 per cent.

Table 2. **Possible change in total hours worked from demographics and labour market trends**

Per cent

	Changes from 2001		
	2010	2020	2040
Total demographic impact on total hours worked, basic scenario	–3.4	–5.0	–11.8
of which:			
Size of working age population	1.3	0.7	–5.0
Composition effects from participation rates (age, gender and origin)	–3.6	–4.5	–5.8
Composition effects from working hours (age and gender)	–1.1	–1.3	–1.0
Additional change with prolonged trends			
Trend towards earlier retirement continues until 2020	–0.8	–1.4	–1.4
Trend towards longer education continues until 2020	–1.2	–2.7	–2.3
Trend towards lower official working hours per year continues until 2020	–1.5	–3.5	–3.5

Source: OECD calculations.

Growth prospects

In the last 40 years, inflow to the labour force has contributed to the expansion of growth potential in Denmark. That is likely to reverse in the future, when demographic changes will put downward pressure on effective labour supply. In addition to the direct impact on growth through labour force developments, the demographic changes may also affect average labour productivity due to differences in productivity between population groups (distinguished by gender, age and origin). But given current evidence, it is unclear what the direction

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and magnitude might be.[10] However, since increasing the proportion of immigrants and older Danes in the labour force involves bringing those with lower average work capacity into work, this could exert some downward pressure on aggregate productivity growth.

While demographic changes are likely to reduce growth in GDP per capita, Danes are unlikely to face a reduction of living standards, since on present trends labour productivity growth will probably be more than sufficient to offset the diminution in labour supply and a possible dilution of average productivity. Based on the basic scenario for the development in total hours worked presented above and a trend growth rate in private-sector (hourly) labour productivity of 2¼ per cent per year, average annual growth in GDP per capita can be estimated at around 1½ per cent in the period up to 2040. However, that is around ¼ percentage point lower than the average for the last 20 years, and public expectations of increasing welfare will have to adjust to these prospects.

These estimates do not take into account a possible relative shift of employment towards the public sector, which might further reduce measured growth in GDP, as national accounts do not record productivity growth in the public sector. Nor is any feedback on growth from developments in public finances included. If taxes on work were raised to curb deteriorating tax revenue and increasing public expenditure due to population changes (see below), choosing leisure would be even more attractive, and a vicious circle could then result. While these effects could have a visible impact on GDP per capita, they also illustrate how these demographic developments present a major challenge for public finances.

Elements and mechanisms shaping the pressure on public finances

The general government primary balance will deteriorate over the next 40 years due to the demographic changes. There are three major effects shaping the developments in expenditure and tax revenue:

- *Public consumption* as a per cent of GDP will increase rapidly after 2020 (Figure 9), primarily reflecting the impact of ageing on individual public consumption, which is about two-thirds of total public consumption.[11] It includes health, childcare and elderly care as well as education, and is therefore highly dependent on the age composition of the population.
- *Public transfers* will increase relative to GDP, as a larger proportion of the population qualify for public income support. This is partly offset by the effect of means-testing certain benefits, primarily the old-age pension supplement and rent allowance, as both the number of people with private pension savings and the average level of these savings will increase.

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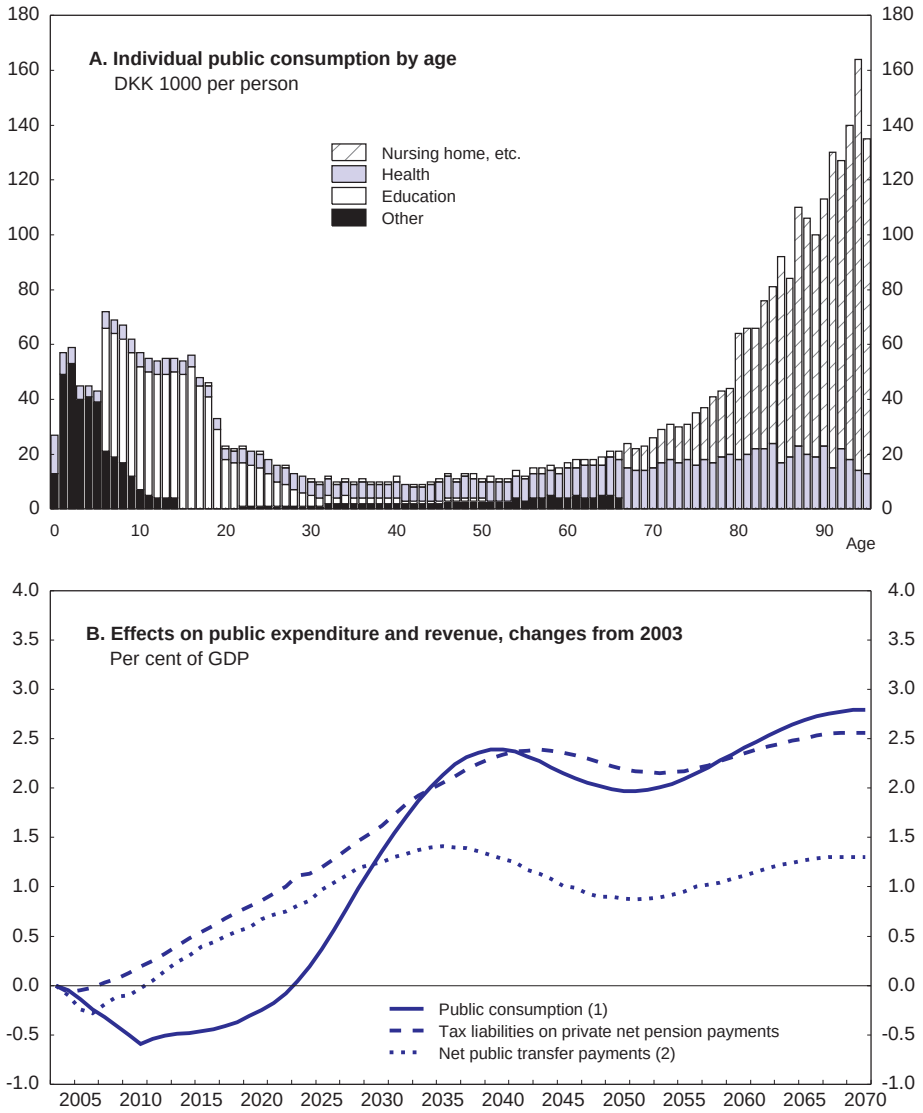
Figure 9. **Elements shaping the pressure on public finances**



1. Including the announced lower growth in real public consumption to offset the government's proposed tax cut.
2. Corrected for taxes and means testing.

Source: Ministry of Finance; OECD calculations.

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– *Revenue from taxable pension payments* will rise substantially, reflecting a shift from net pension *contributions* of almost 3 per cent of GDP in 2003 to net pension *payments* of around 3½ per cent of GDP in 2040 as the private labour market pension schemes mature. In general, pension contributions are exempt from ordinary income tax, whereas pension payments are taxed. The tax revenue foregone in the period before pension systems mature leads to a higher recorded general government net debt, but at the same time an implicit asset is built up in the form of deferred taxes.

Overall, the projected changes in expenditure and tax revenue imply a deterioration in the primary budget balance by around 3¼ per cent of GDP in 2040 (Table 3), notwithstanding an estimated boost of 2¼ percentage points from taxes on private pensions. Also, pressure on public finances is eased by the assumption of a substantial increase in employment in the period up to 2010 in accordance with the government's medium-term objectives (see Annex III). However, the government's tax strategy (*i.e.* the tax freeze and tax cuts to be phased in between 2004 and 2007) reduces tax revenue by close to 1½ per cent of GDP in subsequent years. In addition, the government has been very prudent in assuming that revenue from extraction of oil and gas in the North Sea will gradually diminish to zero, reducing overall tax revenue by almost ¾ per cent of GDP in 2040. Furthermore, continued fiscal surpluses and associated reductions in the public net debt will gradually ease the interest burden.

Table 3. **Central and local government primary balance**[1]
Change from 2003, per cent of GDP

	2010	2040	2070
Expenditure	**–0.6**	**3.7**	**4.1**
Consumption[2]	–0.6	2.4	2.8
Transfer payments after tax	0.0	1.7	1.7
Means testing of public transfers	0.0	–0.4	–0.5
Revenue	**–1.4**	**0.3**	**0.5**
Tax on private pension savings, net	0.2	2.3	2.6
Revenue from North Sea oil extraction	–0.3	–0.7	–0.7
Lower tax from the tax freeze until 2010	–0.7	–0.7	–0.7
Lower tax on earned income from 2004[2]	–0.7	–0.7	–0.7
Impact on primary balance	**–0.8**	**–3.3**	**–3.5**

1. These changes are calculated net of taxes on public transfer expenditure and excluding net interest payments and payments from the labour market supplementary pension fund. Thus, the recorded increases in expenditure and tax revenue in per cent of GDP in national accounts would be somewhat higher.
2. Including the agreed tax cuts from 2004 and offsetting reductions in real public consumption.

Source: Finansministeriet (2002a and 2003); OECD calculations.

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Fiscal sustainability

The projected deterioration of the primary budget balance raises the question whether current fiscal policy is sustainable, *i.e.* whether the government is able to meet its future net obligations (in present value terms), given present public service standards.[12] If this were not the case, fiscal policy sooner or later would have to be tightened. In Denmark, this obligation is generally expressed in terms of the required change in the general government structural balance to ensure that the inter-temporal budget constraint is exactly met. Based on the projections of public expenditure and tax revenue presented above, fiscal policy is sustainable according to the method applied by the Danish authorities, as the required change in the general government fiscal balance is –0.1 per cent of GDP in 2003, *i.e.* the current structural budget balance is actually slightly higher than is needed (Table 4).

Table 4. **Long-term requirements for general government finances**
Per cent of GDP

	2003
Annual constant contribution from 2003 to future net obligations:	
1) *Central and local governments' primary expenditure*	1.7
Consumption[1]	1.0
Transfer payments after tax	0.9
Means testing of public transfers	–0.2
2) *Central and local governments' primary revenue*	0.4
Tax on pension savings, net	–1.4
Revenue from North Sea extraction	0.6
Lower tax from the tax freeze until 2010	0.6
Lower tax on earned income from 2004[1]	0.6
3) *Interest burden on central and local governments' net debt*[2]	1.0
4) Required surplus on central and local governments' structural primary balance = (1) + (2) + (3)	**3.2**
5) Correction for central and local governments' structural net interest payments and the structural balance of public funds	1.4
6) Required surplus on general government balance = (4) – (5)	**1.8**
7) General government structural fiscal balance	1.9
Required change in general government fiscal balance = (6) – (7)	**–0.1**

1. Including the agreed tax cuts from 2004 and offsetting reductions in real public consumption.
2. The interest burden is equal to the part of interest payments that is not eroded by GDP and, therefore, requires financing to prevent the debt-to-GDP ratio from rising.

Source: Finansministeriet (2002a; 2002b; and 2003); OECD calculations.

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These calculations are based on a number of specific policy assumptions. An alternative demographic scenario, which does not include the government's medium-term objectives and assumes constant real public service standards[13] until 2010, would indicate a required increase in the general government surplus of 0.1 percentage point. However, if nominal public service standards are assumed to be indexed to wages (which would imply a small trend increase in real public service standards), then the structural balance today would need to be 1.1 percentage points of GDP higher for fiscal policy to be sustainable.

A number of other studies have evaluated the sustainability of Denmark's public finances, and in general they suggest that higher structural surpluses are required. DREAM finds that either a permanent reduction of collective public consumption of 1.6 per cent of GDP or an increase in the ordinary state tax of 3.9 percentage points is required if nominal public service standards are wage-indexed (Table 5). The Economic Council suggests that the ordinary state tax

Table 5. **Indicators of fiscal sustainability in various studies**

	Required permanent change in:		
	General government balance	Public consumption	Ordinary state tax
	Per cent of GDP		Percentage point
Ministry of Finance			
Policy scenario, medium-term objectives included until 2010	0.1	–	–
Demographic scenario, wage-indexed nominal service standards (public services per user)	–1.1	–	–
Demographic scenario, unchanged real service standards until 2010, wage-indexed service standards thereafter	–0.1	–	–
DREAM			
Demographic scenario, wage-indexed nominal public service standards	–	–1.6	3.9
The Economic Council			
Main scenario, excluding tax freeze	–	–0.4	1.1
Main scenario, excluding tax freeze but with continued trend towards more people on VERB[1]			2.2
Main scenario, including tax freeze			2.4
Demographic scenario, excluding tax freeze	–	–1.9	5.1
Demographic scenario, including tax freeze			6.4

1. Voluntary early retirement benefit.
Source: Finansministeriet (2002a); DREAM (2003); Det Økonomiske Råd (2002a).

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would need to increase by 2.4 percentage points if the government's tax freeze for the period 2003-10 is taken into account, and 1.1 percentage points if it is not. These different assessments are basically the result of different assumptions about future labour market developments, whether or not various policy measures are included, and the assumptions embodied in the projection of public consumption. The variety of results demonstrates that the underlying assumptions play an important role in the assessment of fiscal sustainability.[14] The DREAM projection, which could be interpreted as a neutral policy scenario, indicates that some strengthening of fiscal policy is needed if the present institutional setting and behavioural patterns are unchanged.

Nevertheless, Denmark seems well prepared for the ageing of its population when compared with other OECD countries. A number of studies suggest that substantially greater strengthening of public finances would be needed to counteract the projected increases in age-related net spending in most countries (see, for instance, Dang *et al.*, 2001 and Frederiksen, 2003). There are several reasons why prospects seem more favourable for Denmark than for most other OECD countries:

- Denmark has a higher structural primary surplus. The required improvement in public finances is highly sensitive to the initial position, and minor changes in the primary surplus can lead to substantial differences in the future debt-to-GDP ratio. However, if appropriate corrections are not made, high volatility in capital gains taxes, in particular, could alter the assessment of required fiscal consolidation from year to year. Thus, some caution is warranted when interpreting the quantitative estimates for different countries.
- The public debt-to-GDP ratio is relatively low in Denmark. Thus, the interest burden – and therefore the required primary surplus to service the debt – is correspondingly lower.
- The expected future increase in net expenditure is lower in Denmark. Demographic changes are more moderate, public old-age pensions are modest, and the deferred taxes on private pension savings will increase tax revenue in the future.

However, the favourable initial position largely reflects a high employment ratio in Denmark. Thus, the potential for improving fiscal balances through higher labour market participation and lower unemployment is more limited than in other countries. In addition, the projected increase in life expectancy is lower than in most other EU countries, suggesting that the budgetary impact from population ageing could be underestimated.

Risks to fiscal sustainability

As the discussion above has indicated, the assessment of fiscal sustainability is sensitive to changes in certain basic assumptions. However, public finances in

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Table 6. **Effects on fiscal sustainability of various changes to assumptions**

	Per cent of GDP
Average life expectancy increases by one additional year	–0.2
Average life expectancy increases to the average EU level in 2050	–0.7
Average life expectancy remains at the current level	0.6
Immigration from LDCs per year rises by 5 000 people	–0.1
Immigration from MDCs per year rises by 5 000 people	0.1
Average retirement age increases by half a year	0.3
Medium-term employment objective is not achieved	–1.7
Implemented policy measures have no effect	–0.7
Not yet specified measures are not implemented or have no effect	–1.0
Fall in average working hours by 2¼ per cent	–0.7

Source: Finansministeriet (2002a).

Denmark are affected mainly by changes in the dependency ratio, while the size of the population itself has very little impact (Finansministeriet, 2002a).[15] Thus, changes in fertility or the extent of net immigration will not matter much for sustainability. But changes in life expectancy, average retirement age and integration of immigrants on the labour market could have a substantial impact (Table 6).

This also illustrates how the fiscal sustainability indicator could deteriorate, if the structural improvements on the labour market embodied in the estimations from the Ministry of Finance and the Economic Council are not achieved, and fiscal policy is not adjusted to compensate. The Ministry of Finance's estimates assume an increase in structural employment of around 2 per cent by 2010 brought about by additional policy measures yet to be implemented. If that rise is not realised, fiscal policy would need to be strengthened by 1 per cent of GDP. Therefore the government will put at risk its medium-term fiscal strategy if effective measures to boost labour supply are not forthcoming in the near future. It should be noted that, although weak employment growth in the last two years has not brought the 2010 employment objective any closer, it has not led to a deterioration of public finances that would limit the fiscal room to manoeuvre in the medium term. However, to the extent that recent employment developments reflect a structural problem in the labour market as well as cyclical weakness, it could indicate greater difficulties in raising employment in coming years.

Alternative approaches to ensure sustainable public finances

The Danish strategy of preparing for the future pressure on public finances has until now been predominantly focused on substantial savings in the public sector to reduce the debt and net interest payments before the major impact from the changes in the composition of the population emerge. Some

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changes to the public benefit system to increase the incentive for later retirement have also been implemented, whereas benefit levels have remained more or less unchanged and public service standards have improved somewhat.

The debt reduction strategy implies that 2-3 per cent of GDP is being set aside on an annual basis to finance the increase in net expenditure from demographic changes compared with a scenario of a stable debt-to-GDP ratio. Reducing net debt and turning it into a net asset within the next 25 years places a relatively large part of the financing burden on current generations of working age. The pure debt-reduction strategy aims only at financing the fiscal burden, not reducing it. There are, however, alternatives to this strategy. Other countries have pursued more rigorous policies to reduce future obligations, focusing on entitlement ages and adjustment mechanisms of benefits (Dang *et al.*, 2001):

- The age at which entitlement to public retirement benefits is obtained has been, or will be, gradually raised in a number of countries.
- Temporary or permanent indexing of pension rates to price movements, or a mix of prices and wages, has been implemented in various countries.
- A progressive shift to defined-contribution public pension systems, thereby taking into account changes in expected lifetimes, has been effected in Italy and Sweden.

These kinds of measures reduce the pressure from population ageing, and implicitly place a larger burden on future generations through the reduction of public standards or offsetting increases in life expectancy. Indeed, if life expectancy in Denmark does not increase from its current level, or the average time spent drawing public old-age pension is held constant, the required general government fiscal surplus would be ¼ to ½ percentage point of GDP lower in 2003. While the distribution of the burden between and within current and future generations is basically a political choice, the country might be forced to take a closer look at the current rules and mechanisms to find ways of reducing future net obligations should it fail to achieve the needed increase in employment. The rise in per capita incomes following from productivity growth would still ensure higher living standards to future generations, even if they were to bear a larger part of the burdens. However, some measures, such as higher taxes, would effect labour supply incentives and thereby growth prospects negatively.

Delivering sustainable development

Another challenge facing Denmark is to ensure that economic growth is sustainable and appropriately balanced with environmental and social outcomes. Denmark has been a leader in championing sustainable development, not only in the domestic context, but also at the global level (see Chapter V). Its emphasis on fiscal sustainability is discussed above. Despite some recent scaling-back,

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Denmark provides development assistance to poorer countries at a level exceeding that recommended by the United Nations, which is attained by relatively few countries (OECD, 2003c).

Denmark has also taken a lead on a range of environmental issues (OECD, 2000). Climate change is one of the major channels by which airborne emissions of gases and particles can affect both current and future well-being, and the importance of reducing these emissions in the most cost-effective way is well established. Denmark agreed a major reduction of emissions under the EU burden-sharing arrangements agreed within the overall framework of the Kyoto Protocol, but the difficult task is to not only deliver on these commitments, but to do so in the least-cost way. Policy efforts have brought the required emissions reduction within reach, but some instruments chosen in the past have been unnecessarily costly.

Domestic environmental management has also been characterised by ambitious targets but sometimes poorly co-ordinated measures to reach them, and in some cases insufficient consideration of the costs relative to the benefits of different approaches. Even where economic instruments have been employed, they have been undermined by overlapping directives and special treatment and exemptions for certain groups. Water pollution through discharge of nutrients into groundwater, streams and lakes are one example; another is waste collection and treatment. A more coherent approach in each area and also across the board would help Denmark to achieve a better balance of economic, social and environmental policies in order to deliver sustainable development.

Concluding remarks

Denmark is in the very fortunate position of having established a long track record of macroeconomic stability, which leaves it well placed to cope with short-term fluctuations in activity, while allowing it to concentrate its efforts on enhancing its medium- and long-term prospects. The most important medium-term priority is to find ways of lifting total labour supply. The above analysis points to the areas where policy action could effectively boost participation rates. The policy options to achieve this are examined in detail in Chapters II and III. The government's medium-term objectives and their impact on the sustainability of public finances are examined in Chapter IV, as is the role of the tax system in promoting or discouraging labour supply. Some aspects of sustainable development are covered in Chapter V.

The Danish authorities have a clear understanding of the issues, and progress has been made on a number of fronts since the last *Survey*, although no action has been taken yet in a number of other areas where policy changes were recommended (see Annex IV). Perhaps the biggest challenge overall may be changing public expectations about the future importance of work *versus* leisure in providing the increases in national output necessary to maintain a viable and sustainable welfare state.

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II. Raising labour supply for the longer term

Boosting the proportion of people in the workforce would help Denmark to reduce the effective dependency ratio and thereby limit the degree of income redistribution necessary to attain the desired minimum standard of living for everyone as the population ages. Chapter I illustrated the impact on long-term public finances of raising age-specific participation rates. This chapter focuses on sections of the working age population where participation and employment rates could be raised and considers the measures that could be applied to make work more economically attractive. The special situation of immigrants and their integration into the labour market is discussed in the next chapter.

The government has already put in place the first stage of its strategy to increase employment, with the "More People in Work" package announced in the autumn of last year, and signalled its intention to propose other measures to expand the labour supply. This chapter first examines the measures announced in the package and considers areas where it could usefully be extended. The discussion then turns to the two ends of the working-age distribution, the relatively late entry of young people into the workforce and the early exit to retirement, where policies could be adjusted to sharpen the incentives to spend more years in work. Finally, the chapter considers some other dimensions that would boost the total hours worked each year, notably by reducing temporary absences.

More People in Work

Although Denmark has a high employment rate compared with other European countries, there are still people on the fringes of the labour market. Indeed, it actually has marginally fewer prime-age men[16] in work than on average elsewhere in Europe (Table 7), while almost all of the 14 per cent of that age group who are not working are instead drawing some form of temporary or permanent public income support.[17] This suggests that with a stronger mix of economic incentives and assistance, more of those not working could be coaxed back into employment, even if they are not currently active job seekers. The government's 2002 "More People in Work" initiative endeavours to do this.

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Table 7. **Labour force status of prime-age men**
Percentage of men aged 25 to 54, 2000

	Carries out a job or profession	Unemployed	Student, further training, unpaid work experience	In retirement or early retirement or has given up business	Permanently disabled	Other inactive persons including fulfilling domestic tasks
Austria	89.2	5.1	2.0	2.7	0.3	0.8
Belgium	86.7	5.2	0.9	1.5	3.8	1.8
Czech Republic	90.0	5.5	0.6	0.2	3.3	0.3
Denmark	**86.3**	**4.4**	**3.6**	**0.4**	**4.8**	**0.5**
Finland	83.3	8.1	2.8	0.3	5.0	0.5
France	87.5	8.1	0.8	0.6	2.1	0.9
Greece	87.6	7.0	0.8	1.8	2.0	0.8
Hungary	78.5	9.3	0.6	5.8	3.8	2.0
Iceland	95.2	0.7	2.0	–	1.5	0.6
Ireland	87.7	6.6	1.1	0.6	3.1	1.1
Italy	84.2	8.4	2.9	2.7	1.2	0.7
Luxembourg	92.8	0.8	1.6	1.2	1.8	2.0
Netherlands	90.5	1.6	2.0	0.1	3.3	2.6
Norway	89.3	3.4	1.5	0.2	5.0	0.7
Poland	77.5	10.8	0.5	1.3	6.8	3.0
Portugal	89.6	4.4	1.1	2.2	1.1	1.6
Slovak Republic	79.2	15.3	0.4	4.8	–	0.4
Sweden	83.8	5.9	4.4	0.1	3.5	2.3
Switzerland	95.2	1.5	1.1	1.2	–	0.9
United Kingdom[1]	87.1	5.0	0.7	0.2	4.8	2.1
Unweighted average	**86.6**	**6.1**	**1.6**	**1.4**	**3.0**	**1.2**

1. 1999.
Source: OECD (2002a).

The key components of the package are designed to harmonise and streamline the various administrative systems that had previously applied to different categories of unemployed (see Annex V for details):

- A common labour-market system for all unemployed people, whether they are receiving unemployment insurance or cash benefits under the social assistance provisions.
- A faster and more direct path towards employment, with strengthened contacts with the public employment service and individual plans of action for all unemployed, using a new database containing *curricula vitae* for all unemployed.
- Simplified and more results-oriented activation programmes, using three key tools: guidance and counselling, job training in a firm and wage subsidies.

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- Enrolment in education courses linked to labour-market participation, while those under 30 years of age would receive the (lower) education benefit rate while undertaking such studies.
- Stronger economic incentives through: greater scope to keep earned income while receiving means-tested cash benefits; a slight reduction in the cash benefit for married "bread-winners" after six months; and a ceiling on the total public income support that a beneficiary can receive after the first six months.
- Clearer and stricter rules governing availability and willingness-to-work criteria. Job search should start from the first day of unemployment, and there is an immediate requirement to accept a "reasonable" job offer. The requirements on geographic mobility have also been extended. Sanctions for turning down job or activation offers have been increased.

These changes certainly go in the right direction and reflect a number of best practices identified by the OECD in recent years, particularly concerning efficient management of active labour market policy approaches to help those out of work back into employment (OECD, 2002a).

But the economic incentives to actively search for work remain relatively weak for those at the lower end of the wage spectrum. The net replacement rates for unemployment insurance are the highest in the OECD for those earning two-thirds of the average wage. For all long-term beneficiaries, including those on daily and social assistance cash benefits, the net replacement rates are also very high by international standards, although there is significantly more variation across countries. The recent measures primarily strengthen the economic incentives of cash benefits recipients, whereas for the average beneficiary, the incentives to work only increase marginally. In contrast, simply bringing Denmark's average net replacement rate down to the OECD average would imply a reduction of more than 20 percentage points (Table 8).

A significant cutback in benefit rates is clearly the most direct way to reduce the unemployment trap. But the inevitable trade-off is that strengthening economic incentives in this way would mean a lower income for those who remain out of employment, which appears to be politically unpalatable. However, there is evidence that providing a greater reward to those who do find work is almost certain to encourage more intensive job search among the unemployed and also to strengthen the attachment of women, in particular, to the labour force (Pedersen and Smith, 2001). However, even without trimming the benefit rate itself, other measures could help to trim the effective replacement rate indirectly and improve the incentives for job-search. Options would include shortening the maximum duration of unemployment insurance benefits from the present four years length,[18] tapering benefits as the unemployment spell lengthens (by basing

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Table 8. **Net replacement rates**

Average over 60 months of unemployment for four family types at two earnings levels[1]

	Single	Married couple	Couple 2 children	Lone parent 2 children	Overall average
Australia	39	34	69	53	49
Austria	57	59	75	72	66
Belgium	57	71	75	77	70
Canada	35	51	56	74	54
Czech Republic	46	71	88	83	72
Denmark	**76**	**76**	**84**	**87**	**81**
Finland	61	66	72	77	69
France	47	45	58	59	52
Germany	60	57	69	68	63
Greece	16	16	17	19	17
Hungary	36	37	47	48	42
Iceland	65	58	73	76	68
Ireland	36	52	62	72	55
Italy[2]	4	8	22	18	13
Japan	45	60	77	73	64
Korea	12	18	26	23	20
Netherlands	73	80	82	70	76
New Zealand	48	66	78	72	66
Norway	58	65	76	77	69
Poland	41	59	74	65	60
Portugal	57	62	65	65	62
Slovak Republic	49	75	87	79	73
Spain	44	46	55	53	50
Sweden	69	84	95	70	79
Switzerland	74	84	89	86	83
United Kingdom	56	67	81	74	69
United States	13	19	53	44	32
Average	**48**	**56**	**67**	**65**	**59**

1. 100 per cent APW and ⅔ APW earnings levels.
2. Social assistance (*Reddito Minimo di Inserimento*) is not included in Net Replacement Rates due to its experimental character (on trial in 39 municipalities). Long-term NRRs are based on family benefits.

Source: OECD (2002b).

benefits on income earned over the previous 12 months), and instituting longer work requirements before qualifying for full unemployment insurance benefits.

An alternative approach used in some countries is to boost the returns from working via an earned income tax credit (EITC). Denmark has also just announced a type of EITC (see Chapter IV) applying to all workers and which does not gradually abate as incomes rise, a typical feature of EITC arrangements in other countries. Abatement would be particularly problematic in Denmark's case since phasing out a tax credit would imply a significant rise in effective marginal tax rates for a large number of workers because the wage distribution is relatively compressed: one quarter of all employees earn at most 20 per cent more than what the average person on unemployment benefit receives.

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With weak direct economic incentives to "pull people into work", policies such as activation and tight eligibility conditions that "push people into work" need to shoulder more of the task. Denmark has established a tradition of emphasising active rather than passive labour market policies, and the "right and duty" principle embodied in activation policies has been successful in reducing unemployment rates among the young (OECD, 2000). The latest measures seek to refine and build on the experience of what policies work best and are consistent with established international best practice (Martin and Grubb, 2001). However, their success will depend on the willingness to apply sanctions to enforce the eligibility criteria for benefits, which are a sticking point for many countries (Grubb, 2001), especially where they confront concerns about hardship (as when children are involved). Indeed it is difficult to reliably assess the degree of strictness of sanctions across countries, as there may often be significant gaps between legal provisions and actual practice. But the improved operational procedures embodied in the "More People in Work" package should provide a stronger basis on which to enforce sanctions, if the political commitment is there.

Earlier labour-force entry

Young people in Denmark start basic school comparatively late (at the age of seven) and take a relatively long time to complete their education (OECD, 2002c). The average five year-old in 2000 could expect to spend 18 years in the education system, one year longer than in 1995. More than eight of those years would be between the ages of 15 and 29 years (Figure 10). Many young people do not complete upper secondary school until they are in their early 20s, while almost one-quarter of 25 to 29 year-olds were enrolled in education in 2000. Denmark's unusually high proportion of students who have jobs while studying complicates the picture: little information is available about whether this work is study-related. But the financial support arrangements discourage students from earning significant incomes, and the average income earned by those students in work corresponds to around 10 hours paid work per week (SUstyrelsen, 2002). In total, some 24 per cent of men and 35 per cent of women from this age group who are employed are working less than 15 hours per week. These patterns suggest that there may be scope for raising the labour supply by encouraging students to complete their studies at a younger age, so that they can spend more time in employment.

A *better focussed* 10*th form*

One reason why the number of years of expected education has risen over time has been the increased attendance in the voluntary 10th form, which is normally at the end of compulsory school. Originally, this year was intended to provide for a relatively small number of pupils that were not ready to choose what

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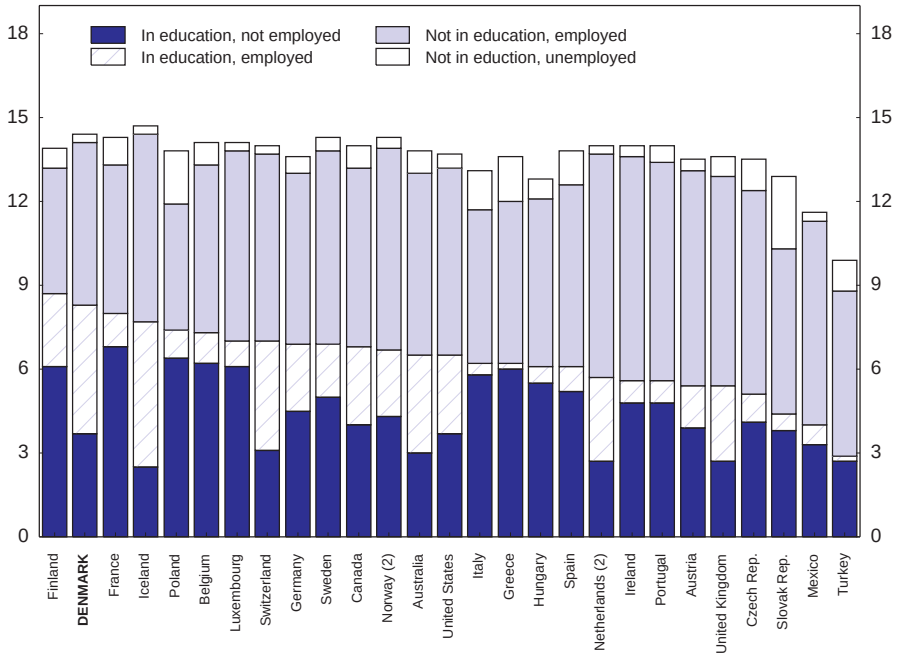
Figure 10. **Expected years in education and the labour force for 15 to 29 year-olds**
Number of years, 2001[1]



Note: Countries are ranked in descending order of the expected years in education of the youth population.
1. The remaining years are spent neither in education, nor in the labour force.
2. Year of reference 2000.
Source: OECD, *Education at a Glance*, 2002.

upper secondary education they wished to pursue or not yet capable of undertaking a higher course of study. However, its popularity has grown and more than 60 per cent of children now attend 10th grade. But there are serious doubts about its instructional value. The government is currently undertaking an evaluation to identify how the resources and pupils' time can be spent to best effect. Without prejudging the outcome of this review, concentrating the resources currently allocated to this year on those in most need of remedial assistance, while sending other children directly to the next level of education, would help a greater proportion to complete higher education at an earlier age and then enter the workforce.

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Better incentives in tertiary education

Danish students face relatively weak economic incentives to start their tertiary studies early or to complete them quickly. Of those who first enrolled in higher education[19] in 1990/91, only two-thirds had completed their original study programme eight years later, and another 9 per cent had completed a different course of study, while more than 12 per cent were still enrolled at university (SUstyrelsen, 2002). On average, only 40 per cent of students who enrolled in 1990 had made normal progress after three years. Younger students were more likely to have progressed at the optimal speed (Table 9). This may indicate an advantage to moving straight from upper secondary school to tertiary education, while study habits are still fresh. But most students do not make such a swift transition[20] and on average students from general upper secondary programmes take a break of between one and two years before tackling further studies, while those from vocationally oriented courses take somewhat less (Figure 11).

These trends may reflect the fact that having additional qualifications gives little gain in earnings compared with many other countries (Figure 12). These low market-based rewards for educational attainment are even further compressed by the high income tax rates that apply to incomes not far above the average wage (see Chapter IV). Denmark charges negligible tuition fees and offers students very generous public income support, compared with other countries. In effect, this partly compensates for the low private, after-tax internal rates of return (Table 10), and makes studying more attractive. Public funding encourages more young people to study than would perhaps otherwise be the case. But they also face little financial penalty for stretching out their studies, compared with say, typical US or French students, who not only have to pay tuition fees and receive less income support while studying, but also forego higher earnings by delaying employment. All else being equal, the rate of return for education is lower if

Table 9. **Study progress[1] by age**

1993, after three years study

Age at enrolment	Minimal progress	Medium progress	Optimal progress
	Per cent		
< 21 years	17	35	48
21 to 24 years	31	33	36
25 to 30 years	44	31	25
> 30 years	56	23	21
All students	27	33	40

1. Study progress is defined as follows: optimal progress implies that after three years enrolment, the student should have credits corresponding to 2½ to 3 years' full-time study. Medium progress implies credits for 1½ to 2 years and minimal progress implies credits less than the equivalent of 1½ years of full-time study.

Source: SUstyrelsen (1997).

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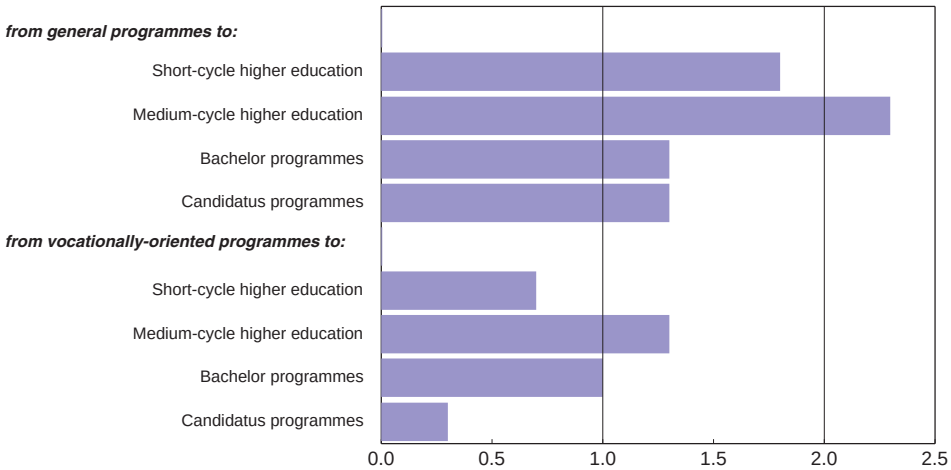
Figure 11. **Median transition time from upper secondary to higher education**

Number of years



Source: Ministry of Education (2002).

students enter the workforce at a later age, simply because they have fewer years of higher earnings available before they retire.

It is relatively difficult to strengthen the economic incentives to undertake and complete higher education expeditiously in the Danish setting. The task would be made much easier if the wage distribution gave greater reward to educational attainment. But this would be hard to engineer in an economic environment that has relatively high effective minimum wages in collective agreements and a social safety net that provides public support with generous replacement rates (see above). Yet if all the adjustment has to come from higher wages towards the top end, this will push up average wages in a manner that may threaten Denmark's overall international competitiveness. Reducing the very high rates of taxation on higher incomes would at least reduce the current compounding effect of progressive taxation on a compressed wage distribution (see Chapter IV).

One way the government could reshape the economic incentives would be to have students bear a greater part of the cost of their education, either through lower public income support and/or tuition fees, so as to encourage them to undertake studies more efficiently. From a purely economic perspective, it would be reasonable to expect individuals to bear some portion of their costs, as it has been widely demonstrated in many countries that higher education generates benefits mainly to the individual, rather than to society as a whole

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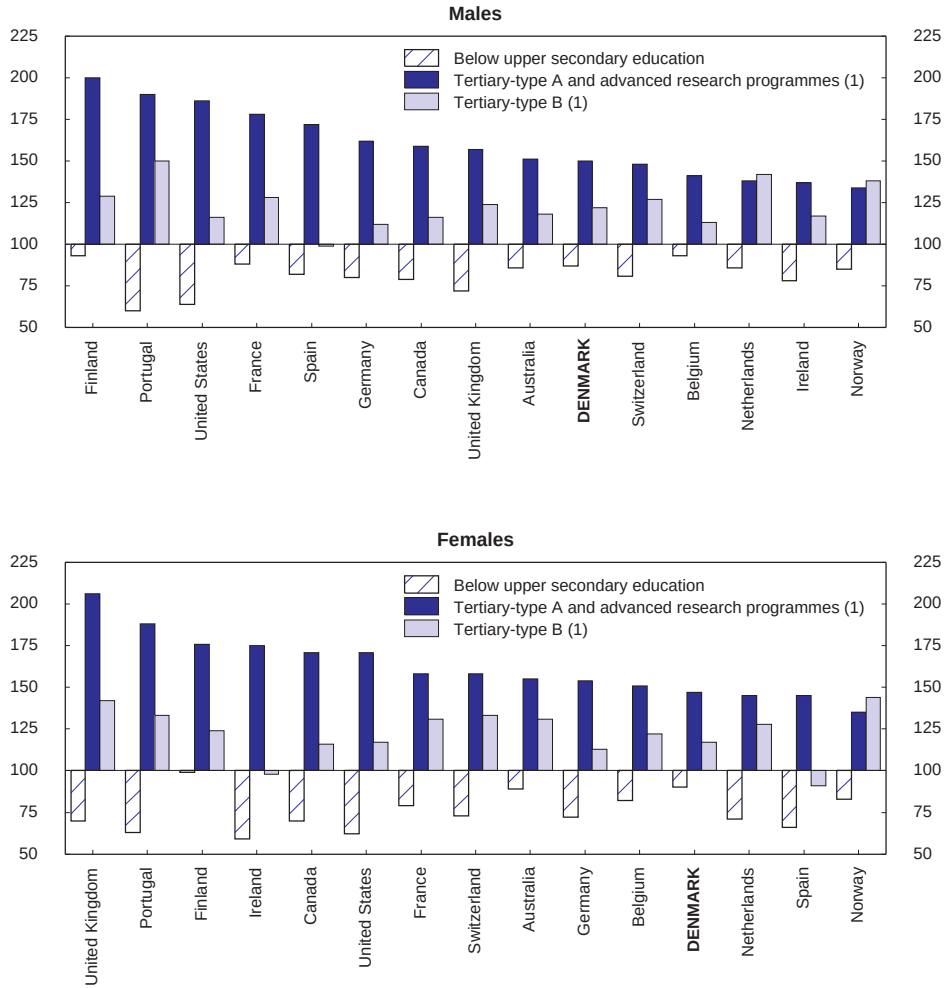
Figure 12. **Relative earnings by educational attainment**

Income from employment

Upper secondary and post-secondary non tertiary education = 100



1. Tertiary type-A programmes are largely theory-based and provide qualifications for entry to advanced research programmes or professions with high skill requirements. They usually involve a minimum of three years full time equivalent study. Tertiary type-B courses are generally shorter and focus on practical, technical or occupational skills.

Source: OECD, *Education at a Glance*, 2002.

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Table 10. **Private internal rates of return to education**

1999-2000, per cent

	United States	Japan	Germany	France	Italy[1]	United Kingdom	Canada	**Denmark**	Netherlands[2]	Sweden	Unweighted average
						A. Men					
Tertiary education											
Return based on pre-tax earnings and the length of studies (narrow rate)	18.9	8.0	7.1	13.3	8.0[3]	18.1	8.4	7.9	11.7	9.4	11.4[4]
Impact of (in percentage points)											
Taxes	–2.3	–0.3	–1.5	–1.6	–	–2.1	–0.5	–2.1	–2.0	–1.5	–1.5
Unemployment risk	0.9	0.9	1.1	2.4	0.3	1.6	1.3	1.0	–	1.2	1.1
Tuition fees	–4.7	–2.0	–0.3	–1.1	–0.8	–2.7	–2.3	–0.1	–0.6	–0.7	–1.5
Public student support	2.1	1.3	2.7	1.3	–	3.6	1.8	4.8	2.9	3.0	2.4
Comprehensive rate	**14.9**	**7.9**	**9.1**	**14.3**	**7.5**	**18.5**	**8.7**	**11.5**	**12.1**	**11.4**	**11.6**
Upper-secondary education											
Return based on pre-tax earnings and the length of studies (narrow rate)	14.4	4.4	10.0	7.5	9.5[3]	12.4	11.9	11.3	6.9	3.9	9.2[4]
Impact of (in percentage points)											
Taxes	–0.9	–0.2	–2.1	–1.0	–	–1.5	–1.6	–2.2	–0.2	–0.6	–1.1
Unemployment risk	2.9	2.6	2.9	7.0	1.7	4.2	3.6	2.2	1.2	3.1	3.1
Comprehensive rate	**16.4**	**6.8**	**10.8**	**13.5**	**11.2**	**15.1**	**13.6**	**11.3**	**7.9**	**6.4**	**11.3**

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Table 10. **Private internal rates of return to education** (*cont.*)

1999-2000, per cent

	United States	Japan	Germany	France	Italy[1]	United Kingdom	Canada	**Denmark**	Netherlands[2]	Sweden	Unweighted average
	B. Women										
Tertiary education											
Return based on pre-tax earnings and the length of studies (narrow rate)	18.8	8.0	7.0	12.1	–	16.4	10.6	6.0	9.4	7.4	10.6
Impact of (in percentage points)											
Taxes	–2.0	–0.2	–1.6	–1.7	–	–2.3	–1.3	–1.1	–1.0	–0.7	–1.3
Unemployment risk	1.4	0.5	0.6	4.8	–	1.3	1.2	0.7	0.7	1.6	1.4
Tuition fees	–6.0	–2.4	–0.6	–1.7	–	–2.5	–2.9	–0.1	–0.7	–0.8	–2.0
Public student support	2.7	1.3	3.0	1.9	–	3.2	2.4	5.6	4.1	3.3	3.1
Comprehensive rate	**14.7**	**7.2**	**8.4**	**15.4**	**–**	**16.1**	**9.9**	**11.1**	**12.5**	**10.8**	**11.8**
Upper-secondary education											
Return based on pre-tax earnings and the length of studies (narrow rate)	10.6	6.6	6.1	10.5	–	–	10.8	8.3	7.9	–	8.7
Impact of (in percentage points)											
Taxes	–1.3	–0.2	–1.7	–0.7	–	–	–1.2	–1.4	–1.6	–	–1.2
Unemployment risk	2.5	3.0	2.6	8.1	–	–	3.1	3.6	2.1	–	3.6
Comprehensive rate	**11.8**	**9.4**	**7.0**	**17.9**	**–**	**–**	**12.7**	**10.5**	**8.4**	**–**	**11.1**

Note: The rates of return to tertiary education are calculated by comparing the benefits and costs with those of upper-secondary education. In the case of rates of return to upper-secondary education, the calculation compares the benefits and costs with those of lower-secondary education. In Sweden, the theoretical length of standard tertiary courses is used in the calculations rather than the average theoretical length of different programmes. Moreover, earnings differentials for women between upper and lower-secondary levels are not large enough to permit a positive rate-of-return calculation. In the United Kingdom, data on earnings of women up to age 30 with lower-secondary education were not available. In Italy, reliable data on earnings for women were not available.

1. 1998.
2. 1997.
3. Post-tax earnings.
4. Excluding Italy.

Source: Blöndal *et al.* (2002).

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(Blöndal *et al.*, 2002). The majority of countries charge some tuition fees, even if these fees are only a relatively modest proportion of the total cost of study, while Denmark stands out for the generosity of income support to students. But there appears to be widespread public attachment in Denmark to the principle that higher education should be free to students. An alternative that might be more politically acceptable would be to tilt the balance of public income support from grants towards loans, since 80 per cent of student income support is currently in the form of grants.[21]

Australia, New Zealand and the United Kingdom have all introduced student loans schemes in recent years, to some extent replacing student grants. This has not had a discernible negative impact on student participation, while it has provided clearer signals to students about the value of completing their studies rapidly (OECD, 2001a). This approach also has implications for equity; free or highly subsidised tertiary education tends to be a rather regressive form of public spending. Since children from relatively better educated, and therefore usually more affluent, families are more likely to attend tertiary institutions in the first place, they benefit disproportionately from the high subsidies and also have higher future income prospects. These effects appear to be less marked in Denmark than in many other countries (Blöndal *et al.*, 2002). But although the Danish system is explicitly designed to ensure that economic circumstances are not an obstacle to further education, children of middle to upper income parents were still five times more likely to complete a long-term higher education course than children from working class backgrounds in the mid-1990s (Hansen, 1997). In any case, equity considerations could be mitigated by introducing means-tested elements into tuition fees and/or to public support schemes for students (Joumard and Suyker, 2002).

The rules governing student support could in any case be tightened to encourage students to choose their courses of study more carefully in the first place and to reduce the tendency for some students to stretch the current rules to take up to a year off their studies while continuing to receive student benefits because they are still enrolled. Students enrolled in a higher education course are entitled to grants corresponding to the prescribed duration of the course of study plus 12 months.[22] Within a total of 70 months' grants, the student can change course. Eliminating the extra 12 months coverage and reducing the financial latitude provided to those who decide to change their study programme would provide stronger incentives to students to complete their studies expeditiously, without reducing the level of support offered to those who choose their courses carefully and maintain a satisfactory academic performance.

Contribution to labour supply

It is difficult to predict the labour supply effects of measures to boost the incentives to complete education more quickly, reduce transition time and focus

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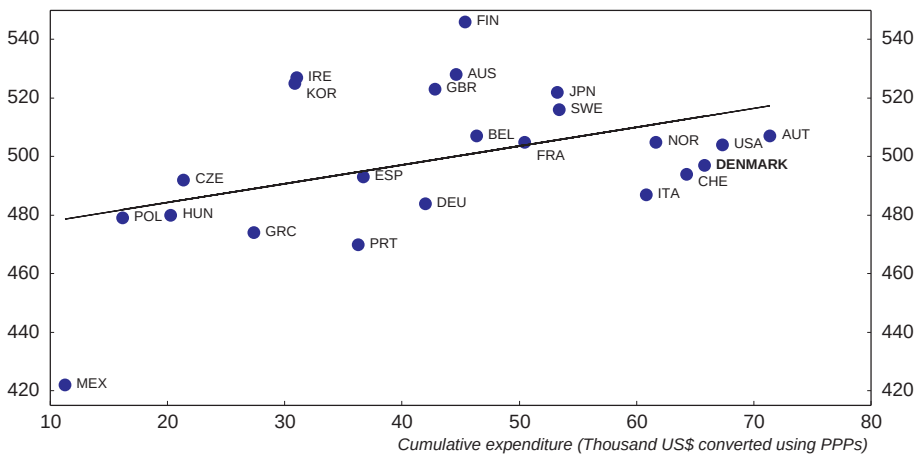
the 10th form on those who really need it. However, if the 10th form were reduced to only 10 per cent of pupils, this would mean around 28 000 students able to enter the workforce a year earlier. If the transition time from upper secondary to tertiary education were shortened by a year then this might add around 25 000 more people to the labour market, assuming that only half of them are working anyway during the transition period.[23] In addition, if tertiary students were to spend on average six months less to obtain their qualifications (by studying more efficiently or making better choices about courses of study), this could boost the labour supply by perhaps a further 15 000.

Improving the efficiency of schooling

The efficiency of the education process is another aspect which is relevant to the quality of new entrants to the workforce, even though it will not affect their numbers. The Danish compulsory education system is one of the most costly per pupil in the OECD, and yet it has delivered lacklustre results: in the OECD's recent PISA assessments of 15 year-olds, reading skills are just below OECD average, and science clearly below, though mathematics was correspondingly above average (Figure 13). This suggests that if other countries can produce better outcomes with fewer resources, Denmark should be in a position to improve the

Figure 13. **Educational performance of 15 year-olds**

Score on the PISA reading literacy scale by expenditure on educational institutions



Source: OECD PISA database.

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efficiency of the education process and lift its standards within the present resource allocation. The government's latest schooling reforms have concentrated on the upper secondary sector, but further efforts may be needed to address the compulsory years as well.

Delaying the effective retirement age

The average age at which people withdraw permanently from the labour force is lower today than in the 1980s in Denmark, as in many OECD countries (Figure 14). In Denmark's case this in part reflects deliberate policy choices, and in particular the introduction of the voluntary early retirement benefit (VERB) scheme in 1979, and the 1994 expansion of the transitional early retirement benefit scheme (OECD, 2000). Early withdrawal not only has negative consequences for the productive capacity of the country, but also puts a double burden on fiscal sustainability, because those outside the labour force in Denmark are almost always drawing public income support instead (Table 11). Indeed, in 2001 more than one in ten 40 to 59 year-olds was receiving a permanent benefit, as were close to two-thirds of those aged 60 to 66 years. The bottom line is that, when combined with later entry into the workforce, adults in Denmark spend fewer years in the labour force than those in many other OECD countries.

The economic costs associated with this shortening of working life are exacerbated by the likelihood that life expectancy will increase over time, so that the proportion of adult life spent in the workforce will gradually decline unless retirement ages adjust upward over time. But at present there seems little willingness among Danish people to contemplate retiring later, and even modest reforms to the VERB scheme, undertaken to encourage later retirement, have been politically very costly. This is unfortunate, given that shorter working lives and longer retirement make the effective dependency ratio higher than it would otherwise be (see Chapter I).

This section reviews policies in three areas that all play a direct role in influencing the effective retirement age: policies towards unemployed older workers, the VERB scheme and the disability scheme. Ways of adjusting the age of entitlement to the public universal old-age pension to take account of increasing life expectancy are also examined.

Getting older unemployed back into unsubsidised work

Older unemployed are likely to withdraw from the labour force between one and two years earlier than those in work: for both men and women this occurs on average a short time before they reach 60 years of age. Older workers do not appear any more likely to become unemployed than younger ones, but once they find themselves out of work, they are far less likely to return to work. The average age of withdrawal from the labour force for the unemployed has actually risen

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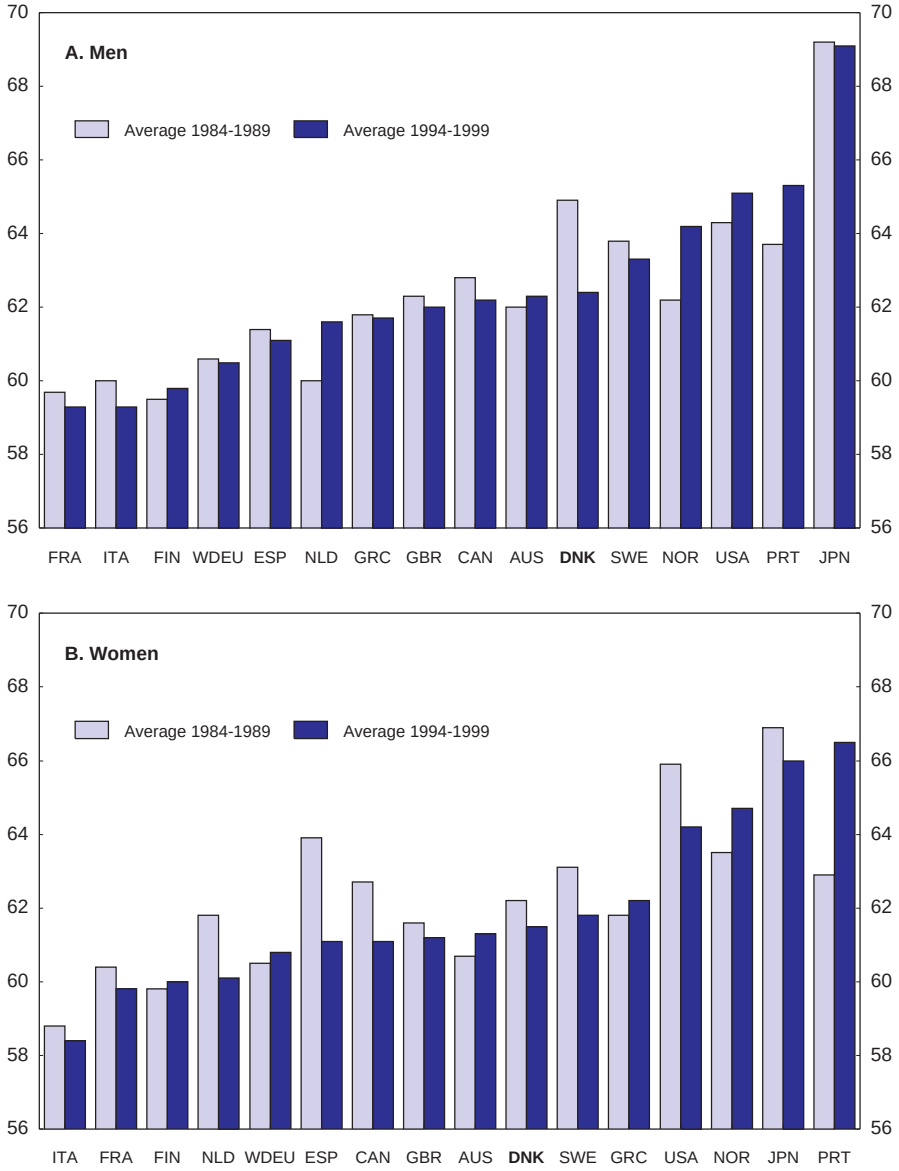
Figure 14. **Average age of withdrawal from the labour force**

Years of age



Source: Scherer (2001).

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Table 11. **Population by age and activity status**
2001

	Employed	Unemployed	Total labour force	Temporarily outside labour force	Retirement pensions	Other pensions	Others	Total population
Men								
30-34	87.0	3.5	90.5	2.5	0.0	1.8	5.1	100.0
35-39	87.3	3.3	90.6	2.3	0.0	2.8	4.3	100.0
40-44	86.7	3.1	89.8	2.0	0.0	4.3	3.8	100.0
45-49	86.0	3.0	89.0	1.7	0.0	6.0	3.2	100.0
50-54	84.8	3.2	88.0	1.3	0.4	7.5	2.7	100.0
55-59	78.3	4.9	83.2	1.0	3.6	9.9	2.3	100.0
60-66	36.4	1.5	37.9	0.2	42.2	14.6	5.1	100.0
67+	13.4	0.0	13.4	0.0	0.0	86.0	0.6	100.0
Women								
30-34	80.3	4.8	85.1	5.4	0.0	1.7	7.9	100.0
35-39	82.8	4.0	86.8	4.3	0.0	2.9	6.0	100.0
40-44	83.8	3.4	87.2	3.1	0.0	4.9	4.8	100.0
45-49	82.9	2.9	85.8	2.3	0.0	7.7	4.4	100.0
50-54	78.8	3.3	82.1	1.9	0.9	10.8	4.4	100.0
55-59	64.5	5.6	70.1	1.4	8.2	15.4	4.9	100.0
60-66	19.2	1.0	20.2	0.1	44.7	27.1	7.8	100.0
67+	3.0	0.0	3.0	0.0	0.0	96.4	0.5	100.0

Source: Statistics Denmark.

quite sharply, but this is primarily because the transitional early retirement benefit scheme for unemployed 50 to 59 year-olds was closed to new entrants in 1996. Nevertheless, public expectations remain that older unemployed workers are incapable of returning to unsubsidised work. This can be seen in the service jobs scheme, which provides an indefinite wage subsidy to public employment for people more than 48 years old and who have been unemployed for at least 18 months, although this scheme was closed to new entrants at the end of 2002. In addition, 58 and 59 year-olds can be exempted from the requirement to participate in activation programmes.

Voluntary early retirement scheme

A considerable part of the trend towards earlier retirement can be explained by the introduction in 1979 of the VERB scheme for members of unemployment insurance funds between 60 and 66 years of age. The scheme gained immediate popularity, as illustrated by a drop in participation rates for men aged 60 to 69 years[24] by 14 percentage points in 1979 (Figure 15). Female participation rates were affected less dramatically at that time, partly reflecting the low level of participation. Since then, the inflow to the scheme has steadily increased

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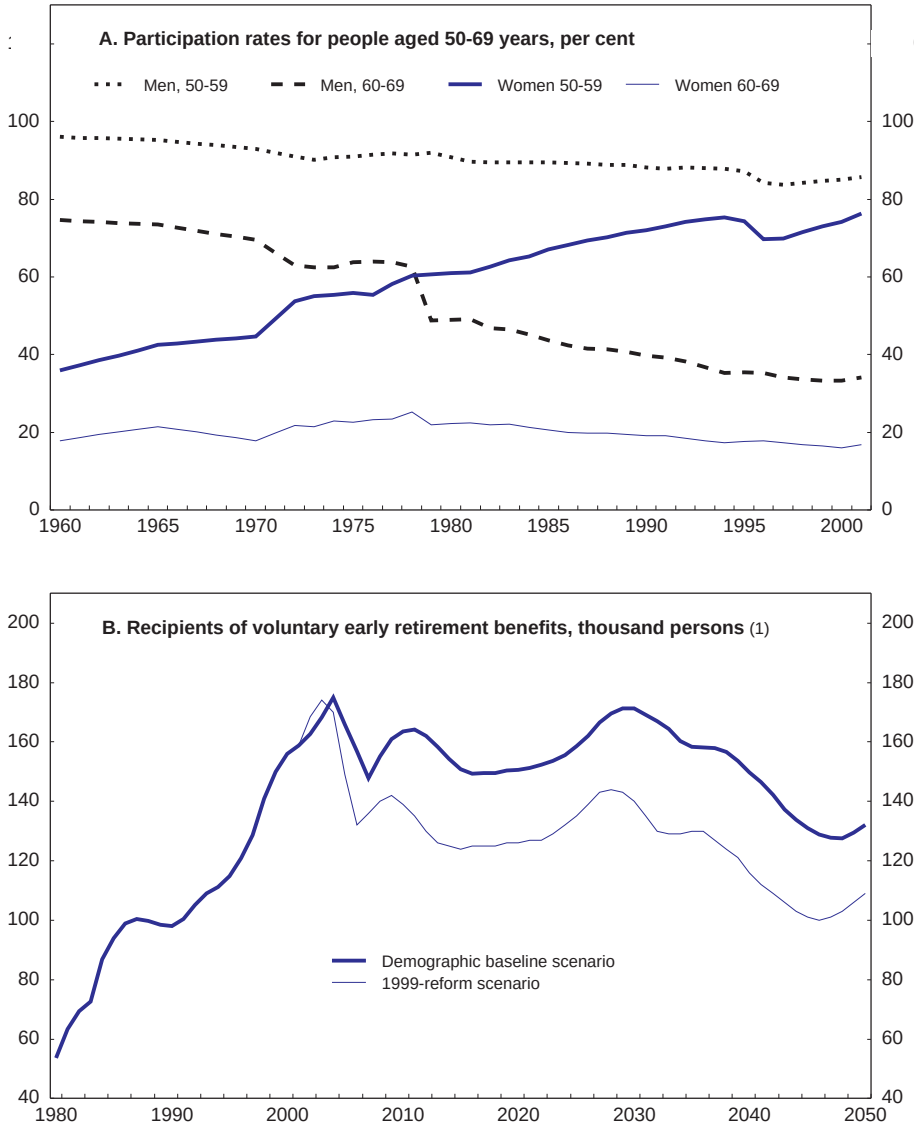
Figure 15. **Participation rates and number of VERB recipients**



1. The steep reduction in VERB recipients in 2005-07 is due to the lowering of the formal pension age from 67 to 65 years starting from 2004.

Source: Statistics Denmark; Ministry of Finance; OECD calculations.

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in line with a growing proportion of 60 to 66 year-olds choosing early retirement from the labour market as well as a rising number of women on the labour market that become entitled to the benefit.[25] New VERB recipients alone have accounted for more than half of the increase in the total number of public transfer recipients (excluding old age pensioners) since 1978, and in 2001 almost 160 000 people received these benefits corresponding to 44 per cent of the population aged 60 to 66 years. The direct cost in terms of public benefit payments amounted to almost DKK 20 billion, which corresponds to around 1½ per cent of GDP. In addition, productive capacity is lower and tax revenue foregone.

The availability of VERB provides a popular exit route from the labour market, and the larger part of Danish early retirees receive this kind of benefits, which is not contingent on reduced working ability. This stands in contrast to the majority of OECD countries, where disability pension schemes provide the dominant path to early retirement (Blöndal and Scarpetta, 1999). To some extent, this could just reflect differences in entitlement criteria, whereas the political perceptions of acceptable reasons for retirement on public support might be more similar between countries. The introduction of the VERB scheme was in part motivated by a wish to offer worn-out (but not disabled) workers a way to leave the labour market before becoming entitled to the old-age pension at the age of 67 years.

An additional and major official motivation behind the introduction of the scheme was the erroneous belief that it was necessary to create room for younger people in the labour market, a belief that is no longer a factor in the Danish debate.[26] Similar initiatives to "share out" employment in periods with low activity have been introduced since then, for instance the transitional early retirement benefit for long-time unemployed persons aged 50 to 59 years and various paid leave schemes. During the later part of the 1990s it was recognised that these schemes were damaging growth potential, and they were either abandoned (the transitional early retirement benefit scheme in 1996[27]) or restricted. However, a correspondingly firm reaction to the continuous net inflow to VERB and the prospects of even higher pressure on the scheme from population ageing was not undertaken. The reform of the scheme in 1999 focussed on changing economic incentives rather than directly restricting access. It involved a mix of carrot and stick measures by introducing additional rewards for postponing retirement and a contribution payment covering part of possible future benefits, while at the same time reducing the formal pension age from 67 to 65 from 2004 (see Annex VI).

The longer-term impact of the 1999 reform has yet to be seen. However, experience from 1999 to 2001 indicates that at least a temporary postponement of retirement has been achieved. The accession rate of 60 and 61 year-olds has decreased, inducing a drop in total inflow, while more people now start drawing benefits at the age of 62 years. This initial postponement effect is reflected in a

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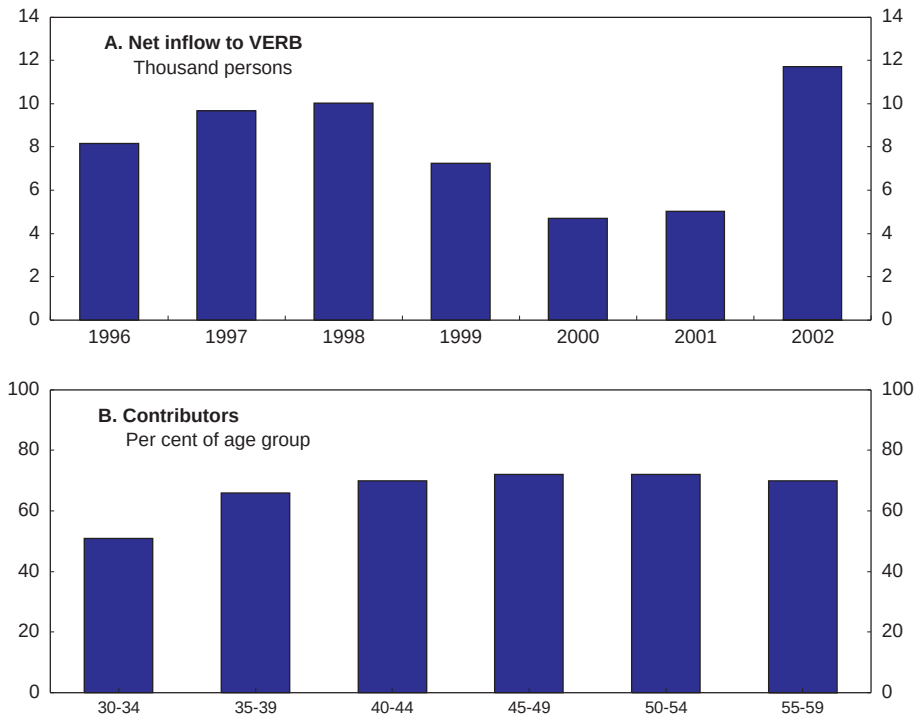
Figure 16. **Net inflow and contributors to VERB**[1]



1. Entitlement to VERB requires contribution payments in 25 out of the preceding 30 years. Thus, if a person plans to retire at the age of 60, starting payment of contributions would not be necessary until the age of 35.

Source: Danish Labour Directorate; Ministry of Finance; Danish Employers Federation.

substantial increase in the inflow to the scheme in 2002 (Figure 16). For a permanent effect to emerge, net inflow would have to return to a lower level in 2003. On the other hand, the new contribution rate has apparently not had the expected deterrent effect on younger generations. The proportion of 35 to 39 year-olds paying contributions is almost as high as the proportion of 55 to 59 year-olds, although the older group will see a substantially higher return on their contribution payments.

Notwithstanding these reforms, the scheme remains a long way from being actuarially fair, where each person would receive in benefit payments the net present value of the contributions they had made. Indeed, contributions only cover around 20-25 per cent of direct future benefits to be paid out under the

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scheme. Furthermore, because the scheme pays flat-rate benefits, the biggest advantages accrue to the low paid, who also have the weakest economic incentive to stay in work. It may seem less obvious why younger workers continue to pay their voluntary contributions, despite the fact that opinion polls reveal widespread scepticism that benefits will be available to them when they retire. However, *having* the option of drawing these benefits and *using* it are two different things. The net return from paying contributions but staying in the labour force until the age of 65 is still positive, due to the tax-exempt reward paid for not using this option of early retirement. In reality, workers enrolling in the scheme cannot lose, since it is possible to withdraw at any time and have previous contributions transferred to another pension scheme, and politicians have promised repayment of contributions if the scheme is ever abandoned.

The official estimate of the impact of the reform is a gradual net contribution to the labour supply of 9 500 people from 2000 to 2010. Thus, even if this effect is actually realised, the reform makes a relatively small contribution to the increase in employment needed to achieve the government's objectives to 2010 (see Chapter I). In contrast, if the trend during the last ten years towards lower participation rates for 60 to 64 year-olds were to continue, the workforce would be reduced by around 24 000 people by 2010 compared with a scenario where participation rates for this age group were unchanged. This estimate does not take into account a possible positive impact on participation rates for over 60 year-olds due to the termination of the transitional early retirement scheme for 50 to 59 year olds.[28]

These prospects indicate that additional reforms of the VERB scheme are likely to be necessary in order to achieve the government's long term objectives. However, there seems to be a strong preference at the political level for continued subsidisation of early retirement, even though the potential gains from making the scheme less attractive or abolishing it all together are significant. The Danish system already fully covers those not able to find work (through unemployment benefits) and people with reduced work capacity can receive an early retirement (disability) pension. Thus, the VERB scheme has primarily provided subsidies to early retirement for persons fully capable of working. In the absence of the scheme those with strong preferences for early retirement could still exercise them, but their decision would be based on their own private financial positions alone, rather than being influenced by implicit public subsidies.

Not all current VERB recipients could be expected to have stayed in work if this early retirement option had not existed. Absenteeism due to sickness would most likely be higher, and relatively generous benefit rates in the scheme have probably sheltered inflow to the co-existing early retirement (disability) pension. Based on a rough estimate, Finansministeriet (2002a) indicates that around one-third of the increase in the net accession rate to VERB between 1990 and 1999 might be due to substitution from early retirement (disability) pension.

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Using this number as a proxy for the total substitution effect between the two schemes would, however, still imply more than 100 000 people available to work if they had not been offered these early retirement benefits. Thus, even if some groups would continue to withdraw from the labour market before the formal pension age, and if sickness-related absenteeism were to increase, there would still be a significant potential for raising participation and employment.

There are several ways to achieve a substantial contribution to boosting labour supply through modifications to the VERB scheme:

- The most effective solution would be to totally abandon the scheme and pay back contributions. This has already been proposed by the Economic Council (Det Økonomiske Råd, 2002a). It would be the most straightforward way of inducing more people to stay attached to the labour force until they qualify for public old-age pensions or to finance earlier withdrawal entirely through their savings. However, efforts would be needed to minimise the extent to which people seek to use other channels to early retirement instead.
- The next-best solution would be to convert the scheme to a defined-contribution basis. This would eliminate the distortion to retirement behaviour, since it would amount to people drawing down their own savings to finance earlier retirement. However, since pension savings schemes are already provided in the private sector, there is no economic reason why an actuarially-fair government-administered system would need to be maintained.
- A less radical – but also less effective – alternative would be to tweak the rules further to push the entitlement age from 60 to 62 years. The 1999 reform clearly intended to raise the effective age of benefit take-up, but a further adjustment might be needed to obtain the hoped-for effects. A similar approach could endeavour to lift effective retirement ages higher still, although experience indicates that such piecemeal reforms may be politically costly.
- Another option might be to index the early retirement option to the statutory retirement age and raise the official retirement age (see below). The early retirement rules could be redesigned to limit the number of years of early retirement, effectively lengthening time spent in the labour force.

Contribution to labour supply

The effects from reforms of the VERB scheme can be estimated only with some uncertainty. Calculations by the Ministry of Finance indicate that an increase in the labour force by 20 000 people as a result of higher participation rates for 60 to

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62 year-olds reduces the public sector surplus required to ensure fiscal sustainability by ¼ per cent of GDP (Finansministeriet, 2002a). The reduction in the required surplus would be somewhat less if the same increase in the labour force originated from higher participation by 60 to 64 year-olds because average annual working hours are lower for the group of 63 to 64 year-olds than for 60 to 62 year-olds.

Using this estimated relationship and assuming that one-third of current beneficiaries would find other forms of public income support, a rise in the entitlement age from 60 to 62 years could increase the labour force by around 30 000 people and reduce the required surplus by 0.4 per cent of GDP (Table 12).[29] Under the same assumptions, totally abolishing the VERB scheme could yield around 100 000 more people in the labour force and a reduction of the required surplus of around 1¼ per cent of GDP. The effect of changing to a defined-contribution scheme is more difficult to quantify. However, removing the public subsidy would reduce net replacement rates significantly. The estimated impacts from these reforms do not include counteracting effects on net replacement rates from maturing private labour market pension schemes.

Table 12. **Effects from different changes to the VERB scheme**

	Effect on:	
	Labour force (1 000 persons)	Required general government fiscal balance (percentage point)
Total abolition of VERB	+100	–1.3
Raise entitlement age to 62 years	+30	–0.4

Note: A rule of thumb is used implying that the required surplus is reduced by ¼ percentage point when the labour force is increased by 20 000 persons. This is inspired by calculations in Finansministeriet (2002a).
Source: OECD calculations.

It should be emphasised that changes of the kind mentioned above will require a certain phase-in period, since many people in their 40s or 50s have planned their working lives on the basis of access to the scheme from the age of 60. Policy action is therefore required in the very near future just to see some effect on labour supply in the years ahead. However, the estimated impacts on fiscal sustainability shown above are contingent on immediate implementation and full effect of changes to the VERB by 2010. Thus, the longer a reform is postponed, the smaller the positive contribution that it will make to fiscal sustainability.

Disability as a means for early exit

It is too early to assess the recent reforms to the early retirement (disability) scheme that came into force at the beginning of this year, and in any case their

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success depends heavily on the way in which the new eligibility criteria are interpreted in practice (see previous *Survey*). But the benefit trap will, for some recipients, be larger than before, since a proportion of those who qualify under the new rules will receive higher income support than they would have received under the old rules. In any case, it will be important to ensure that efforts to tighten other forms of public income support, especially VERB, do not result in using the disability pension as an alternative channel for permanent exit from the labour force.

Outflow from disability and back into work is low in virtually all OECD countries, in part because of weak economic incentives to return to work, but also because of various rules and expectations that discourage re-entry. In practice in Denmark only a marginal amount of part-time work is permissible without putting pension entitlement at risk. Denmark has some 260 000 working-age people (equivalent to almost 10 per cent of the current workforce) receiving disability pensions, many of whom have less than two-thirds disability and therefore might not have qualified if the new rules had applied (Table 13). Those benefits were

Table 13. **Recipients of early retirement (disability) pensions**

	Early retirement pensions			Total
	Highest	Intermediate	Ordinary/increased	
	Number of recipients			
Recipients, total	**61 645**	**111 166**	**86 765**	**259 576**
18-29 years	4 342	3 988	678	9 008
30-39 years	8 624	10 353	4 609	23 586
40-49 years	13 650	21 850	15 390	50 890
50-54 years	9 551	18 357	13 141	41 049
55-59 years	12 025	24 124	17 915	54 064
60-66 years	13 453	32 494	35 032	80 979
Men, total	**33 217**	**50 849**	**29 010**	**113 076**
18-29 years	2 532	2 261	379	5 172
30-39 years	4 909	5 316	2 236	12 461
40-49 years	7 300	9 924	6 725	23 949
50-54 years	4 968	8 031	4 937	17 936
55-59 years	6 350	10 717	5 898	22 965
60-66 years	7 158	14 600	8 835	30 593
Women, total	**28 428**	**60 317**	**57 755**	**146 500**
18-29 years	1 810	1 727	299	3 836
30-39 years	3 715	5 037	2 373	11 125
40-49 years	6 350	11 926	8 665	26 941
50-54 years	4 583	10 326	8 204	23 113
55-59 years	5 675	13 407	12 017	31 099
60-66 years	6 295	17 894	26 197	50 386

Note: Compiled 1st January.
Source: Statistics Denmark.

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awarded on a permanent basis and thus may be seen as an "acquired right". However, an extensive OECD cross-country review of disability and relevant policy approaches (OECD, 2003c) suggests that a renewed emphasis should be placed on allowing and encouraging the disabled to work, with a shift away from the "giving up" attitude that permanent benefits imply (see Annex VII). Concretely, two particular policy adjustments would seem pertinent for the Danish situation:

- Allowing a greater possibility for those already awarded pensions to take up working again, with easier capacity to suspend their pensions while "trying out" work. Assistance with job search to help them reattach themselves to the labour market would also send a signal that their participation in the workforce is sought and valued.
- Periodic review of disability status, taking into account, *inter alia*, medical developments and rehabilitation opportunities that may have become available since the pension was first awarded.

Keeping time spent in retirement constant

Another element that makes a difference to the impacts of an ageing population is raising the age of entitlement to public old-age pensions in line with increases in life expectancy. Denmark is unusual in having actually lowered its age of entitlement from 67 to 65 years, as part of the 1999 reforms to VERB. This stands in sharp contrast to the upward trend in official retirement ages across OECD countries (Dang *et al.*, 2001). Some countries have gone further and instituted arrangements designed to encourage the age of retirement to shift upwards automatically as life expectancy increases: Sweden's notional defined contribution and market-based premium pension approach does this.[30] Danish private pension arrangements also provide an automatic upward adjustment for life expectancy, because they are defined-contribution schemes. Such pensions automatically embody an incentive to work longer as life expectancy increases, if the individual wishes to receive the same replacement rate on their private pension. But their role is complicated by the statutory old-age pension.

The Danish public old-age pension is universal, with needs-tested supplements, designed to ensure a minimum standard of living to all over 65 year-olds.[31] The average after-tax replacement rate is around 40 per cent while other pensions and capital income provides another 30 per cent. It, thus, plays a significant role in the individual's assessment of retirement income, and can undermine the incentives embodied in the defined-contribution pensions.[32] Furthermore, as pointed out in Chapter I, increasing life expectancy will undermine Denmark's fiscal sustainability, as public old-age pensions will be paid out for a longer period, unless the age of entitlement is adjusted upwards as well. Such an adjustment might not seem very relevant when the effective retirement age currently remains well below the current official retirement age, and those not working are almost

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invariably receiving some other form of public income support. But it would help to allow the actuarial adjustments available in private pensions to play a stronger role. It would also send a clearer signal about expectations that people should plan to rely more heavily on their own resources (earnings from work or savings) for a longer period and reinforce the point that state funding of earlier retirement should be restricted to those genuinely incapacitated. Such adjustments would also be consistent with improvements in the average health status of older workers.

Raising average hours worked

A number of factors contribute to the trends in average working hours, identified in Chapter I. Part of the explanation lies in patterns of average weekly working hours, which are closely related to effective marginal tax rates (see Chapter IV). But there are also a number of reasons why people are temporarily absent from the labour force and typically also drawing public transfers at the same time, notably due to sick leave and maternity leave.

Sick leave

Sick leave taken in Denmark corresponds to the equivalent of around 5 per cent of the labour force being absent from the workplace at any point of time, and publicly-funded sickness benefits were paid to the equivalent of around 70 000 persons (*i.e.* around 2½ per cent of the labour force) over a full year in 2001. As in most countries, almost half of the instances of sick leave last less than two weeks, the period during which payments are the responsibility of the employer. But in 2001, about 45 000 people were on sick leave for between 26 and 52 weeks, while some 15 000 were on sick leave for more than one year (Beskæftigelsesministeriet, 2003), although sickness benefit duration is notionally limited to 12 months, and municipalities must bear full costs of sickness benefit beyond that period.[33] Furthermore, under the old disability rules, some 9 per cent of those exiting long-term sick leave (more than one year) were awarded disability pensions (see above), while only around one third return to the labour market. The proportion of beneficiaries shifting into rehabilitation programmes rises markedly after one year, whereas the findings of the OECD cross-country review of disability (OECD, 2003c) point to the value of early and strenuous rehabilitation efforts in raising the likelihood of return to work. It has also been estimated that the probability of sick leave extending beyond one year actually doubles if the municipality has not carried out its obligation to follow up the case at the latest by the end of the first eight weeks (Danish Employers Federation, 2002). A further concern is that with the new disability arrangements, some of those who might previously have gained a pension may now remain on sick leave instead for much longer periods.

Clearly many sickness absences are for entirely legitimate reasons, although surprisingly few absences are supported by a doctor's certificate, even

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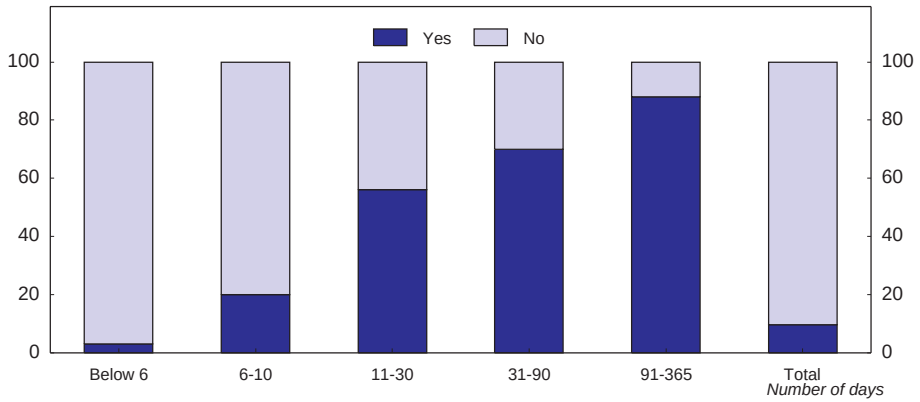
Figure 17. **Duration of sick leave and medical certificates**
Per cent, 2002



Note: Respondents were asked whether a doctor's certificate was obtained for their most recent absence from work.
Source: Megafon (2002).

after the first 10 days of a sickness spell (Figure 17). But differences in sickness rates between sectors, municipalities and individuals could suggest some scope for gains in aggregate hours worked and savings in public expenditures (Danish Employers Federation, 2002). Sick leave is higher among employees in the public sector than in the private sector,[34] after adjustment for different demographic characteristics, such as gender and age.[35] There are also marked differences in long-term sick leave across municipalities, again after adjusting for demographic differences. Short-term sick leave also appears to be disproportionately high in the Copenhagen area for all types of work, the largest gap being for female manual workers whose rate of short-term sick leave is 50 per cent higher than in other parts of the country. Nevertheless, this apparently has very little to do with the working environment, which more than 80 per cent of Danish workers classify as "good", or with workplace accident rates, which have fallen since the mid-1990s.

Increasing the labour supply by reducing sick leave may require careful efforts, especially to avoid penalising those genuinely in need of assistance. Nevertheless, there are several steps which could be taken to tighten up the system, particularly to address the lack of follow-up and timely rehabilitation efforts. These elements could include:

– Making a doctor's certificate obligatory for receipt of public sickness benefits (*i.e.* after the first two weeks of sickness) and for the continuation of benefits after a certain interval. Using independent medical

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officers to make assessments in certain cases and tightening administrative controls would help to minimise the moral hazard risk.

– Enforcing the 12-month limit and requiring the municipalities to assume full costs after a shorter period of time. This could reinforce their incentives to carry out their obligations for follow up more assiduously and to deliver better results.

Parental leave

Denmark has generous provisions for maternity leave by OECD standards, as parents have the right to the equivalent of 12 months of full-time income benefits, which can be shared between the parents, taken part time and partly postponed until the child is 8 years old. In 2002, the number of days with state-financed maternity leave benefits amounted to the equivalent of 40 000 full-time persons. While Danish arrangements have a number of attractive features, especially in terms of equity, they also have significant drawbacks (OECD, 2002d). The OECD review of Denmark's policies to reconcile work and family life concluded that paid leave should cover only a short period of child rearing and that a disproportionate amount of policy attention was accorded to leave arrangements, when the real task was to get a better balance throughout childhood. It also pointed out that long leave periods can be harmful to women's future earnings prospects, a finding reinforced by Danish research (Gupta and Smith, 2001). This analysis suggests that perhaps Denmark has gone beyond the length of time where the benefits outweigh the costs.

Summing up

Overall, there are a number of measures and options available to increase labour force participation in coming years. The main focus of attention should be on extending working life at both ends; by speeding transition from education into work and delaying exit. In a number of areas, the task of boosting employment would be made easier if stronger economic incentives made working significantly more attractive relative to drawing public income support than is currently the case. This requires making tough choices about how generous the welfare system should be. But weak economic incentives make the burden of achieving the same results through other channels more difficult. They could also indicate a degree of policy incoherence between the clear political objective and economic requirement to raise labour supply and an apparent political and public reluctance to use the most effective mechanisms available to achieve those ends. Taking all these factors into account, some recommendations are summarised in Box 1.

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Box 1. Recommendations on policies to increase labour supply

Boosting employment

- Continue to emphasise the self-reliance message, and look at ways of reinforcing it with stronger economic incentives. Reduce the effective net replacement rates for those who become unemployed in the future, for example by tapering benefits over the length of the unemployment spell. Shorten the currently long duration of unemployment insurance and increase the working time required to qualify for full unemployment insurance benefits.
- Monitor the results of the "More People in Work" strategy, and stand ready to tighten income support eligibility rules further and to impose stricter sanctions for non-compliance.

Earlier entry to the labour force

- Refocus the pedagogical objectives and content of the voluntary 10th form so that it becomes a remedial option only for those students whose academic performance impedes them from moving directly to upper secondary education.
- Consider introducing tuition fees for higher education.
- Reduce the period for which students in higher education can receive study grants and shift the balance of income support from grants towards a greater proportion of student loans. Make continued assistance more closely linked to academic performance.

Raising effective retirement age

- Abolish the voluntary early retirement benefit scheme or make it a defined-contribution scheme. Alternatively, adjust the rules to push up the effective retirement age.
- Raise the qualifying age for public old-age pension in line with projected increases in life expectancy.
- Make it easier for those on early retirement (disability) pensions to try out working and provide them greater assistance in their efforts.
- Make all disability pensions subject to periodic review, to take into account, *inter alia*, new medical and rehabilitation opportunities.

Sickness benefits and maternity leave

- Make a doctor's certificate obligatory after the first two weeks of a sickness spell in order to establish entitlement to public sickness benefits. Provide scope for independent medical assessment and tighten administrative controls to minimise moral hazard.
- Shift more of the costs of sickness benefit onto the municipalities at an earlier point in the sickness spell.
- Examine whether the costs of the long duration of maternity leave might outweigh the benefits, and consider whether some shortening might be warranted.

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III. Migration and integration of immigrants

Immigration could offer one way for Denmark to expand its labour supply, thereby lowering the dependency ratio, at least for some time, and easing the task of ensuring fiscal sustainability (see Chapter I). However, these beneficial effects are obtained only if immigrants are in work. Yet a significant proportion of immigrants have found it quite difficult to get work in Denmark, while the country has been relatively unattractive to high-skilled foreigners. Furthermore, the structure of the economy not only makes it difficult for low-skilled foreigners to gain a foothold in the labour market, but also provides generous social benefits that have caught many of the least skilled immigrants in a benefit trap. A heightened appreciation of these problems, including a tighter focus on the economic situation of the immigrants already present, have underpinned the main changes in policies on immigration in recent years.

This chapter first sets out the trends in immigration since the early 1960s and the key demographic characteristics of the foreign population. The next section reviews the entry policies that currently apply for those immigrants from outside the European Economic Area. The recent changes affecting each main type of entry – asylum seekers, family reunification and highly skilled immigrants – are set in an international context and linked back to the authorities' wish to better integrate those immigrants that are already in Denmark. The challenges of integration are discussed in the following section. The government's ultimate goal is that foreigners participate in economic activity on an equal footing with native Danes. Such an ambitious goal will obviously not be reached quickly or easily, and the authorities consider the critical intermediate objective to be getting more immigrants into jobs as a prerequisite to their future economic success. The measures already put in place go in the right direction, and the concluding section makes some suggestions as to how policies might be improved further.

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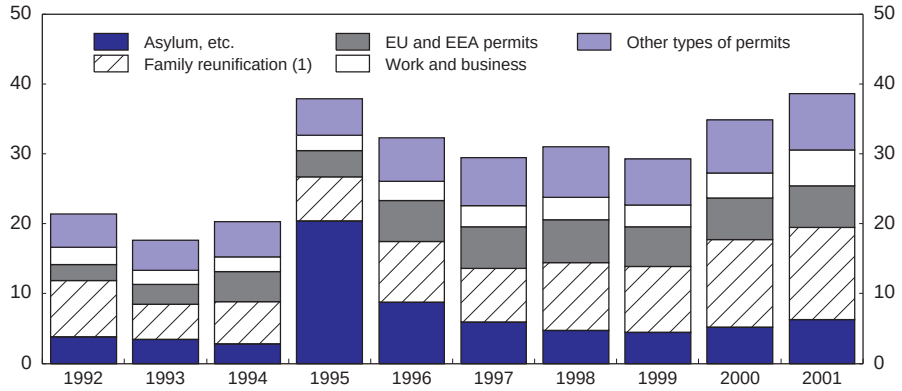
Immigration patterns

Denmark has not traditionally been a popular destination for immigrants: immigration is a relatively recent phenomenon and has involved several distinct waves:

- Before 1960, the few foreign-born residents came almost exclusively from other Nordic countries, Germany and the United Kingdom.
- From 1960 to around 1974, Denmark permitted companies to recruit a significant number of workers from abroad, mostly from Turkey, Yugoslavia and Pakistan, in response to labour-market shortages.
- Since 1985, *de facto* refugees and asylum seekers have become significant. They have come especially from Sri Lanka, Iran, Iraq and Lebanon, and also in the 1990s from the Balkan states, Afghanistan and Somalia.
- As the earlier groups have become more established, entry for family reunification purposes has expanded: in the past few years these entrants have exceeded the flow of asylum seekers (Figure 18). Around half of the family reunification permits issued in 2001 involved spouses, of which almost one-quarter provided entry for people under the age of 24.

Figure 18. **Residence permits granted**
Thousands



1. From May 2000, all children born to parents holding permanent residence permits are required to hold a separate residence permit. It is estimated that this technical change accounted for around an additional 2 500 permits in 2000 and 2001.

Source: Statistics Denmark.

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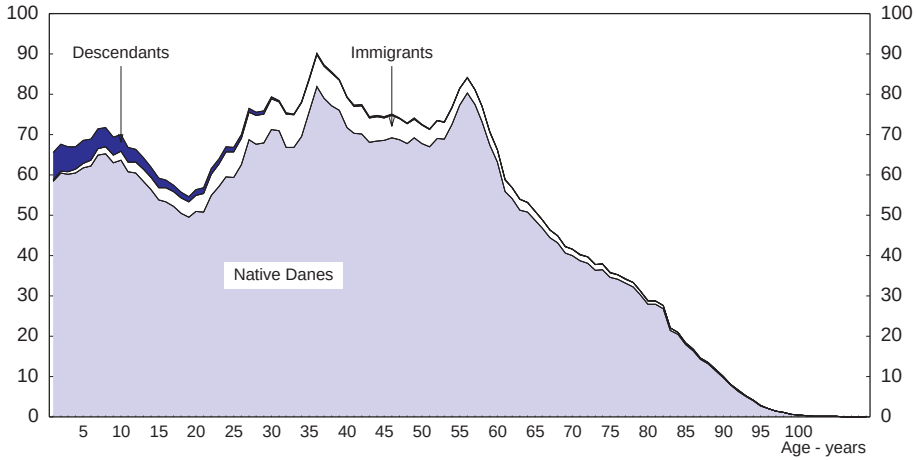
Figure 19. **Population by age and origin**
Thousand persons, 1 January 2002



Source: Statistics Denmark.

As of 1 January 2002, there were almost 332 000 foreign born ("first generation") immigrants and 99 000 descendants ("second generation") living in Denmark, amounting to close to 8 per cent of the total population. Three-quarters of them came from less developed countries (LDCs), with more than half of these having arrived in the last 10 years. Most immigrants are aged between 25 and 49, while most descendants are younger than that (Figure 19). The current age structure, along with significant differences in fertility rates, means that the proportion of immigrants and descendants is projected to almost double over the next 20 years, according to estimates prepared by the DREAM group (see Chapter I) for the Task Force on Integration established by the Ministry of the Interior and Health and now placed under the Ministry of Refugee, Immigration and Integration Affairs. Overall, the growing proportion of immigrants and descendants over time suggests that any associated difficulties are likely to become more important if not addressed. Denmark's immigration trends can also be set in an international perspective. As a small country, it admits significantly fewer foreigners in absolute terms than many larger countries, but foreigners as a percentage of the population is around the middle of the OECD range while the inflow of migrants per 1 000 inhabitants increased significantly between 1985 and 2000 (Figure 20).

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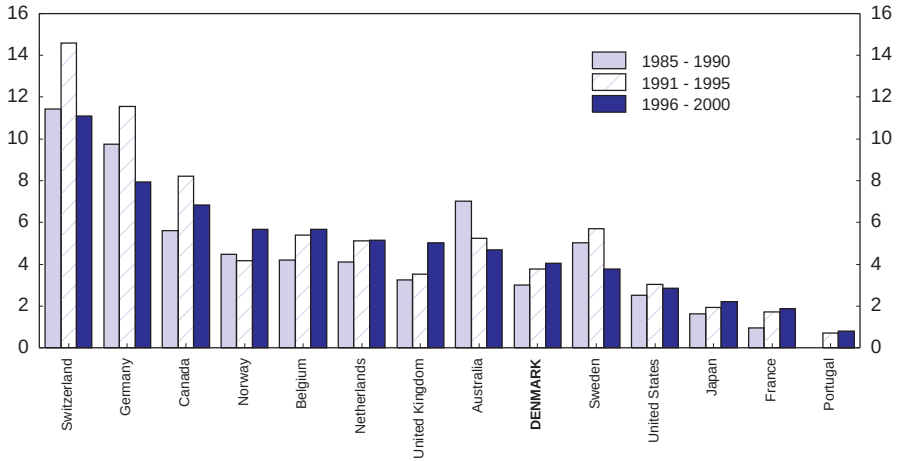
Figure 20. **Inflows of foreigners to selected OECD countries**
Per 1 000 inhabitants



Source: OECD, *Trends in international migration*, 2002.

In contrast to a number of other European countries (see, for example, OECD, 2003d), Denmark does not appear to have a significant number of illegal immigrants and illegal immigration is generally assessed to be almost entirely limited to the construction and entertainment sectors. This is probably for two main reasons:

- It is generally difficult for illegal immigrants to find work, as the social partners monitor, and if necessary, sanction employers using illegal workers or under-paying legal immigrant workers (Hjarnø, 1996). Also, the widespread use of the unique civil registration number in public and private sector administrative systems makes it quite difficult for people to function in Denmark without one.
- The traditional ease with which someone arriving in the country could claim financial assistance means that there is no economic advantage in remaining clandestine.

Labour force participation is a key indicator of progress towards becoming integrated into society. Denmark is well known for its high overall participation rates, especially amongst women, but immigrants are significantly less likely to participate in the labour market (see Chapter I). This gap stands in contrast to a significant number of other OECD countries, where foreigners are either more

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Table 14. **Activity rates in selected OECD countries**
Per cent, 1999-2000 average

	Men			Women		
	Nationals	Foreigners	Difference	Nationals	Foreigners	Difference
Australia (August 2000)[1]	75.3	67.3	–8.0	58.9	49.1	–9.8
Austria	80.5	86.1	5.6	63.1	63.4	0.3
Belgium	74.1	73.0	–1.1	58.2	40.7	–17.5
Canada (1996)[1]	73.8	68.4	–5.4	60.2	52.9	–7.3
Czech Republic	80.4	88.6	8.2	64.4	61.6	–2.8
Denmark[2]	**79.2**	**63.0**	**–16.2**	**75.5**	**48.3**	**–27.2**
Finland	79.8	81.1	1.3	74.4	58.0	–16.4
France	75.6	76.4	0.8	63.5	48.5	–15.0
Germany	80.1	77.9	–2.2	64.8	49.9	–14.9
Greece	78.9	89.3	10.4	50.3	57.9	7.6
Hungary[1]	67.9	73.0	5.1	52.5	53.2	0.7
Ireland	81.1	76.1	–5.0	55.7	54.4	–1.3
Italy	74.8	89.0	14.2	46.3	52.1	5.8
Luxembourg	75.5	77.9	2.4	47.3	56.7	9.4
Netherlands	84.8	67.2	–17.6	66.4	44.6	–21.8
Norway	86.0	84.5	–1.5	77.7	70.7	–7.0
Portugal	83.7	81.3	–2.4	66.7	68.5	1.8
Slovak Republic	76.6	79.5	2.9	62.6	63.9	1.3
Spain	77.2	83.8	6.6	49.8	57.3	7.5
Sweden	80.5	65.1	–15.4	75.3	59.4	–15.9
Switzerland	93.0	89.6	–3.4	74.8	68.4	–6.4
United Kingdom	84.9	76.2	–8.7	69.2	56.0	–13.2
United States[1]	73.4	79.6	6.2	61.6	53.7	–7.9

1. Data refer to persons born abroad or in the host country.
2. As at 1 January 2000.

Source: Statistics Denmark, *Labour Force Surveys*, data provided by Eurostat and by the Australian Bureau of Statistics, 1996 census, Statistics Canada, *Current Population Survey*, US Bureau of the Census.

likely to work than nationals, or where their participation rates are only slightly lower (Table 14). A closer look at the data for Denmark illustrates how much this depends on country of origin: well under half of some nationalities are active on the labour market (Table 15). Participation rates are higher for groups that have already spent a longer period in Denmark. Indeed, those who have spent more than 10 years in Denmark have a participation rate around double that of those who have been in the country two years or less. However, the significant shifts in the mix of migrants over the last decade caution against expecting that time alone will produce the same improvements for more recent arrivals.[36] Another key element in determining the degree of labour force participation is age at arrival. This relationship probably reflects increased exposure to the Danish education system and stronger language skills, as well as a greater flexibility and willingness to adjust to the Danish working environment among young people (Figure 21).

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Table 15. **Participation rates among different immigrant groups**

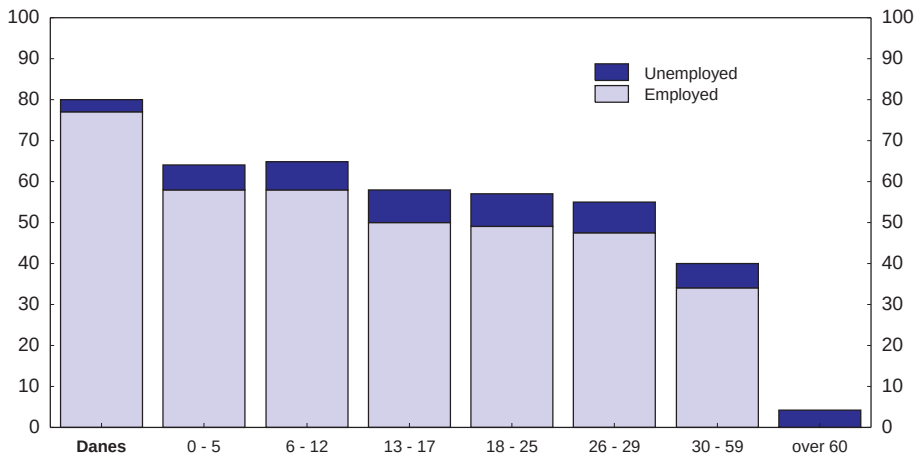
Per cent, as at 1 January 2000

Country of origin	Male	Female	Total
All countries	**63**	**48**	**56**
United Kingdom	79	64	73
Sweden	74	70	72
Germany	72	56	64
Vietnam	71	53	62
Turkey	71	46	59
Oceania	69	49	61
Poland	68	62	64
Pakistan	68	34	53
Chile	67	56	62
Norway	66	62	64
Iran	62	41	54
North America	60	52	56
Ex-Yugoslavia	58	42	51
Unknown and stateless	46	27	39
Lebanon	46	17	33
Iraq	33	15	26
Somalia	25	10	18

Source: Ministry of the Interior and Health.

Figure 21. **Labour force participation by age at arrival**

Immigrants from less developed countries, aged 16-66 years
Per cent, 1 January 2001



Source: The Ministry of Refugee, Immigration and Integration Affairs' database in Statistics Denmark.

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Table 16. **Working-age population outside the labour force**

	Less developed countries		EU, Nordic countries and North America		Native Danes	
	Men	Women	Men	Women	Men	Women
Participation in the workforce	59.0	43.0	71.0	62.0	83.0	75.0
Students in further education	4.5	5.1	2.9	3.8	2.9	3.5
Temporarily outside the labour force (activation, cash benefits, maternity, etc.)	18.0	26.8	3.2	3.8	2.2	4.0
Early retirement schemes (including disability)	7.8	5.7	6.7	12.1	9.7	13.8
Others, not in labour force	10.7	18.8	16.0	17.9	2.2	3.8
Total outside workforce	**41.0**	**57.0**	**29.0**	**38.0**	**17.0**	**25.0**
Total number of persons outside labour forces	**37 200**	**50 900**	**11 900**	**15 000**	**289 800**	**403 600**

Source: SOPEMI report for Denmark, 2001.

There are also quite different patterns of reliance on public transfers for those who are of working age but outside the labour force (Table 16). Almost one in five males is receiving income support through activation measures, cash benefits etc., and more than one in four women are receiving this type of support or maternity pay, while a dramatically smaller fraction of native Danes and immigrants from more developed countries is drawing such benefits. This has a double fiscal effect: not only are these people not adding to tax revenues, but they are also drawing public expenditures.

Entry policies

The strong increase in immigration over the past decade, together with relatively weak integration of the earlier cohorts of immigrants from some LDCs, has led to a reconsideration of entry policies. The shift in policies was motivated in large part by three factors:

- a belief that Denmark had become a magnet for low-skilled asylum seekers because of both easier entry and more generous public income support than provided by most other countries (see Box 2);
- a concern that entry procedures made it particularly difficult for firms to recruit highly skilled immigrants;
- a realisation that economic and social integration of the foreigners needed improvement and that until the backlog of poorly integrated migrants had been effectively dealt with, adding more newcomers with a low probability of success would make the task more difficult.

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Box 2. Welfare assistance for asylum seekers

Denmark has traditionally provided generous welfare assistance to all residents, including new arrivals. Asylum seekers are lodged in special reception centres – most of them run by the Red Cross, under contract – while their applications are considered. They are provided with not only food, lodging and health care, but also with pocket money amounting to around 30 euros per week, significantly more than received by their counterparts in Austria, Belgium, the United Kingdom, Germany or Sweden, although slightly less than is paid in Norway (Ministry of Foreign Affairs, 2002). However, in January 2003 the government put forward a bill that lowers the welfare assistance to asylum seekers significantly. It was adopted by the Parliament in April 2003 and enters into force on 1 July 2003.

Furthermore, until the changes in mid-2002 (which imposed the residency requirement of seven out of the previous eight years for full access to social welfare benefits), asylum seekers recognised as refugees were entitled to receive full cash benefits immediately. Full welfare benefits are generous, both by international standards and especially relative to the income from work those unskilled immigrants could hope to obtain (see below). It is hard to know to what extent this generosity acted as a magnet to asylum seekers, although a number of countries operate some sort of waiting period before granting immigrants full access to the social welfare system. However, the Aliens Act of 2002 entitles the Danish Immigration Service to deny asylum seekers cash benefits if they fail to meet certain requirements.

In response, the rules governing entry have been changed incrementally since around 1998, with a series of legislative amendments to tighten both entry and integration requirements.[37] In contrast to a number of other countries, Denmark does not operate quotas for any category of immigrants, nor does it make more than a technical distinction between residency and work permits:[38] for practical purposes most legal residents can work.

Asylum seekers

Faced with the sharp increase in asylum seekers over the 1990s, Denmark enacted a new law on entry from 1 July 2002 (Annex VIII). Reflecting this, there was a sharp drop of around 50 per cent in the number of applications for asylum in Denmark in 2002, whereas overall applications to the European Union remained stable, and countries such as Sweden and the United Kingdom registered strong increases. Denmark also stood out as having the highest rate of recognition (*i.e.* granting refugee status to asylum seekers) of all industrialised countries over the decade to 2001 (Figure 22), dramatically higher than the European Union

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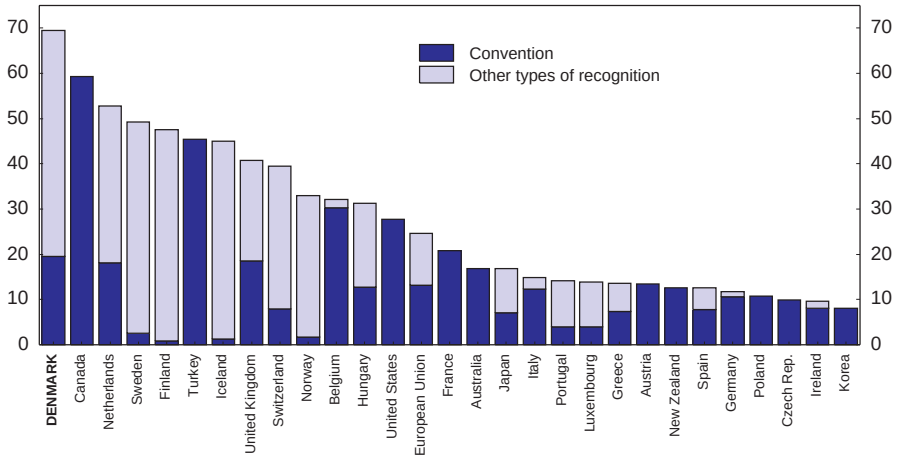
Figure 22. **Recognition rates for asylum seekers**
Per cent, 1992-2001



Source: UN Human Rights Commission (2002), Statistical Yearbook 2001.

average. The new rules adopted in the Aliens Act 2002 bring Denmark more closely into line with other countries' practices, even as a number of other OECD countries are also tightening their rules. In a further development, the European Union has now reached agreement on a common set of rules to be applied concerning asylum, both to provide clear protection to those who genuinely need it and to prevent "asylum shopping" by those seeking to take advantage of variations in treatment between different EU destination countries.

Another element that has changed in Danish policy towards refugees is increased emphasis on the option of sending refugees home if the situation in their country of origin becomes safe once again. For refugees from Afghanistan, a common concrete repatriation plan had been adopted by the European Union during the Danish presidency. The Danish Board of Immigration had concluded in mid-2002 that for many Iraqi asylum seekers it was no longer dangerous for them to return to Iraq,[39] although this judgement has since been suspended. These two countries alone accounted for around 40 per cent of asylum seekers between 1998 and 2001. At the same time, new rules on granting permanent residency (which once granted, protects the immigrant from being repatriated) operate to provide a stronger incentive for refugees to make every effort to integrate. It is thus a requirement for obtaining permanent residence that the immigrant completes an introduction programme, passes a Danish language test and has no overdue debt

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to public authorities. Furthermore, in March 2003 the government put forward a bill which *inter alia* implies that permanent residence can be granted earlier than after seven years of legal residence in Denmark, if the applicant is well integrated into Danish society and can support himself. On the other hand, the Repatriation Act 1999 allows for the possibility that the cultural and economic gap to be bridged for some immigrants and refugees is so large that their best option may lie in returning to their country of origin, and it therefore provides financial assistance for voluntary resettlement.[40]

Family reunification

Family reunification has been a somewhat contentious aspect of Danish immigration amid concerns that some ethnic groups[41] were disproportionately using these rules to expand overall immigration from their countries. It may slow the integration process of the sponsor, most particularly for young people who bring in a spouse. Another concern has been whether these rules may have facilitated forced marriages in some cases, although it is difficult to assess how significant this problem is, as arranged marriages are a relatively common feature in some ethnic groups in any case.

Virtually all countries operate family reunification policies, and the strictness of Denmark's rules appears to fall within the range of policies that are applied elsewhere (OECD, 2001b).[42] The requirements for sponsorship that apply in different countries show many common elements, reflecting that they address similar concerns. Sponsorship is generally required to ensure that those coming into the country do not become reliant on state income support. Adequate housing must also be provided. For Denmark, the sponsor must – unless exceptional reasons make it inappropriate – take financial responsibility for the family member until he or she obtains permanent residence – otherwise the residence permit may be revoked. The main difference in the latest changes, brought into force in mid-2002, is that the same rules now apply to Danish citizens who seek to sponsor a foreign citizen under family reunification provisions, as well as immigrants. Previously Danish citizens were exempt, leading to claims of discriminatory treatment against immigrants. Other changes such as tightening the "ties to Denmark" provision, further limiting unification by marriage for those under the age of 24 years, and increasing the self-support requirements primarily serve to reinforce the policy changes made in May 2000. In any case, the total number of permits granted for reunification for spouses and partners has remained relatively stable in recent years, while the number of applications from residing immigrants, apart from refugees, has actually fallen since the peak in the mid-1990s (Table 17). In any case, only one-quarter of reunification permits granted for spouses in 2001 involved a partner less than 24 years old and it is unclear to what extent the tightening of these provisions will lead to postponements of such marriages or to a

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Table 17. **Family reunification permits granted and denied**

	1996			2001		
	Granted	Denied	Per cent granted	Granted	Denied	Per cent granted
Spouses and partners	6 122	895	87.2	6 499	2 043	76.1
of which:						
To residing refugees	1 338	193	87.4	1 694	297	85.1
To residing immigrants	1 325	344	79.4	443	807	35.4
To residing Danish or Nordic citizens	3 449	358	90.6	4 362	939	82.3
Under age children	2 429	431	84.9	6 422	684	90.4
of which:						
To residing refugees	1 309	172	88.4	3 777	314	92.3
To others residing in the country	1 120	259	81.2	2 645	370	87.7
Parents over the age of 60	186	565	24.8	266	559	32.2
of which:						
To residing refugees	71	379	15.8	71	370	16.1
To residing immigrants	38	137	21.7	0	9	0.0
To residing Danish or Nordic citizens	77	49	61.1	195	180	52.0
Total	**8 727**	**1 891**	**82.2**	**13 187**	**3 286**	**80.1**
of which: To residing refugees (per cent)	31	–	–	42	–	–

Source: Ministry of Refugee, Immigration and Integration Affairs.

higher tendency to marry Danish residents instead. However, new data show that applications for family reunification have fallen since the latest changes brought into place in mid-2002.

Highly skilled immigrants

Recent policy changes have made it somewhat easier for highly skilled immigrants to come and work in Denmark. European Union and Nordic citizens are entitled to move to Denmark anyway, and the latest modifications to the rules are intended to grant easier access to those of other nationalities with needed skills. The job-card scheme means that foreigners from other countries who have a concrete job offer within one of the sought-after occupational categories will be able to get work and residence permits within 30 days, as long as the salary and terms of employment conform to standard Danish employment conditions.

Even with these changes, Denmark's approach is still quite a long way from the approach taken by countries such as Australia, Canada and New Zealand, all of which operate a "points" system. Under this approach, these countries encourage anyone who wants to immigrate to apply and then select the "best" according to an evolving set of criteria. The emphasis is on selecting those most likely to be able to establish themselves economically, rather than trying only to

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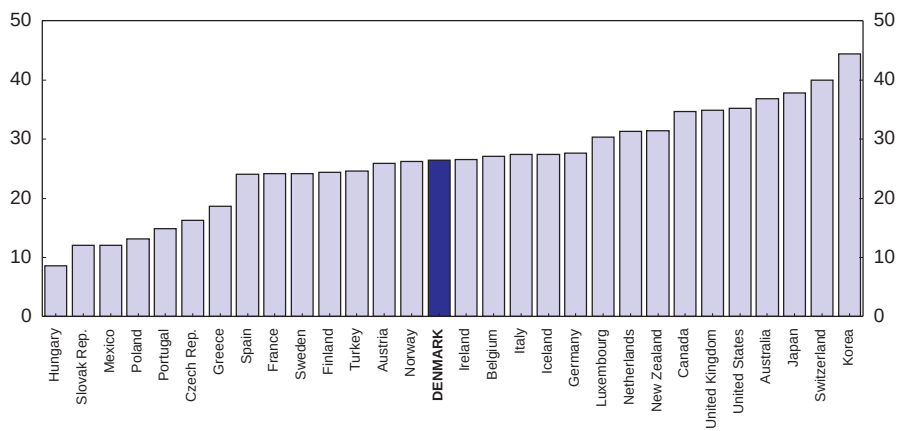
match particular skills or requiring concrete job offers. Indeed, evidence from Canada indicates that job offers are an unreliable indicator of integration potential, and, thus, this factor is no longer given any weight in selection.

In competing to attract highly qualified migrants, Denmark faces two disadvantages: language and relatively low after-tax incomes. Almost all highly skilled migrants already have some English language skills, giving English-speaking countries a significant advantage as hosts that Denmark can do relatively little about. Furthermore, despite easing entry requirements, it may remain a less attractive destination than those countries where wages reflect more closely the international value of such human capital and taxes are less onerous (Figure 23). The special tax relief scheme for foreign researchers and key employees recruited abroad[43] has the effect of raising their after-tax incomes.

Another possible consequence of relatively low after-tax incomes compared with other countries is that this might encourage well-qualified Danes to move abroad. This would be facilitated by the scope for free movement of labour within the European Economic Area, while most well qualified Danes also have strong English language skills. Although there are a significant number of Danes living abroad, and some commentators have pointed to a risk of a "brain drain", it is rather difficult to conclude that a significant problem currently exists. Thus far, it would appear that the majority of exits are temporary: after six years,

Figure 23. **Annual after-tax income for high-earners[1] in selected OECD countries**
Thousand US dollars, 2000, current PPPs



1. Single person earning 167 per cent of the average production worker wage.
Source: OECD, *Taxing Wages, 2000-2001*.

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three-quarters of expatriate Danes have returned home (S.B. Nielsen *et al.*, 2002). It could be argued that such international experience is a plus for Denmark. Nevertheless, some concerns about the risk of a future "brain drain" persist.

The broader issue is the extent to which the population move either temporarily or permanently in order to minimise their lifetime tax payments while maximising their lifetime access to, or use of, the generous public services and income support offered in Denmark (*i.e.* "social shopping"). This is an inevitable risk in a system where graduates have received extensive public funding to acquire their education and where access to many parts of the social welfare system is largely disconnected from contributions[44] (see Chapter II).

Improving the integration of immigrants from less-developed countries

Improving the economic integration of immigrants from LDCs is a challenging task and has become a priority for the Danish government. The Danish Think Tank on Integration was set up in November 2000 by the then Minister of the Interior to clarify the integration of foreigners in Danish society, analyse future trends in the number of foreigners and address the social consequences. It established seven criteria for successful integration (Think Tank on Integration in Denmark, 2001): Danish skills and education; employment; economic independence; lack of discrimination; contact between foreigners and Danes; participation in political life; and fundamental values and norms. The Think Tank considered that education, employment and economic independence are crucial, and participation in the labour market is the most important precondition to success. These seven criteria were accepted by the government as appropriate benchmarks. The government's current strategy is particularly focused on the economic integration of foreigners, and this chapter also addresses this aspect of integration.[45]

The importance of a job

Even in the best of circumstances, full economic integration cannot be achieved overnight and may take even longer than a generation, but the objective for policy is to speed up the process as much as possible; indisputably, the critical step along the path is getting a job.[46] Immigrants' participation in the labour force is significantly lower than for native Danes, and unemployment amongst foreigners is also considerably higher (Figure 24). Getting migrants into work depends on both the availability of jobs that correspond to their skills, and the willingness of migrants to seek work. At the most basic level, the former requires either providing more low-skilled jobs, or raising the skills of immigrants to the point where they can all compete with native Danes for the jobs that are typically available.

In some countries, low-skilled work is relatively abundant, albeit at very low wages, and immigrant workers tend to be concentrated in certain sectors that employ a relatively large share of unskilled workers, most often agriculture and services such as hotels and restaurants.[47] In Denmark, these low-skilled jobs are

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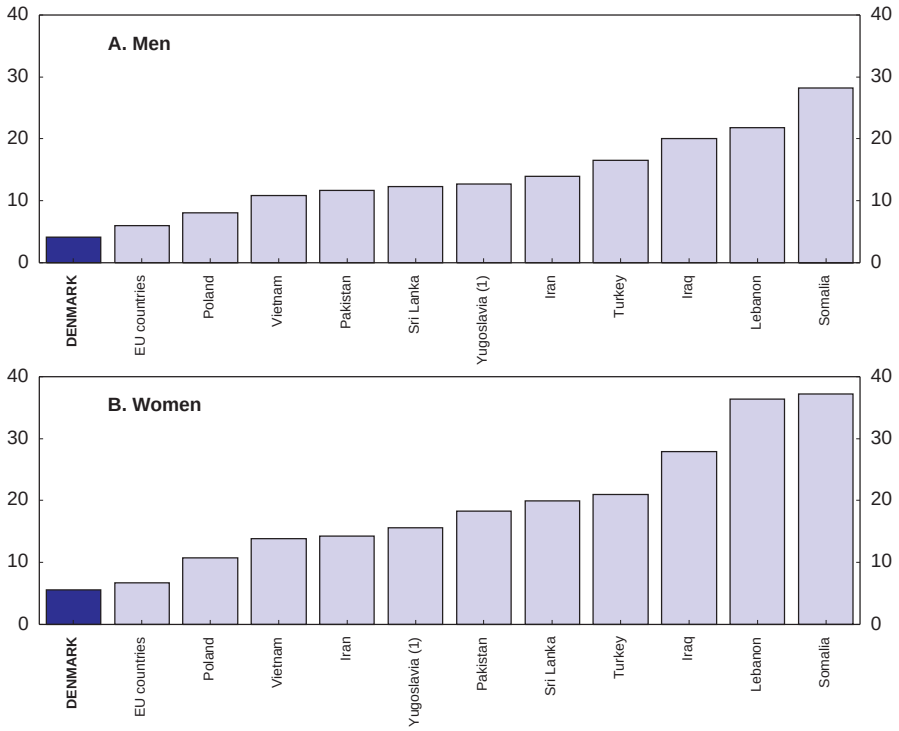
Figure 24. **Unemployment rates by country of origin**
Per cent, 2001



1. Former Yugoslavia.
Source: Statistical Yearbook 2002.

comparatively scarce (Figure 25). The economic structure has evolved into one where wages at the bottom end of the wage scale are relatively close to the average; the lower quartile is paid up to 83 per cent of the median hourly wage, while the ratio of the bottom decile to the median wage is one of the highest in the European Union.[48] This wage compression at the low end would be a problem for low-skilled Danes as well, but with extensive public funding for higher education and skills acquisition, their wages and skills are better matched, resulting in high employment. But low-skilled migrants have a tougher time in this environment to find jobs that they are capable of doing, since generous income transfers and high effective minimum wages in collective agreements keep wages from falling to levels where they could price themselves into employment.

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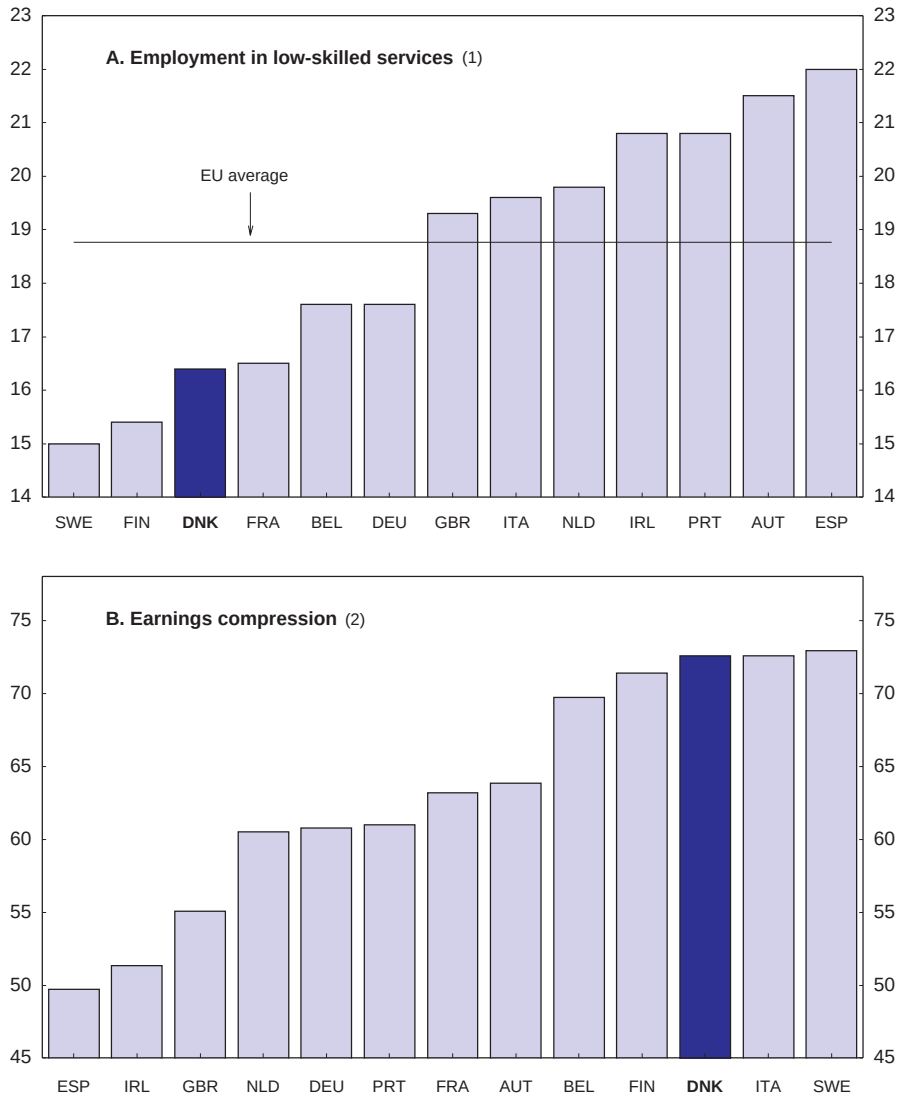
Figure 25. **Low-skilled jobs and wage compression**

Per cent



1. Wholesale and retail trade, repair of motor vehicles, hotels and restaurants.
2. Earning of lowest decile as a percentage of median, 1998 or latest year available. Data are based on gross earnings of full-time workers.

Source: Eurostat; OECD Earning database.

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Furthermore, with a relatively high tax wedge on labour income, households have an economic incentive to reduce their working hours and carry out more domestic chores and home maintenance themselves, even if their comparative advantage and personal preferences might lead them to hire someone else if the tax wedge were smaller. Once more, this trade-off seems to be relatively well accepted by native Danes, who undertake a significant proportion of major house repairs themselves (Brodersen, 1998). But it reduces the opportunities for low-skilled immigrants to find suitable work providing household services, such as cleaning and gardening, where immigrants are typically over-represented in most countries. Alternatively they may be undertaking work in the black economy, but there is little precise knowledge about the extent of this.[49]

Wage subsidies – bridging the gap between wages and productivity

One policy response to the gap between wages and immigrant productivity would be to develop wage subsidies of some sort, targeted on not yet skilled migrants to enable them to gain employment in the ordinary labour market. A new scheme has been announced that provides gradually diminishing wage subsidies for newly-hired immigrants, while providing in-the-workplace language training as well as job-specific skills (see Box 3). In principle, this approach has a great deal to recommend it, especially as it integrates training in the workplace and is designed to take immigrants to the point where they can continue working, without requiring ongoing subsidies. Since getting the job may be the most difficult step of all, this could be all the help those immigrants need to achieve economic independence. It is too early to judge the likely success of this strategy, which may depend on careful selection of those immigrants best equipped to make good use of the experience. It could also prove costly unless a clear and rapid phase-out path for each participant is adhered to. However, the immigrants with the weakest skills to start with may not even be capable of entering such a programme. They will still be able to take courses in Danish and build up their skills to the point where it is possible to introduce them to a workplace.

Self-employment

One way of bringing hourly wages for immigrants down to levels that make employment viable is self-employment, where effective minimum wages no longer act as a constraint. Self-employment is much higher among immigrant groups, although earnings are much lower than for those same immigrant groups in dependent employment (Table 18). In contrast, Danish entrepreneurs on average earn almost one-quarter more than Danish wage-earners, but they tend to be better educated and are much less likely to be concentrated in low value-added business sectors (Bager and Rezaei, 2001).[50] Self-employed immigrants work very long hours, with only around 10 per cent working 39 hours or less per week, while

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Box 3. Integration through the workplace

In May 2002, the social partners, municipalities and the government signed a new agreement, "The Integration Package for Immigrants". This focuses on rapid introduction to working life and acquisition of Danish language skills through the workplace itself. It is motivated by the belief that the workplace is the best environment for immigrants to learn the skills they need, including Danish language, for successful labour market participation. It also reflects a realisation that the workplace provides excellent opportunities for closer contact between immigrants and Danes. The necessary legislation was introduced to Parliament in March 2003.

This agreement, *inter alia*, proposes a new model for workplace integration, involving three stages after an introduction to Danish language and the labour market:

Stage 1. Workplace introduction phase

- Immigrants undergoing workplace introduction will not be covered by an employment contract, and the enterprise will not pay them any wages. However, they will receive public income support (social assistance or introduction benefit).
- Danish language teaching will be the primary focus and will take place on site. It will be paid for by the municipalities, who will also cover the cost of other skills training.
- The workplace introduction phase will last up to 13 or up to 26 weeks if special needs apply, depending on the specifics of the programme. People with very special difficulties might even participate for a longer period if this is assessed to be relevant.

Stage 2. Workplace training phase

- Immigrants are recruited by the enterprise, and are paid according to collective agreements.
- The immigrant's time will be split between working and training, and employers will pay only for the periods when the employee is actually working. Specific job-training might also be used as an instrument, during this stage.
- The public authorities will pay the costs of continuing Danish language and other skills training and to some extent subsidise wages.

Stage 3. Ordinary employment

- The immigrant will be in ordinary employment and receiving hourly wages from the company employing him or her, like any other employee. There is a scope to reach agreement to pay lower wages to persons not yet completely skilled for the job. However, such employment may be part time, and combined with Danish courses or skills training, paid for by the public authorities.

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Table 18. **Gross income for self-employed and wage-earners**
18-59 year olds, 1996

Country of origin	Self-employed DKK	Wage earners[1] DKK	Gap DKK	Self-employed as per cent of active labour force
Denmark	**285 200**	**233 200**	**52 000**	**8.7**
Iran	104 600	183 400	–78 800	29.3
Pakistan	136 600	167 900	–31 300	30.7
Ex-Yugoslavia	176 500	190 800	–14 300	7.0
Turkey	114 300	166 300	–52 000	18.7
China	126 500	181 000	–54 500	21.7
Other countries	189 300	222 600	–33 300	13.2
Total	**277 300**	**232 300**	**45 000**	

1. Includes unemployed.
Source: Statistics Denmark; Bager and Rezaei (2001).

the majority work 70 hours or more. Low returns to business ownership would seem to imply that those who could get a job working for someone else would prefer to do so. This suggests a very low hourly productivity and implicit hourly wage. Thus, low-skilled immigrants tend to opt for self-employment for lack of alternatives (Clark and Drinkwater, 1999).[51] This is in quite sharp contrast to the experience of immigrant entrepreneurs in the United States: there, the self-employment rate for immigrants was around 12 per cent in 1990, only slightly more than that of natives, and their earnings were sharply higher than those for immigrants in dependent employment and tended over time to converge with, and then surpass, the earnings of both dependent and self-employed natives (Lofstrom, 1999). Thus, self-employment has played a quite different role in the integration of immigrants in the United States than in Denmark.

Indeed, it is hard for most immigrants in Denmark to use entrepreneurial activities to break out of traditional sectors, and it rarely happens. This may be in large part because, despite long hours and hard work, the experience does little to broaden their skills or offset their lack of qualifications. The skills gap with Danish entrepreneurs is huge – two-thirds of all native entrepreneurs have technical or advanced education, while less than one-third of Iranian entrepreneurs, the most qualified immigrant group, have equivalent qualifications. A lack of language skills is also seen as a handicap by many ethnic entrepreneurs, although they often rated even lower their skills on such basic business tasks as marketing and advertising, negotiating with authorities and staff administration (Bager and Rezaei, 2001). But a growing number of family businesses are now being handed on to better-educated descendants: already in 1996 around one in five Turkish-owned and Pakistani-owned businesses were owned by those aged between 18 and 25 years old. These younger owners might, with access to financial resources

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from their parents, and/or better skills at tapping external sources of finance, find it easier to either raise productivity and earnings or use their business as a springboard to a wider range of opportunities.

The Government has announced three initiatives concerning immigrants in its recent general plan to improve entrepreneurial activities:

- Developing local networks for ethnic entrepreneurs and strengthening guidance services for them.
- Making entrepreneurial activities a theme in the courses for asylum seekers and the introduction programmes for aliens.
- Conducting a survey concerning the barriers faced by ethnic entrepreneurs in obtaining financial support.

The aim is to improve and broaden entrepreneurial activities among immigrants.

Addressing the skills gap

Another approach to addressing the skills gap for immigrants from LDCs is to invest in upgrading their skills, although careful attention needs to be paid to the cost-efficiency of such efforts. The distribution of education among immigrants from LDCs is quite noticeably different from that of native Danes (Figure 26), with large variations among different ethnic groups. Overall, half of these immigrants had no more than basic education compared with a third of native Danes, while significantly fewer had vocational education. For men, tertiary education rates were around the same as for Danes, although rates for women were significantly lower. Iranian and Polish immigrants are strikingly better educated on average. However, only around half of those with higher education when they arrived actually have jobs. This may be due to language difficulties, but also partly be due to lower, or even impossible to assess, quality of some foreign higher education, notably when obtained in some LDCs, and employers may find it difficult to evaluate these foreign qualifications. To assist them and to make it easier for skilled immigrants to make proper use of their qualifications in the Danish job market, the government has allowed the Danish Centre for Assessment of Foreign Qualifications to make binding assessment of foreign qualifications, permitting entry to regulated professions. The government also wants to make it easier for immigrants to have their actual level of skills verified, either through business colleges or on-the-job testing, as is already done by a group of companies located in Funen who issue their immigrant workers with written certificates of their demonstrated skills.

Having some Danish education is associated with higher labour market participation rates at all levels of educational achievement (Table 19). Few immigrants from LDCs have Danish post-compulsory education, and even after 15 years of residence in Denmark, around two-thirds still do not have any Danish

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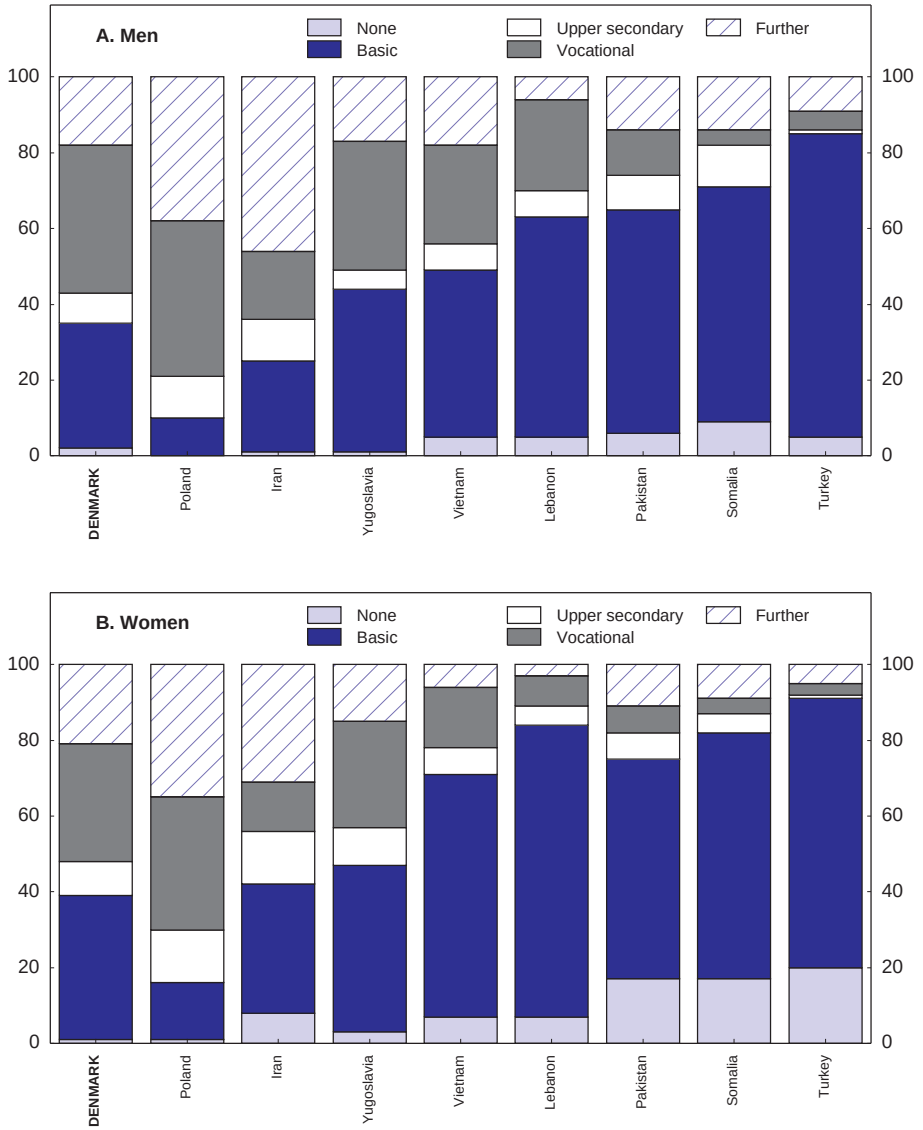
Figure 26. **Educational attainment of Danes and immigrants**[1]
Per cent, 2001



1. For immigrants, education attained either in Denmark or abroad.
Source: Rockwool Foundation.

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Table 19. **Education and labour market participation**

Level of education attained as at 1 January 2001, 16-66 year olds

	Elementary school	Secondary school	Basic vocational education/ apprenticeships	Shorter-term advanced studies (< 4 years)	Medium-/ long-term university studies (> 4 years)	Not stated/ no education
	Per cent					
Immigrants from third countries – education only acquired in the country of origin						
In work	41	42	51	49	50	28
Unemployed	10	9	8	8	7	4
Not in the labour force	50	49	41	43	43	67
Total	100	100	100	100	100	100
Immigrants and descendants from third countries – at least some Danish education						
In work	51	66	76	75	80	40
Unemployed	5	3	8	9	6	7
Not in the labour force	44	31	15	16	14	52
Total	100	100	100	100	100	100
The remaining part of the population						
In work	63	82	84	87	92	52
Unemployed	5	3	3	3	2	4
Not in the labour force	32	14	13	9	6	44
Total	100	100	100	100	100	100

Source: The Ministry of Refugee, Immigration and Integration Affairs; Statistics Denmark.

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Table 20. **School-to-work transition**
18-35 year olds, 1997

	Ethnic Danes		Descendants	
	Men	Women	Men	Women
Average age	26.8	26.8	20.7	20.7
Education, years	11.8	12.1	10.7	11.1
Employed, per cent	69.0	56.0	46.0	28.0
Unemployment, rate	6.8	10.8	15.3	16.2
Completed a qualifying education, per cent	64.7	61.7	17.7	22.6
Duration of waiting time spell before first job, months	2.7	3.9	3.6	5.1
Duration of first employment spell, months	19.9	19.2	12.8	11.7

Source: H.S. Nielsen *et al.* (2002).

education. Younger immigrants and descendants are more likely to participate in post-compulsory education than their parents, but it is still a lower proportion than for young native Danes (Table 20). Both men and women aged between 18 and 35 are significantly less likely to have completed a qualifying education, with predictable consequences: a longer wait for their first job, shorter tenure in their first job and lower wages (H.S. Nielsen *et al.*, 2002).[52] These results are not unique to Denmark, and similar outcomes can be observed for descendants from less-developed regions in all destination countries, including the United States.[53] Nevertheless, there is actually greater improvement in educational attainment among the children of immigrants compared with their parents than can be seen for native Danes, with second-generation immigrants finding it easier, on average, to overcome the disadvantages of low income and parental education.

Several key factors have been identified as making a difference to education attainment for immigrants and descendants from LDCs:

– *Level of parental education and positive attitudes.* Many immigrants who arrived relatively young and went through the Danish education system had mothers with very low levels of education (Jakobsen and Smith, 2002). This particularly affects Turkish-born offspring: on average Turkish mothers who arrived during the "guest worker" period had only 1.4 years of formal education. This probably made it difficult for those parents to help their children with their education, especially in a school setting they were unfamiliar with. For descendants, the education level of their parents was not a factor affecting educational attainment, but still made a difference to their success on the labour market (H.S. Nielsen *et al.*, 2002). A positive parental attitude towards their children's education also makes a significant difference, especially for girls.[54]

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– *Marrying young*. Danes marry for the first time on average at the age of 31 years, Turkish immigrants at around 19 years, and Pakistanis and ex-Yugoslavians typically at around 22 years. Women tend to marry younger than men for all three minority groups and are less likely to complete their education as a result.[55] Early marriage is a particular handicap, given that in Denmark formal education is generally completed at a later age than in many other countries (see Chapter II). For second-generation immigrants, marriage to an ethnic Dane was associated with an increased likelihood that a female descendant will complete a qualifying education. But very few young women from LDCs actually marry outside their ethnic group (Pedersen, 2000).[56]

– *Danish language proficiency*. This is obviously critical to obtaining a higher level of education and plays the most significant role in determining the individual dropout risk. This factor is clearly linked to both age at arrival and may also be affected by bilingual education practices and ethnic concentration (see below).

A key feature of the Danish education system is the extensive opportunities for adult education. This provides a second chance for Danes and foreigners to obtain basic and advanced skills, and attendance is free in most cases. Indeed, Danes show high rates of participation in adult education. In principle, this existing structure should make it not only possible but relatively easy for motivated immigrants to boost their qualifications and thereby improve their prospects. It is unclear why so few immigrants appear to be using this channel and might be worth examining more closely. In any case, the recent strengthening of the framework of activation plans could help to raise awareness among immigrants of the value of undertaking additional education (see Chapter II).

Language skills

Language skills are clearly important to labour market success in Denmark, as in many countries.[57] Lack of Danish language proficiency remains a major handicap for many immigrants (Table 21).[58] Even among descendants, some 12 per cent are not fluent. Around DKK 1 billion of public funds per year are spent on providing free Danish language courses for adults, but the quality of some of these programmes may be variable, and absenteeism is around 25 per cent. It is not possible to identify to what extent these two factors have contributed to the weak language performance of immigrants. Nor is it clear what strategies might incite long established immigrants to make more effort to become fluent in Danish, if they have managed to get by in daily life without it thus far.

A greater awareness of the importance of language skills for successful labour market integration led the authorities to emphasise Danish language as part of the new Integration Act in 1999. This Act applies to all immigrants arriving

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Table 21. **Danish language skills**[1]

Per cent

	Fluent	Good	Average	Poor	Total
Men					
All immigrants and descendants	28	29	26	16	100
Immigrants, aged 13 or more on arrival	15	31	33	21	100
Immigrants, aged 12 or less on arrival	62	29	7	2	100
Descendants	88	10	2	0	100
Women					
All immigrants and descendants	28	21	25	25	100
Immigrants, aged 13 or more on arrival	13	23	31	32	100
Immigrants, aged 12 or less on arrival	69	22	8	2	100
Descendants	88	9	2	1	100

1. Interviewers' assessments of respondents' level of knowledge of Danish in 2001.
Source: Larsen (2002).

from outside the European Union or Nordic countries, whether as refugees or under family reunification schemes. Under this legislation, responsibility for integration was handed to the municipalities, on the grounds that they would be better placed to implement comprehensive and co-ordinated measures to facilitate integration. They are obliged to offer all arrivals (except from Nordic or EU countries) a three-year introduction programme. This should be designed as an individual contract for each immigrant, after assessing his or her individual abilities and background, and be aimed towards eventual participation in the labour market or further relevant education. The programme involves Danish language lessons and classes in understanding Danish society and also involves activation for those eligible for the introductory benefit. For such immigrants, the introduction programme must be at least 30 hours per week.

The first introduction courses have only recently been completed, and it is perhaps too early to judge their success. However, despite the general good sense of this strategy, two aspects need to be carefully monitored. The first risk is that the length and intensity of the course (and associated income support) might actually keep some immigrants out of the job market longer than they would otherwise be, or discourage them from looking for work and achieving self-reliance. The present government has stressed that it does not believe that foreigners have to learn Danish before getting a job, and it is looking to find ways to organise language training so that it fits better the needs of individuals and companies. The new wage-subsidy scheme for immigrants, which incorporates workplace language courses, is one approach along these lines. The second potential shortcoming is that local case officers may be reluctant to exercise the sanction on non-attendance provided, namely reducing the introduction benefit, or such penalties might prove ineffective in changing behaviour.

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Bilingual education and performance

Language acquisition is also critical for the successful integration of immigrant children. But those children currently within the school system are clearly performing less well than Danes, and the gap for reading literacy among 15 year-olds is more marked in Denmark than in many other OECD countries. It is more surprising to find that the gaps on reading are worse for descendants than for immigrant 15 year-olds tested at the same time (Figure 27). Overall, this suggests that the education system is unsuccessful in overcoming the disadvantages ethnic minorities face in acquiring basic education and skills. Identifying the educational shortcomings and addressing them is beyond the scope of this *Survey*, but one factor which has been called into question is bilingual education.[59] Until 2002, municipal authorities were obliged to provide bilingual instruction to all bilingual children, including descendants, and the majority of ethnic-minority pupils were enrolled in such education programmes. This requirement has now been abolished, but a number of municipalities continue to offer instruction in both languages spoken by those children.

Whether bilingual education *per se* helps or hinders educational attainment and integration more generally is a contentious issue in a number of countries (see, for example, Duignan, 1998).[60] In one study of 12 OECD countries, most had established the goal of having immigrant children mainstreamed into regular classes within three months to three years (Glenn and de Jong, 1996). But although the countries followed quite a range of different strategies, and sometimes changed tack, none had achieved clear success in overcoming the weaker overall scholastic performance of immigrants relative to natives. The same study also pointed out that policies adopted were generally not grounded on research and evidence of a link to educational attainment.[61] Given the lack of solid evidence demonstrating a strong role for mother-tongue education in improving educational attainment or assisting economic integration, and the weak Danish skills of immigrants, it seems reasonable to ask whether bilingual education might be retarding the integration process.

In this context, the government's recent decision to extend the opportunities for immigrant children to receive special Danish language exposure and stimulus programmes to 3 year-olds as well as 4 and 5 year-olds should help to get those children off to a better start in Danish, as well as redress the slightly lower participation of young immigrant children in childcare activities. Evidence from the US Head Start pre-school programme, designed to boost school readiness among disadvantaged children, suggests this would be a useful strategy (Currie and Thomas, 1999).

Enclaves

Ethnic concentration plays a complex role in integration. In principle, network effects may make it easier for new arrivals to become more quickly

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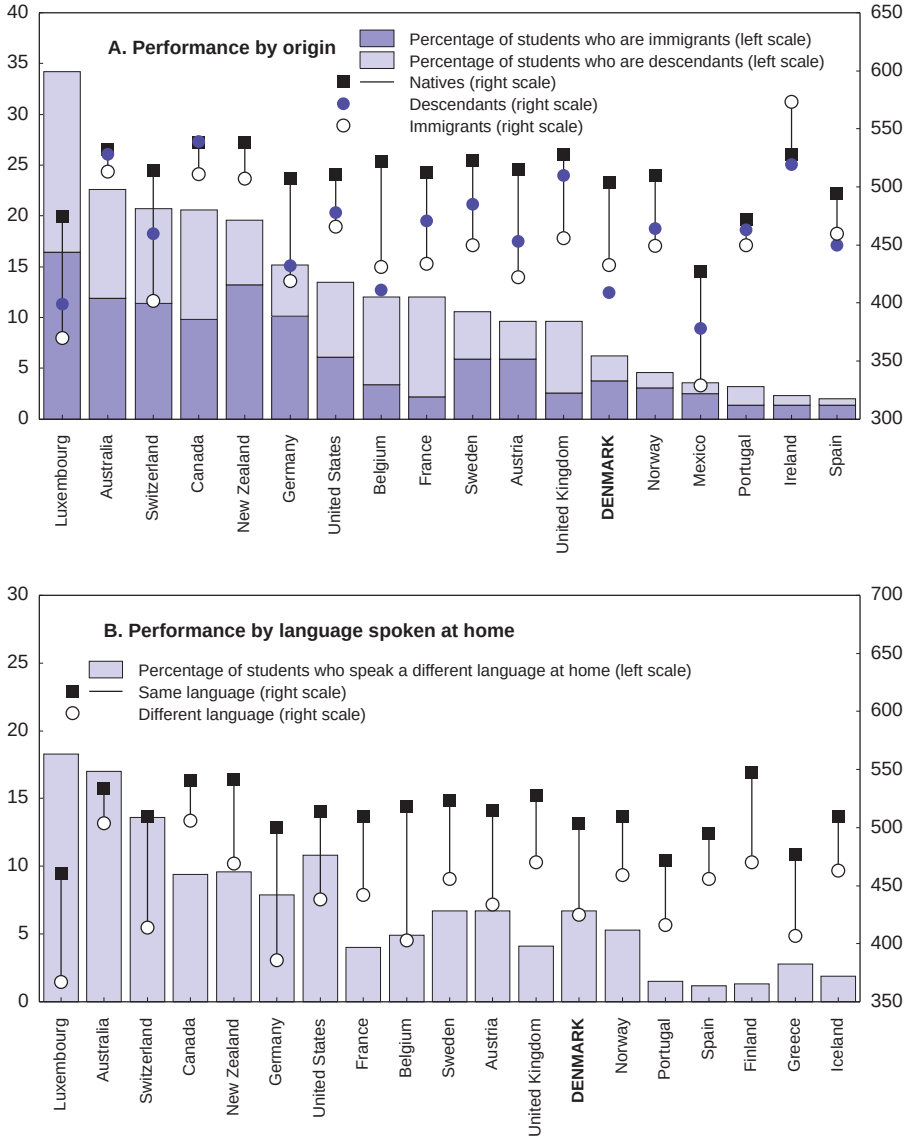
Figure 27. **Reading literacy performance**
15 year-olds, 2000



Source: OECD, *Education at a Glance*, 2002.

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established in housing, as well as find jobs, through the established contacts of other immigrants. However, international evidence indicates that immigrants living in an enclave have weaker host-language proficiency, with both less incentive and fewer opportunities to improve language skills (Chiswick and Miller, 2002; Bauer *et al.*, 2002). They are also less likely to get involved in mainstream daily life and activities. In Denmark some 8 per cent of immigrants and descendants surveyed indicated they had no contacts at all with Danes, while another 19 per cent were on no more than nodding terms with any Dane (Schultz-Nielsen, 2001). The lack of contact with Danes among the unemployed, course participants and others outside employment is significantly higher. All else equal, including language proficiency, associating with native Danes and reading Danish newspapers regularly appears to boost male immigrants' chances of finding employment. Contact with Danes was also important for women's employment prospects. This raises questions about the extent to which ethnic enclaves may be hindering the integration process for adults and also providing a less favourable language environment for children, especially for those in bilingual schools.

Most immigrants from LDCs are concentrated in a few municipalities, primarily around Copenhagen, Århus, Odense and Aalborg (Table 22). Although Copenhagen has the greatest number, four smaller authorities in the greater Copenhagen area actually have higher concentrations. The probability of finding

Table 22. **Concentration of immigrants by municipalities**
As at 1 January 2002

Municipality	Total population	Immigrants and descendants			
		From less developed countries		From EU, Nordic and North American countries	
			Per cent		Per cent
Copenhagen	500 531	69 110	13.8	20 663	4.1
Århus	288 837	25 451	8.8	6 076	2.1
Odense	183 628	15 738	8.6	3 066	1.7
Aalborg	162 264	7 990	4.9	2 558	1.6
Frederiksberg[1]	91 322	7 455	8.2	3 881	4.2
Brøndby[1]	34 542	6 870	19.9	609	1.8
Høje Taastrup[1]	45 947	6 227	13.6	709	1.5
Albertslund[1]	29 130	5 396	18.5	566	1.9
Ishøj[1]	20 987	4 974	23.7	323	1.5
Hvidovre[1]	49 674	4 877	9.8	1 012	2.0
Total of above municipalities	1 406 862	154 088	11.0	39 463	2.8
Rest of Denmark	3 961 492	157 281	4.0	64 499	1.6
Total	**5 368 354**	**311 369**	**5.8**	**103 962**	**1.9**

1. Municipalities within the Greater Copenhagen area.
Source: Statistics Denmark.

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work appears to be lower for immigrants who live in these enclaves (Ministry of Refugee, Immigration and Integration Affairs, 2002), which is consistent with the above findings on the importance of contact with Danes. There has also been a clear trend towards segregation within all the larger towns and cities in Denmark: this problem has been exacerbated by the way in which municipalities have primarily allocated refugees to non-profit (social) housing estates and to dwelling blocks where there is already a larger-than-average proportion of Danes with social problems.

The government became concerned about the concentration of immigrants around the beginning of the 1990s and established a committee on Metropolitan Affairs in 1993 to find ways to discourage ethnic enclaves. This group proposed a number of measures to combat social problems in "distressed" urban areas. In 1996, the committee changed tack and focused on a new strategy, through an urban regeneration programme (Kvarterløft), aimed at making such areas more attractive without trying to directly influence the social composition of tenants. This approach has been extended further with the present government's programme of "inclusive cities".

The concentration of immigrants in social housing is probably made worse by the rigidities of the Danish housing market (OECD, 1999). The majority of Danes own their own home, with only 15 per cent residing in social housing, while the proportions are almost exactly reversed for immigrants and descendants from LDCs. Rent controls for private rented dwellings and co-operatives keep the rents under market value but produce long waiting lists and low turnover, which again makes access more difficult for immigrants. Turnover is highest in the least attractive social housing, and municipalities have a right to allocate a part of this housing stock,[62] which tends to reinforce the concentration of immigrants in specific areas. The present government's announced reform to make it possible for social housing tenants to purchase their homes has been criticised for making it less likely that a more diversified mix of occupants would result over time (Groes, 2003). Policy changes to improve the way the housing markets function overall (as recommended in previous *Surveys*) would help to ease access for immigrants,[63] even though such measures must be implemented carefully and would not produce instant results.

Newly-arrived refugees are dispersed across the country, in accordance with the Integration Act, in order to distribute the responsibility among the municipalities and also to promote contact with native Danes. However, there is a tendency in Denmark, as in most countries, for migrants themselves to gravitate towards others of their ethnic group, even though it does not seem to be in their economic interests. This is an area where further research is needed, especially as international experience indicates that it is difficult to identify policies that can successfully and durably disperse immigrants more widely (Ministry of Refugee,

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Immigration and Integration Affairs, 2002). This suggests that, for the meantime, a more relevant target for policy might be to ensure that other measures minimise the present negative outcomes associated with enclaves: for example, through greater efforts to boost Danish language exposure.

The benefit trap

One factor that discourages immigrants from working is the low economic return, compared with living on public income support. The Danish welfare system is generous: so-called "short-term" benefits such as unemployment insurance and social assistance are intended to provide people with a bridge back to supporting themselves (see Chapter II). For native Danes, the system appears to fulfil this role, in sharp contrast to the situation of migrants from LDCs (Table 23). Migrants from non-western countries are particularly vulnerable to finding themselves in a welfare trap because their average earnings prospects are so much lower than for native Danes.[64] Seen in that light, it is remarkable that around one in five immigrants and descendants in work would be financially better off drawing unemployment insurance, while another third gain less than DKK 500 per month by working (Table 24). The situation for these low paid workers does not show any

Table 23. **Years on social assistance, municipal activation or retraining programmes**
Period 1996-2000

	Native Danes		Non-western immigrants and descendants	
	Men	Women	Men	Women
Not involved	89.2	87.5	47.7	48.5
Up to 1 year	6.2	7.0	20.1	15.6
1-2 years	1.6	1.8	8.7	6.7
2-3 years	1.0	1.2	7.6	7.1
3-4 years	0.8	0.8	6.2	6.8
More than 4 years	1.1	1.7	9.8	15.4
Proportion of each immigrant group who received support for more than 4 years				
Ex-Yugoslavia	–	–	9.4	14.6
Iran	–	–	13.5	23.3
Lebanon	–	–	30.5	56.2
Pakistan	–	–	2.8	9.6
Poland	–	–	3.9	8.4
Somalia	–	–	29.4	36.6
Turkey	–	–	2.9	9.4
Vietnam	–	–	8.3	19.8
Other non-western countries	–	–	8.9	11.9
Western countries	–	–	1.4	1.7

Source: N. Nielsen (2002).

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Table 24. **Disposable income gaps between work and unemployment**[1]
18-59 year olds, per cent

	Whole population, 1996			Immigrants and descendants, 1998/99		
	Men	Women	Total	Men	Women	Total
Childcare costs						
Included:						
Negative "gap"	6	16	10	18	26	21
"Gap" under DKK 500	12	28	20	34	41	36
Excluded:						
Negative "gap"	5	7	6	14	21	17
"Gap" under DKK 500	11	22	16	26	35	30

1. Proportion of employed members of an unemployment insurance fund who have a smaller disposable income from full-time employment than from full-time unemployment.
Source: Schultz-Nielsen (2001).

improvement until they have been working around five years. Indeed, it has been estimated that while Danish men benefit from a significant positive boost to earnings for five years work experience, all else equal, the gains to experience are marginal for immigrants (Husted *et al.*, 2001).[65]

For those immigrants and descendants who are not members of an unemployment insurance fund, and thus qualify for social assistance instead, the incentive problems are even worse, because if one spouse gets a job, the social assistance to the partner is withdrawn. This means that if only one spouse enters the workforce, the household would experience a net loss unless he or she could earn more than DKK 28 000 per month, which is well above the median wage. This creates a benefit trap for any family in the same situation, but it affects immigrants disproportionately for three reasons: they are more likely to be married with dependant children than Danish recipients; they are more likely to find themselves in a situation where neither partner has a job; and their earnings potential is likely to be lower than that of native Danes.

To address these incentive problems, the government's "More People in Work" strategy includes a number of measures to make work relatively more attractive (see Chapter II). In addition, for new arrivals, it has introduced a residence requirement of seven out of the past eight years to qualify for full access to social welfare benefits. During those first seven years, those in need of social assistance will receive less generous introduction benefits, albeit within the range of net replacement rates that apply in other OECD countries (Figure 28). In every case, as a minimum, those receiving these benefits will receive at least as much as the benefits paid to students in Denmark. It is too early to assess the extent to which these adjustments to incentives will boost employment among new arrivals, but as already noted, it has probably reduced the risk of Denmark being considered a "soft touch" by some low-skilled asylum seekers.

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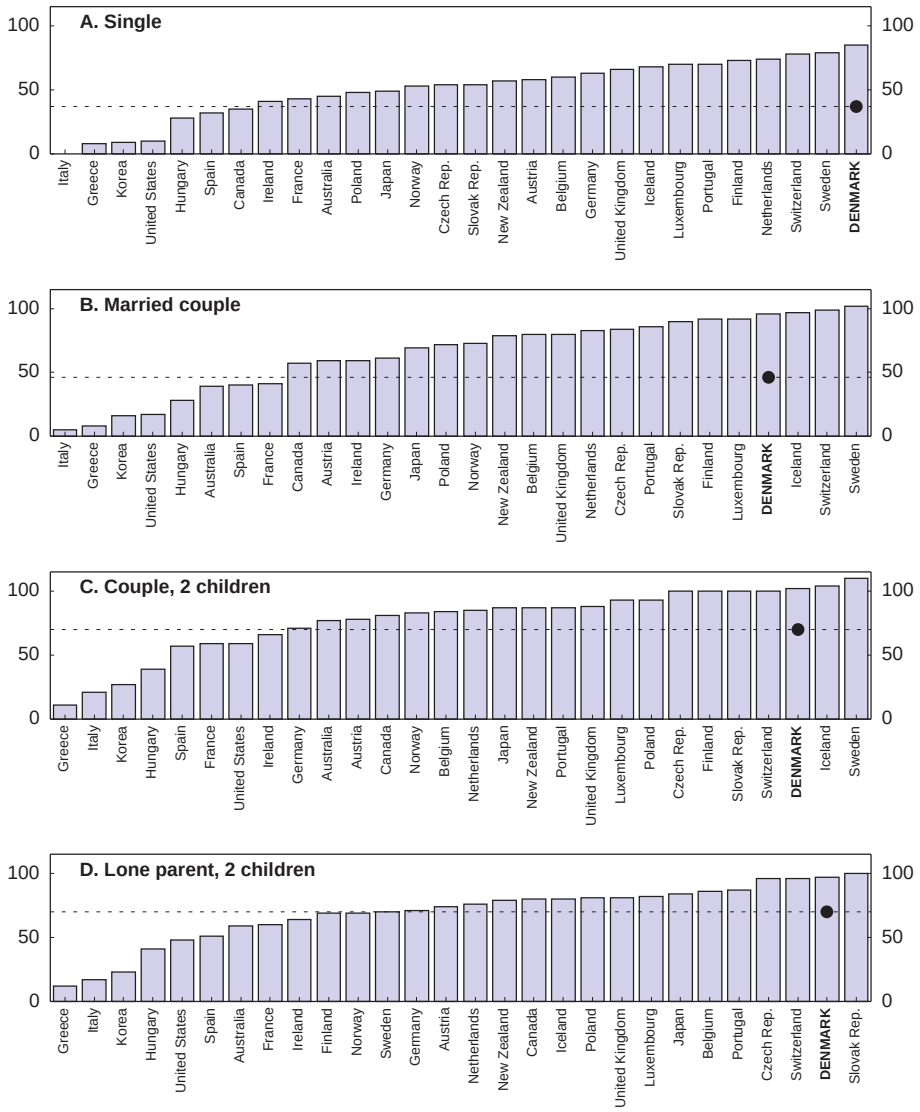
Figure 28. **Net replacement rates for four family types**

1999, at 66.7 per cent of APW earning level

After tax and including family and housing benefits for long-term benefit recipients



Note: The dots in the columns for Denmark indicate the replacement rate for introduction benefits.
Source: OECD, Tax-benefit models.

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Conclusions

Denmark has made considerable progress in putting in place a more coherent strategy for addressing the economic aspects of immigration. However, it still has to address the situation of those foreigners who arrived under earlier entry policies and who have not achieved economic integration. The economic consequences of those earlier policies are significant, given the gap between the economic prospects of immigrants from LDCs and the nature of the Danish economy. Low employment rates and high reliance on public income support among some immigrant groups has meant that, overall, immigration has not significantly boosted the country's economic output (see Chapter I). It has at the same time added to the effective dependency ratio and entailed redistribution of resources from working Danes to non-working immigrants. Furthermore, some immigrants from LDCs have had visible difficulties in integrating into Danish society, which has created some social tensions.

The logical first steps in addressing the integration difficulties facing immigrants from LDCs are to avoid adding unduly to the problem and to do more to ensure that those who arrive in future can more quickly become self-supporting. Denmark has gradually tightened restrictions on entry over the past several years. It has also endeavoured to distinguish more clearly between providing safety and freedom from persecution for genuine refugees and restricting entry for those migrating for economic reasons but with poor prospects of achieving success in Denmark. The government has also delivered the clear message that anyone coming to Denmark will be expected to work and to make every effort to integrate. This message of self-reliance has also been delivered to Danes, along with measures to boost employment (see Chapter II).

The current challenge is to improve the integration of immigrants, especially those from LDCs with very low skills, some of whom arrived many years ago. With few low-skilled jobs in Denmark, high effective minimum wages and high net replacement rates on public benefits, these migrants face several obstacles to obtaining a job. These difficulties could be reduced by scaling back welfare benefits and allowing lower wages to be paid for low-skilled work. The social partners have announced that they will consider the possibility of making agreements that allow lower wages for persons not yet completely skilled for the job. Such agreements have already been made in parts of the industrial and financial sector. Furthermore, the unions and employers have collaborated on a scheme to place migrants in the workplace, with temporary wage subsidies linked to on-site training and work experience. This could be a promising approach as long as a rapid subsidy phase-out is enforced, although it may not reach enough migrants to make a major difference. Another helpful element would be to reduce the tax wedge on labour incomes (see Chapter IV), making it cheaper for households to employ low-skilled workers to provide household services, as they do in a number of other countries.

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Raising the skills of immigrants is another way of achieving better labour market integration. Speeding up progress for young immigrants and descendants may be difficult, since the government may be able to exercise only limited influence over some of the factors that make a difference to educational performance, such as mothers' education. However putting more stress on the importance of education for economic integration may help to strengthen aspirations and counter negative parental attitudes; greater wage premia for educational qualifications would also make education more rewarding (see Chapter II). Persuading young immigrants to concentrate on completing their education and gaining a foothold in the labour market before marriage may also yield considerable dividends although it is difficult to see how exactly this could be done. In addition, special efforts to reduce the gap in reading performance shown for immigrants and descendants still at school would seem to be a prerequisite to improve educational attainment for immigrants in general. Furthermore, the role of bilingual education in explaining this poor performance needs to be clarified. In any case, efforts to reinforce Danish language skills at all levels of the school system would help to reduce the current performance gap.

Evidence on the importance of host-country language skills for successful economic integration is overwhelming, not just in Denmark, but in other OECD countries as well. As a result, most countries have tightened their expectations that immigrants should make an effort to quickly acquire these skills, while facilitating, or requiring, attendance in language courses. Denmark instituted a three-year introduction programme for immigrants assigned to local authorities in 1999. Careful evaluation of these courses would help to signpost where improvements can be made and how the link to job readiness can be strengthened, especially for women. Attendance requirements need to be more effectively enforced, which to some extent depends on effective and consistent case management at the local level. Sharing of best practices for achieving results across municipalities might provide useful insights into how attendance can be boosted. Another approach would be to apply stronger economic sanctions on those who fail to make reasonable efforts to progress.

The benefit trap in which immigrants from LDCs too often find themselves is an additional obstacle to labour force participation. The government has already instituted a number of measures to limit the disincentives to work arising from relatively generous welfare payments. The most notable change is the lower introduction benefit for new arrivals, which should significantly increase their incentives to obtain employment. But for immigrants already in Denmark and drawing social assistance benefits, the government has limited room to manoeuvre. Tighter work requirements and effective activation measures that quickly lead participants to real jobs may thus be the primary way in which this unemployment trap can be overcome. But if this approach cannot be made to work, then the government will either have to consider more radical changes to the incentive structure faced by

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immigrants or accept paying social benefits to a high proportion of immigrants indefinitely. Specific policy recommendations arising from this chapter are presented in Box 4.

Box 4. Recommendations on policies for immigrants

Entry

Policies concerning entry to Denmark need to take into account a range of non-economic factors, but the government should in any case consider the following:

- Monitor the entry rules to ensure that Denmark receives in-flows of migrants that correspond to its objectives.
- Reassess the role that compressed wages and high taxes play in making it harder to attract more highly-skilled immigrants and also expose Denmark to the risk of a "brain drain". Shifting more of the cost of higher education onto the individual, beneficial in any case, would reduce the economic cost of such emigration.

Integration

Policies concerning integration overlap to some extent with the policy recommendations from Chapter II to boost employment, since the primary focus is on raising immigrants' participation rates.

- Ensure that policies to overcome the skills gap and the benefit trap, such as wage subsidies and stricter activation policies are clearly time-limited and deliver results. If these do not generate significant improvements in integration, consider the scope for scaling back welfare benefits and reducing effective minimum wages to facilitate the emergence of more low-skilled jobs, which low-skilled immigrants could use as an entry point into the labour market.
- Facilitate the emergence of greater wage premia for education and skills to send clearer signals about the value of obtaining additional human capital, which would be beneficial in any case.
- Pinpoint and address the weaknesses in the school system that result in relatively poor results for immigrants and descendants. Put more emphasis on Danish language skills, and monitor programmes to identify and extend best practices.
- Monitor closely the results obtained from the introduction programme, and stand ready to modify it to make it more effective. Ensure that sanctions for non-attendance are applied effectively at the local level.

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IV. Refining the medium-term fiscal framework

Denmark is preparing for the predictable effects on public finances of future demographic changes by aiming for structural general government surpluses of around 2 per cent of GDP. This enables public debt to be paid off quickly and the interest burden reduced. This cornerstone of the Danish medium-term fiscal strategy for the period until 2010 has been pursued for several years by successive governments and has broad political support.

The other main elements of the strategy are: *i*) low growth in real public service standards; *ii*) lower taxes on earned income; and *iii*) a significant increase in employment. Together, these elements are broadly neutral to public finances; thus the gain from higher employment is almost offset by the increase in service standards and lower taxes. Put differently, if the government does not succeed in raising employment, there will be no room for improvements in public service standards or lower taxes unless various expenditures are reconsidered or the debt reduction strategy is compromised and the fiscal balance is allowed to deteriorate.

Within this framework the government has introduced a number of measures to achieve its objectives:

- The "tax freeze".
- Sanctions on local governments that do not comply with the tax freeze.
- Gradual cuts in taxes on earned income from 2004 to 2007.
- A package of initiatives to increase employment, "More People in Work" (see Chapter II).

The government's recent tax initiatives take up most of the fiscal room to manoeuvre until 2010, given the assumed increase in employment (Table 25). Lower taxes on earned income and "More People in Work" should make it easier to achieve the employment objective. So will the reforms of the voluntary early retirement benefit scheme in 1999 and the early retirement (disability) pension implemented from 2003. But they will not be sufficient: additional measures will be required (see Chapter II). Furthermore, ensuring low growth in real public consumption will be a major challenge. The assumed increases to 2010 are modest compared with the past decade, and, given a history of weak expenditure

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Table 25. **Contributions to required structural fiscal surplus in 2003**[1]

	Per cent of GDP
Tax freeze[2]	0.6
Lower tax on earned income	0.6
Real growth in public service standards	0.3
Implemented measures to raise employment[3]	–0.7
Not yet specified measures to raise employment	–1.0
Total effect on required surplus from medium-term objectives	**–0.2**

1. The structural fiscal surplus that is required for the general government to meet its future net obligations (see Chapter I).
2. The tax freeze in itself lowers tax revenue by ¾ per cent of GDP (at the 2002 level) by 2010. It implies a permanent net revenue loss of 0.6 per cent of GDP when converted into a fixed annual amount (net present value) in 2003, compared with a scenario where excise taxes, etc., are indexed to prices.
3. These include the termination of the transitory early retirement benefits in 1996; the 1999-reform of the voluntary early retirement benefits; the reform of maternity leave in 2002; the reform of early retirement (disability) pension in work from 2003; and More People in Work.

Source: Finansministeriet (2002a and 2003).

control in the past 10 years, a strong budgetary framework is needed (see previous *Survey*). The tax freeze and sanctions on counties and municipalities in case of non-compliance, along with the requirement that local governments do not run budget deficits, are the government's primary instruments to ensure expenditure control at all government levels.

This chapter deals with the tax and spending policies applied to achieve the government's medium-term objectives. The first two sections assess the power of the tax freeze as an instrument to lower taxes as a share of GDP and as a spending-control mechanism. In the following section, the newly implemented sanctions on local governments are described. The agreed earned income tax cuts from 2004 are subsequently assessed, and then some policy recommendations are summarised.

The tax freeze

The level of taxation in Denmark is among the highest of all OECD countries, in particular when measured in terms of the effect on incentives and income distribution (Finansministeriet, 2002a). The level has been rising steadily for several decades along with the expansion of the welfare state. The ratio of tax revenue to GDP might at first glance appear to have changed primarily due to cyclical developments in the past 20 years, but there have been significant underlying contributions to higher taxes from economic policy. In order to avoid further rise in taxes the government imposed the tax freeze upon its election in the autumn of 2001. The government's tax freeze commits it to abstain from raising any tax or duty rate, either as a percentage rate or as a nominal amount, according to

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how the tax is legally specified. Included in the definition of the tax freeze is a nominal ceiling on the property value tax. If there are compelling reasons to introduce or raise a tax or duty, another tax or duty must be reduced so that overall revenue is not increased. The definition of the tax freeze also includes restrictions on fiscal fees. Furthermore the government's fee and user charge policy imposes restrictions on the use of non-fiscal fees and user charges (Box 5).[66]

The nominal principle for property value tax and excise taxes set in *kroner* implies that these taxes are not adjusted in parallel with inflation. Revenue from these sources will, therefore, be effectively eroded by approximately ¾ per cent of GDP between 2002 and 2010.[67] In fact, the tax freeze is equivalent to a gradual tax cut, as a decrease of the tax ratio will be the inevitable outcome in the longer run. This is part of the government's motivation for implementing it as excise duties in particular have risen considerably over the last 10 years.[68]

Strictly complied with, the tax freeze along with the commitment to maintaining fiscal surpluses may provide an efficient framework for putting the burden of adjustment on expenditures. It will prevent expenditure slippage being accommodated through tax increases and if fiscal surpluses look like falling below the medium-term target of 1½ to 2½ per cent of GDP, adjustments must take place solely on the expenditure side. Similarly, if the economy is overheating and fiscal contraction is warranted this can only be achieved by trimming outlays. This marks a laudable break from recent trends.

However, it should be possible to retain these benefits while refining the tax freeze approach to make a more positive contribution to boosting labour supply, another crucial government objective. Indeed, a more supply-oriented approach would in itself have a virtuous feedback on public finances. By putting first priority on the nominal principle in the tax freeze, the government has chosen instead to use much of its room to manoeuvre within the medium-term fiscal framework to make specific tax reductions that less directly address the most important issues in the medium and long term: the downward trend in average working hours and the demographic pressure on the labour force. Better economic outcomes would probably be obtained by adjusting property value tax and excise duties in line with price increases and applying the additional revenue to a gradual reduction of earned income taxes, even though the reduction of excise duties and, to a lesser extent, the property tax also have some positive indirect impact on the labour supply decision.[69] This would still leave Danish property tax revenues well below the OECD average (Figure 29). Such adjustments of the tax system are not excluded by the tax freeze *per se*, as the definition provides scope for revenue-neutral reforms if there are "compelling reasons" to raise a tax or a duty. However, statements from key ministers emphasising the transparency of the present system for the individual tax payer indicate that a strict interpretation is being applied that rules out this type of reform for the present.

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Box 5. Principles of the tax freeze

The tax freeze is effective for any decisions made by the government formed by the Liberal Party and the Conservative People's Party in November 2001.[1] Changes to taxes and excise duties passed by Parliament during the term of the previous government are not affected, including the last stage of the Whitsun Package from 1998 that was implemented in 2002.

The basic principles of the tax freeze are (Finansministeriet *et al.*, 2002):

- No tax or duty rate can be raised. If the rate is set in percentage terms, the percentage cannot be raised; if the rate is set in *kroner* per unit, that amount cannot be raised.
- If there are compelling environmental or other reasons to introduce or increase a tax or a duty, the extra revenue must be used to reduce another tax or duty. If decisions in the European Union or international agreements require Denmark to lower a tax or a duty, the corresponding loss of revenue can be offset by raising other taxes or duties to keep the total effect on revenue neutral.
- A nominal ceiling is imposed on residential property value tax so that an increase in the value of property will not lead to higher tax payment for the homeowner. A fall in the property value will reduce the tax payment accordingly, but subsequent rises cannot lead to tax payments exceeding the level corresponding to the value of the property on 1 January 2001 plus 5 per cent, or the value on 1 January 2002, whichever is lower.

The tax freeze also applies to local governments. It does not prevent individual municipalities and counties from adjusting tax rates on income and land, but tax rates set either by municipalities or counties cannot on average exceed the level in 2002. Imposition of the tax freeze on local government has become a central feature in the annual negotiations on funding between the central and the local governments. If local governments' budgets imply a breach of the tax freeze, central government taxation will be lowered correspondingly to keep total public-sector tax revenue unaffected, while lower central government revenue will be reflected in reduced block grants to local governments through the sanctions mechanism introduced in addition to the tax freeze.

Fees with a fiscal element, *i.e.* fees that exceed the costs associated with the service in question, are similar to taxes. Therefore, they are included in the tax freeze and frozen nominally from 2002. User charges, *i.e.* non-fiscal fees that partly or fully cover the cost of producing a service, are not taxes and hence not included in the tax freeze. However, the application of user charges is subject to the government's fee and user charge policy, which implies that:

- User charges can follow the increase in costs so that the charge-to-cost ratio is maintained for individual services.
- The charge-to-cost ratio of existing services can be raised or new user charges introduced only if: *i*) the extra revenue from increased user payments is offset by reductions in other taxes or duties; and *ii*) the introduction or increase of a user charge concerns a service where users are free to choose.[2]
- User charges on new services can be introduced exceptionally and only after approval by the government's economic committee.[3]

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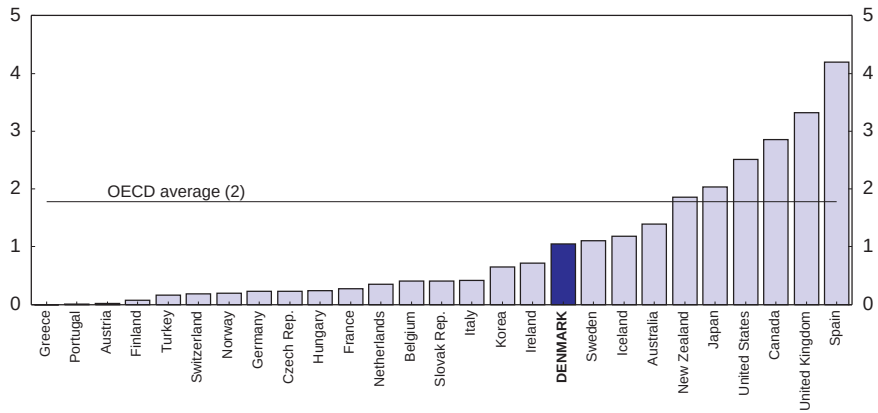
Box 5. **Principles of the tax freeze** (*cont.*)

Expenses on subsidised activities and goods and services sold under market conditions are not affected by the tax freeze. This includes, for instance, payments to public supply services (electricity, gas, heating) and radio/TV licence fees.

1. The tax freeze was first put forward in 1999 by the Liberal Party in its upcoming election campaign. It was chosen as a self-imposed credibility test of the promise not to raise taxes that could be easily monitored by the public. When the Liberal Party and the Conservative People's Party took office in November 2001, the tax freeze became part of the government's programme.
2. If the user charge is concerning a government control or inspection assignment, the "free to choose" condition does not apply.
3. The committee has three permanent members: the Minister for Finance, the Minister for Economic and Business Affairs, and the Minister for Taxation. Other ministers can participate on an *ad hoc* basis. The committee deals with issues relating to the government's economic policy.

Figure 29. **Recurrent taxes on immovable property**

Per cent of GDP, 2000[1]



1. 1999 for Australia and United States.
2. 27 countries; weighted by the 1997 GDP at PPPs.

Source: OECD, *Revenue Statistics*.

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The cost of maintaining this strategy may increase because international mobility of financial and human capital is likely to exert downward pressure on tax bases and revenue in the future, thus highlighting a need to reduce taxes that are relatively high by international comparison and subject to international competition. In fact, most EU countries would be better off by shifting a higher proportion of the tax burden to immobile factors, *i.e.* property and land (Joumard, 2001). But there is a widespread reluctance at the political level in Denmark to do so, and by applying a strict interpretation of the tax freeze, the government has locked in a tax structure that could be improved.

Reinforcing expenditure control and efficiency

While the strict interpretation of the tax freeze does not address a number of the challenges facing the tax system, it could be effective in dealing with the well-identified problem of maintaining control of public expenditures (see previous S*urvey*). In itself, the tax freeze will not exert any pressure on the expenditure side, but when combined with the fiscal surplus target it will serve as a budget rule restricting the long-term development in expenditures. For a given year, the tax freeze does not dictate a particular expenditure level, since the requirement of fiscal sustainability allows the general government balance to move within an interval of 1½ to 2½ per cent of GDP. But as the tax freeze establishes the value of future revenue, the value of future expenditure is automatically determined as a residual. In that sense, the tax freeze is an indirect rule to control expenditures, in contrast to direct rules such as expenditure ceilings.

As a budget rule, the tax freeze has the appealing feature of being easy to monitor and, thus, less subject to manipulation compared with expenditure ceilings, which can be vulnerable to changes in accounting practices[70] (Box 6). Furthermore, the tax freeze has become enforceable at all government levels by the introduction of a sanctions mechanism on local governments, thus adding to its credibility. However, the credibility of maintaining the tax freeze throughout the period until 2010 is strongly connected to whether the government's other medium-term objectives are perceived to be achievable. Lowering growth rates in real public consumption to ½ per cent per year and raising employment will both require clear breaks in recent trends (Table 26). Comparatively low growth rates in real public consumption were seen for a period in the 1980s and the early part of the 1990s, although during this time demographics helped to restrain public consumption growth, in contrast to the projected demographic changes in the rest of the current decade. If the government does not succeed in raising employment, it will be forced to reconsider its priorities; either to compromise the fiscal surplus target, forego some of the increase in public service standards, reduce the extent of benefit schemes or raise taxes. A similar prioritisation of objectives will be inevitable if expenditure slippage occurs. In this situation, cutting back on public

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Box 6. Using the tax freeze as a budget rule

The tax freeze addresses imperfections in the budget process at both the central and local government level, which tend to drive up spending and taxes, instead of forcing prioritisation within existing resources (Skatteministeriet, 2002). Successive Danish governments' lack of success in expenditure control is due mainly to the fact that most responsibilities involving public expenditure are devolved to local governments, making direct control by the central government difficult. Local government autonomy prevents the central government from directly determining the level or the use of expenditure, and the agreements negotiated between the central government and local governments' associations are not legally binding on the individual local government. However, there has also been a tendency to upward drift in public spending due to central government promises of improved public service standards in areas under local government control – spending that the local governments typically have financed by raising taxes.

Even if the major part of expenditure were not devolved to local governments, it would be practically difficult to set ceilings for total expenditure developments. A large part of overall spending is on entitlements (for instance, public transfers), which cannot be capped directly if *ex ante* projections of expenditures are exceeded *ex post*. Offsetting entitlement expenditures by cuts in discretionary outlays may even result in procyclical behaviour by overriding automatic stabilisers.

This is one of the reasons why the government has turned to a tax freeze to ensure declining tax and spending burdens. For the tax freeze to be an effective budget rule, it should optimally meet a number of criteria (OECD, 2002e). The tax freeze is clearly a *sufficient* measure to achieve a decrease in the tax-to-GDP ratio, since that is a direct implication. However, it will ensure spending control only in combination with the fiscal surplus target. Hence, the strength of the tax freeze as a spending rule is tied to the government's commitment to sustain fiscal surpluses around 2 per cent of GDP in the medium term.

Discretionary stabilisation policy (on the spending side) is still possible and automatic stabilisers will still operate. But the tax freeze does prevent fiscal contraction through higher tax rates. In general, a cautious approach to using fiscal policy as a tool for managing aggregate demand is advisable. But in the event that Denmark should find itself both overheating and significantly out of step with the euro area, it will no longer be able to raise taxes to choke off excess demand as it did with the Whitsun package in 1998. Besides expenditure cuts, only increases in Special Pension contributions remain available as an instrument to curtail households' disposable income, and exercising this option will work only if some households are facing credit constraints.

The government has strongly emphasised its commitment to the tax freeze, and it has ensured *enforceability* on all government levels by implementing a sanctions mechanism on local governments. In that sense the tax freeze could be deemed *credible*. However, it is less credible in the economic sense of compatibility with the structural policy challenges from external and internal pressures, even though lowering excise duties does contribute indirectly to reducing the internal and external pressure. The strict interpretation of the tax freeze prevents the government from reforming the tax system to shift burdens to immobile tax bases to address both external pressure on capital and corporate tax bases from globalisation of markets and internal pressure on earned income tax bases from population ageing.

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Box 6. Using the tax freeze as a budget rule (*cont.*)

The tax freeze, *per se*, is *consistent* with other policy objectives, even though it restricts the range of instruments available to achieve a stable economic environment and deliver fiscal surpluses. However, the *credibility* of the tax freeze is strongly connected with the credibility of other important government objectives: fiscal sustainability, tax cuts and limited growth in real public service standards. *First*, all these objectives depend crucially on a substantial increase in the labour force until 2010. This will require additional policy measures that have not yet been announced (see Chapter II). *Second*, the objective of only very limited growth in real public service standards is a challenge in itself. This could partly be offset by increased efficiency in the public sector but it will undoubtedly require better expenditure control than what has been seen earlier. Clear breaks in labour market participation and real public consumption trends are required to meet these objectives, and the real test of credibility will be to ensure only minor increases in real public service standards.

The strongest point of the tax freeze is probably that it is *simple to understand* and easily communicated. It relates to clearly visible concepts, such as tax rates and nominal duties, which are not affected by the actual position in the business cycle and can be readily verified *ex post*. In that sense, the tax freeze is *well defined* and *transparent*. However, one non-transparent element is the requirement of stable or falling tax rates in local governments on average. For an individual residing in a municipality that raises taxes, recognising an offset made elsewhere and making the connection to the total impact of municipalities' behaviour is probably not straightforward.

Table 26. **Key assumptions in the government's medium-term survey**

	1993-2002	2003-10
	Average annual changes, percentage point	
Participation rates[1]		
15-29 year-olds	–1.7	–1.8
30-59 year-olds	–2.2	1.7
60-64 year-olds	–3.7	6.7
	Average annual growth rates, per cent	
Private employment	0.3	0.3
Public employment	0.6	0.4
Real public consumption[2]	2.2	0.5
Real public consumption per capita[2]	1.8	0.3

1. 1993-2000.
2. Including lower growth in real public consumption in 2004 and 2005 to finance larger tax cuts than previously planned for.

Source: Ministry of Finance; Statistics Denmark; OECD calculations.

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investments instead might be a tempting alternative for the government. However, it would be a very short-sighted policy if real public capital was allowed to run down as the reinvestment burden would just be carried forward. For this reason the tax freeze necessitates a more careful prioritisation of expenditures.

When looking at the 2003 budget process at different levels of government, it is not evident that the introduction of the tax freeze has strengthened expenditure discipline although the adopted budgets at all government levels should keep real public consumption growth at the targeted rate. At the central government level, an "elderly cheque" of maximum DKK 5 000 before tax was provided to low-income old-age pensioners (at a net cost after tax of DKK 260 million). In principle, it only applies in 2003, but past experience suggests that this kind of spending initiative has a high probability of becoming permanent. While the effect on total expenditure is minor, there is a risk that other groups of public transfer recipients will press for similar improvements. It is exactly this kind of stepwise increase in public transfers that is especially unhelpful with the prospect of population ageing, and it adds pressure on other spending areas.

At the local government level, counties breached the tax freeze in 2003. However, the subsequent implementation of a sanctions mechanism should effectively put a lid on overall spending increases as local governments are not allowed to run budget deficits. Nevertheless, the underlying upward expenditure pressure facing local governments from population changes will increase, raising the risk of additional future breaches if local governments are not allowed to fund themselves or are not given higher grants from the central government. The central government seems to be aware of these possible tensions, but it remains to be seen whether it will be willing to provide for the necessary extra funding to meet these obligations. Anyway, the combination of expenditure pressure from population changes and the shaving of tax revenue by the tax freeze is likely to result in cutbacks on lower-priority spending items, either at the central or the local government level.

A better allocation of public funds could be facilitated through more appropriate application of user charges. The previous S*urvey* pointed out that the mix of user charges in Denmark was sub-optimal and based on inheritance from the past. There would be scope for improvements in overall efficiency of public expenditure just by re-balancing the existing arrangements. In general, a broader application of user charges and more autonomy for local governments to set fees were recommended, as well as an alignment of user charges with the likely price elasticities of demand for services and a closer link between charges and individuals' actual consumption.

The tax freeze and the government's fee and user charge policy will probably impede such rearrangements due to the restrictions on applying new user charges or increasing already existing charges. The requirement to make offsetting reductions in other taxes or duties makes sense for ensuring effective

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expenditure control. It also prevents charging users for supplementary services produced in the public sector that would anyway be more appropriately undertaken by the private sector (see previous S*urvey*). But individual authorities lack incentives to suggest changes to the structure of user charges within their area, as their appropriations are reduced in parallel with the expected revenue from additional user charges, without regard for reductions in other user charges. Also, the requirement that changes be made only in service areas with free choice is likely to hinder rearrangements towards a more efficient structure of user charges.

Strengthening inter-governmental relations

Inter-governmental relations in Denmark are characterised by a system of formalised budget co-operation between the central government and local government associations. The annual negotiations normally result in agreements on aggregate expenditure levels and tax rates as well as the size of the block grant to local governments. However, the agreements are not reached directly with individual counties and municipalities, and breaches at the aggregate level have been the rule rather than the exception in past years. Local governments tend to raise taxes rather than lower expenditure when faced with a budget shortfall and raise expenditure rather than lower taxes in the opposite case. This ratcheting-up of both sides of the account explains a significant part of the increase in the average tax rate of local governments in the last 25 years (Figure 30).[71] The government and parliament have indirectly consented to non-compliance with agreements by largely refraining from taking action to punish this behaviour, despite the resulting upward drift in taxes on earned income that has undermined the government's efforts to boost labour supply.

The 2003 agreements on the economy for counties and municipalities (negotiated in June 2002) included the overall requirement that the tax freeze be observed at the local government level. It implied that the average income tax rate, land tax rate and contribution duty on commercial real estate should not increase at either the county or the municipality level. This meant that an increase in tax rates by some local governments should be counterbalanced by lower tax rates in other local governments at the same level. However, the budgets adopted for 2003 involved a breach of the tax freeze for the counties as the West Zealand County decided to raise its income tax rate by 0.4 percentage point without offsetting reductions being made by others. Consistent with its announcements, the government has set aside an amount equivalent to the extra tax revenue at the county level (DKK 120 million) for reductions in central government taxes, thereby ensuring that the tax freeze held at the general government level.

A sanctions mechanism was then developed to ensure a neutral impact on central government fiscal balances (see Annex IX). The sanctions model is designed to support the system of negotiations between the central government

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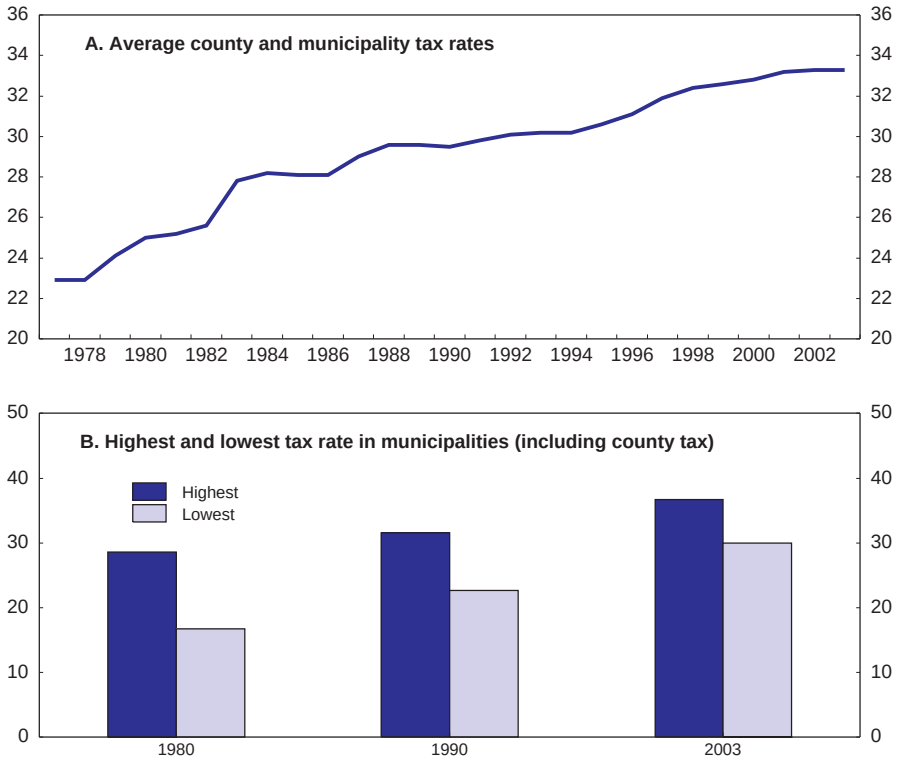
Figure 30. **County and municipality tax rates**

Per cent



Source: Ministry of Taxation.

and local governments by strengthening incentives for co-operative behaviour at the lower levels of government.[72] It also aims to increase incentives for municipalities and counties to maintain or lower taxes, while still allowing room for each individual local administration to make longer-term structural adjustments to taxes as long as the average tax rate does not increase.

The sanctions will be activated if agreements are not kept or if taxes are raised when no agreements have been reached. The size of the block grant will be established in the negotiations with local governments under the assumption of unchanged average tax rates. Any excess revenue from higher average tax rates at the county or municipality level will be neutralised by an adjustment of central

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government grants at the local government level where the tax freeze is breached, thereby keeping total revenue of local governments at the agreed level:

- In the first two years after a breach of the tax freeze, 50 per cent of the extra revenue is called in individually by reducing the central government's grant to those local governments that have raised taxes. On the other hand, local governments that have lowered tax rates will be compensated for 50 per cent of their loss of revenue through an increase in their grants. The remaining overall net extra revenue to local governments will be offset by a general reduction in the block grant.
- From the third year after a breach and onwards, the total amount of extra revenue at the local government level will be offset by a general reduction in the block grant.

In general, the sanctions mechanism should provide the enforcement that had been missing from the earlier fiscal arrangements and thereby make it possible for the government to rein in local public spending. But it implies less flexibility for local governments to adjust to structural developments in their territorial areas, while still meeting their obligations to provide certain service standards. It is also not clear whether sanctions will be lifted if average tax levels are reduced below their initial level in the years following a breach. Furthermore, even though the collective element of the sanctions mechanism and the refund system (in case of a breach) induce those local governments with room in their budgets to lower their tax rates, it is highly uncertain whether they will actually do so and, thus, provide scope for others to obtain additional funding through tax hikes.

As a reaction to local governments worrying about their situation if they lowered tax rates in 2003, the government refined its stated intention to neutralise any breach of the agreement on unchanged taxes in counties and municipalities in 2004 before local budgets for 2003 were adopted and the sanctions mechanism was introduced. Local governments were promised exemption from potential individual sanctions in 2004 if they lowered their taxes in 2003 and subsequently raised them again in 2004, conditional on overall compliance with the tax freeze in 2003 and that the tax increases in 2004 would not exceed the reductions made by the same authorities in 2003. Since counties breached the tax freeze in 2003, Copenhagen and Bornholm – who lowered taxes in 2003 – will not be exempted from possible sanctions on the county level in 2004. However, the promise will be effective for municipalities lowering taxes in 2003. It is not clear how the central government will act if the exemption from sanctions at the municipal level comes into force in 2004. It could impose a correspondingly higher reduction of the block grant or let the effect on central government revenue be less than neutral, *i.e.* by adjusting its own expenditures to keep the general government surplus from deteriorating.

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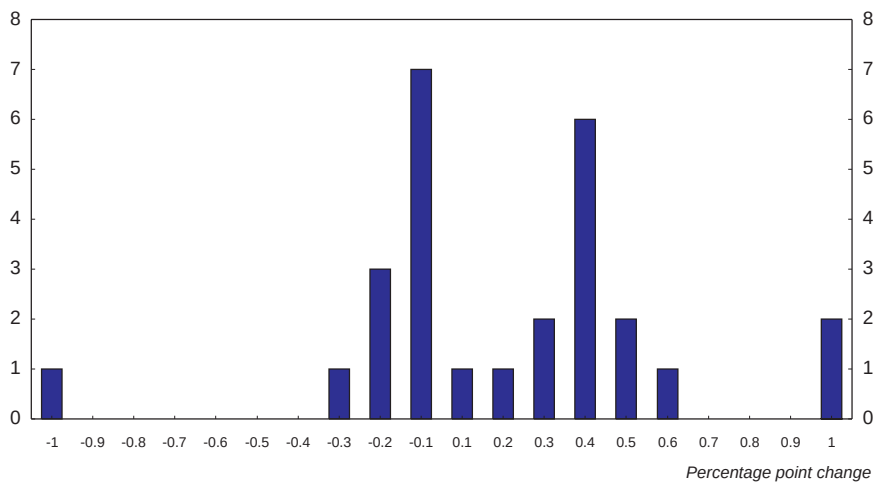
Some uncertainty among local governments about the effects of their actions will probably persist even after implementation of the sanctions mechanism, as the outcome for individual local governments will depend heavily on actions by other local governments.[73] The result could be that local government tax rates become locked-in. Although around 10 per cent of local governments changed their tax rates in 2003, it is still relatively few compared with previous years, suggesting a development in that direction (Figure 31). However, it is clearly premature to judge from a single year's experience where the outcome is affected by the promise of exemption from sanctions and by uncertainties about the design of the sanctions mechanism. Nevertheless, one alternative approach has already been proposed (Det Økonomiske Råd, 2002a): using a trading scheme on top of the tax freeze to make it possible for local governments wanting to increase spending to buy permits to raise taxes from other local governments that are willing to lower tax rates (Box 7).

Figure 31. **Changes in tax rates of local governments, 2003**[1]



1. Number of municipalities changing their rates by a certain percentage point. Farum municipality, (exempted from the tax freeze due to severe economic difficulties), which raised its tax rate by 3.2 percentage points, as well as municipalities keeping their rates unchanged are not included. The governments in Copenhagen, Frederiksberg and Bornholm (with status as both counties and municipalities) are included. Except for West Zealand County raising its tax rate by 0.4 per cent, no other counties changed their income tax rates in 2003.

Source: Ministry of Interior Affairs and Health.

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Box 7. A trading scheme alternative for managing local government taxes

By implementing a sanctions mechanism, the central government now has the instrument to ensure that the tax freeze will hold at the aggregate level. But concerns have been raised that this arrangement might lock in current variations in tax and service levels across local governments. It might also make it more difficult to adjust to changes in individual local governments' tax bases and underlying expenditure needs from demographic shifts. On top of differences in efficiency of providing services, variations in the trends and levels of local tax rates also reflect incomplete equalisation and the less-than-complete reimbursement they receive for their transfer outlays – factors that are mostly beyond local government control.

One alternative approach that would add flexibility to the system is to establish a market for local governments to trade tax increases and cuts among themselves (Det Økonomiske Råd, 2002a and 2002b). The concept of trading schemes is well known from environmental markets. For instance, a CO_2 emissions trading system for electricity was introduced in Denmark from 2000 with total quotas for emissions set and emissions allowances assigned to eight power companies (see previous *Survey*). The general idea is that in the presence of real world imperfections, a market for pollution permits could achieve the target reduction in pollution at lowest cost, because trading takes place until marginal abatement costs are equalised. The analogy to other areas has been drawn in Casella (1999), where a system of tradable budget deficit permits is proposed as an efficient mechanism for implementing the fiscal constraints of the Stability and Growth Pact in the European Monetary Union. Under that proposal, individual countries would be allocated a number of deficit permits each year, which could be freely traded, as well as banked for future use. If a country did not hold enough permits to cover its deficit in a given year, a steep fee would be charged for its non-compliance.

From a theoretical point of view, setting up a trading scheme for local government tax changes might be appealing, as it could ensure that tax revenue is raised in the localities where it is most needed. Local governments could then buy "permits" from each other allowing the buyer to raise tax rates corresponding to a certain amount of extra revenue, while the seller would be required to lower its tax rate to reduce its revenue by an equivalent amount. That could help to strengthen co-operative behaviour between local governments and enforce the credibility of local government associations in the negotiations with the central government. Since direct restriction of individual local governments' right to set tax rates does not seem to be a feasible policy option, some sort of sanctions mechanism must accompany the trading scheme to ensure a positive price on permits. If added on top of the current sanctions mechanism, the trading scheme would certainly increase the reward for lowering taxes (in situations with excess demand for tax revenue and thus a positive price on tax cut commitments). But the collective element in the present sanctions mechanism reduces the transparency of the system, as the cost to individual local governments of raising taxes above the level for which they hold permits is not clearly identified. Instead, sanctions could be imposed only on local governments raising taxes in excess of their "permits", which would clearly signal the cost of breaches and also provide a ceiling for the price of permits sold by those local governments committing themselves to cut taxes.

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Box 7. **A trading scheme alternative for managing local government taxes** (*cont.*)

While the theoretical gains are straightforward, practicalities could make implementation of a trading scheme quite difficult. The design of such a system would therefore need to be considered carefully and take into account the fact that the initial levels and desired changes in tax rates across local governments reflect more than just local preferences.

The implementation of sanctions and the nominal ceiling on the property value tax will lead to erosion of local governments' revenue. Together with expenditure pressure from population ageing, this implies that a higher proportion of local government funding must come from central government grants. That could increase the central government's ability to decide spending priorities and the extent of activity-contingent funding of specific key policy areas. For example, the central government set aside DKK 1½ billion in the budget bill for 2002 to reduce waiting lists for certain types of hospital treatment. Counties could then apply for funds by documenting increased output in these areas. Extending this kind of micro-managing of local governments' activities would certainly align the allocation of additional funds and activity at the local government level to the wishes of the central government. It is in line with the recommendation in the previous *Survey* to have the central government take over financial responsibility by directly reimbursing local governments in those areas where the central government specifies standards, such as maximum waiting periods for hospital treatment, and/or imposes direct regulation.

Tax cuts on earned income

The government's medium-term fiscal framework provided for reductions in taxes on earned income from 2004 corresponding to ½ per cent of GDP. In April 2003, the government found parliamentary support for the detailed implementation of these tax cuts during the period 2004 to 2007. There are two elements in the agreement on tax cuts:

- An increase in the threshold for paying intermediate state tax by DKK 12 000 per year over the four years, raising it by DKK 48 000 in total by 2007.
- An earned income tax credit of 2.5 per cent of the taxable income below the new threshold for paying intermediate state tax and a fixed deduction per year of DKK 6 800 (in 2007) for people with income above the threshold.[74]

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Furthermore, the agreement includes additional funds for social purposes in the budget for 2004 and onwards, as well as a mechanism to neutralise the impact on tax revenue from possible extraordinary increases in retail car prices due to the harmonisation of car prices following from strengthened competition within the EU. The total revenue effect of the agreement is estimated at 0.4 per cent of GDP in 2004 rising to ¾ per cent of GDP in 2007. Thus, it is a bit larger than the part of the fiscal room to manoeuvre that was initially set aside within the medium-term framework, and the government has therefore announced its intention to further reduce growth in real public consumption in 2004 and 2005.

The cuts in earned income taxes are steps in the right direction and in line with previous recommendations. Taxes on earned income are high by OECD standards, with the highest marginal rate starting at incomes only around 6 per cent above the average wage (Figure 32). However, the design of the package does not seem optimal from a labour supply, and hence public finances, perspective. The increase in the threshold of paying intermediate state tax will certainly lower marginal tax rates for around 740 000 people currently paying the tax, but nearly 40 per cent of full-time prime-age employed would still be paying the top state tax rate implying a marginal tax rate of almost 63 per cent.

Furthermore, an earned income tax credit may be one way of making work relatively more rewarding, but it would have been better in terms of public finances to look at the scope for scaling back public income support levels, which are among the most generous in the OECD (see Chapter II). The particular design of the earned income tax credit means that all working people benefit, not just those on low wages. The absence of a gradual phase-out of the tax credit avoids imposing higher effective marginal tax rates on middle income earners, but it also makes it more costly in terms of foregone tax revenue. Thus, the package is effectively providing more after-tax income in the hand to all employed, including many people for whom the incentive to work is unaffected.

The long-run impact on effective labour supply is estimated to be around 0.35-0.40 per cent, of which around one-third is from higher employment and the rest from an increase in working time. This effect will probably emerge only gradually, and the cuts therefore contribute little to the targeted growth in employment in the period up to 2010. In fact, without further initiatives the government would lack a substantial part of the 2¼ percentage point rise in structural employment that is required to maintain the pace of debt reduction given the other objectives in the medium-term fiscal framework.

Overall, it seems that the government has put significant weight on the distributional effects of the tax cuts at the expense of stronger incentives to increase labour supply. However, the distribution of total consumption across disposable income deciles is fairly flat in Denmark, and absolute, as well as relative, poverty is almost non-existent as illustrated in the previous *Survey*.

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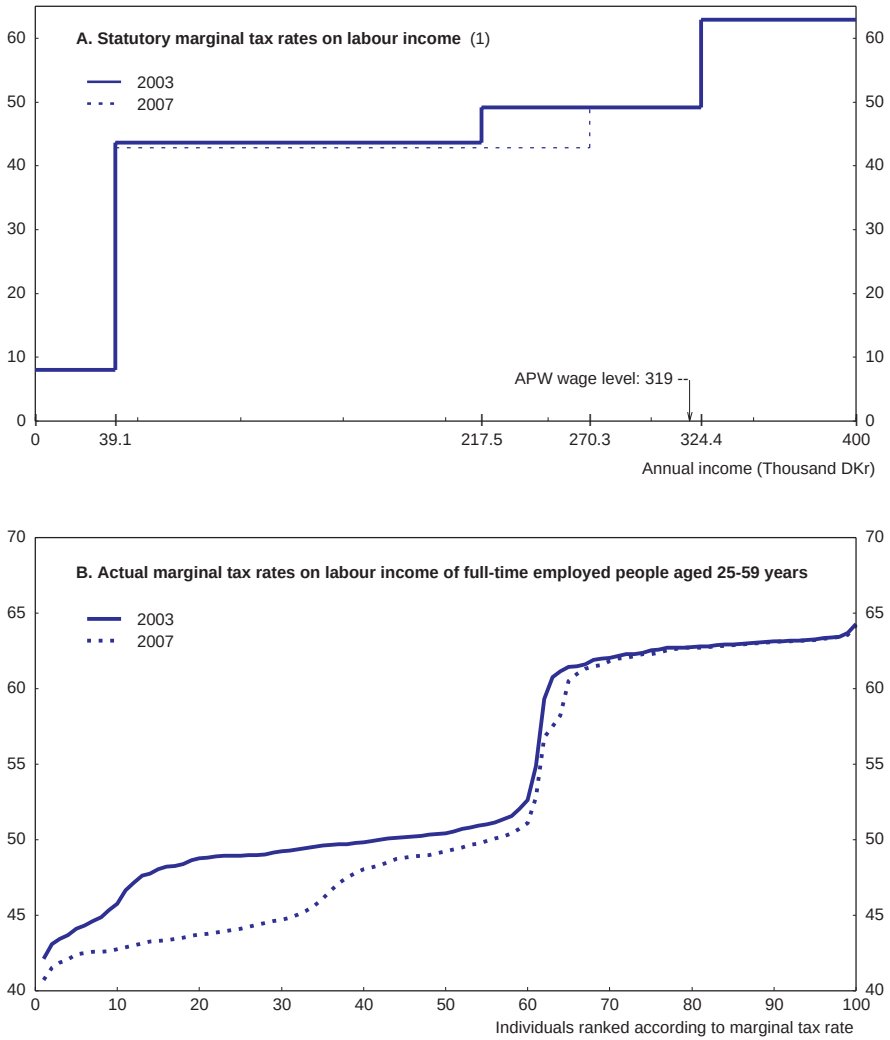
Figure 32. **Marginal tax rates**
Per cent



1. The labour market contributions (8 per cent) and Special Pension contributions (1 per cent) are deductible in all other tax components. To calculate the marginal tax rates, the statutory rates have therefore been multiplied by 0.91. Also, statutory threshold is corrected for deductable components to get actual thresholds on annual income.

Source: Ministry of Taxation; Ministry of Finance; OECD calculations.

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Continued high priority to immediate distribution concerns, thereby hindering tax reforms that would effectively boost labour supply, might be increasingly costly in terms of foregone fiscal room to manoeuvre. Hence, it could eventually give rise to adverse distribution effects if other main government objectives within the medium term fiscal framework have to be compromised.

In general, the possible (long term) structural effects on labour supply from tax reforms based on estimated degrees of self-financing should be interpreted with great caution due to uncertainties surrounding parameters such as the elasticity of substitution. Fiscal sustainability could be compromised if wider tax reforms were partly financed by expected structural effects that failed to materialise. However, a ranking of different tax-cut designs based on the degree of self-financing could give useful information on relative efficiency. Based on an elasticity of substitution of 0.1, a recent study (Finansministeriet, 2002c) found that: *i*) reducing the top state tax rate, *ii*) increasing the top state tax threshold; and *iii*) reducing the intermediate top state tax rate would deliver stronger labour supply and self-financing responses than both elements of the agreed tax cuts (Table 27).[75] On the other hand, the effect on income distribution would have been stronger.

If the government wished to address the top marginal taxes, fiscal room could be available if the tax freeze were redefined or interpreted differently. Given the prospects of an ageing population and substantial pressure on public finances, a

Table 27. **Degree of self-financing from various possible reductions in earned income taxes**[1]

	Average marginal tax rate (percentage point)	Change in labour supply (per cent)			Degree of self-financing (per cent)
		Working hours	Participation rates	Total	
Increase in basic personal allowance	–0.1	–0.04	–0.02	–0.06	0.00
Reduction of ordinary state tax rate	–0.6	0.09	0.01	0.10	0.18
Increase in intermediate state tax threshold	–1.3	0.24	0.06	0.30	0.33
Reduction of intermediate state tax rate	–1.4	0.33	0.05	0.37	0.51
Increase in top state tax threshold	–1.5	0.35	0.02	0.37	0.50
Reduction of top state tax rate	–1.1	0.28	0.01	0.29	0.52
Earned income tax credit[2]	–0.7	0.11	0.18	0.28	0.40

1. The direct revenue cost of each alternative is DKK 5 billion. A lump-sum tax is assumed to cover revenue that is not regained from higher labour supply. The degree of self-financing is the ratio of revenue regained from higher labour supply to the direct revenue cost of the tax cuts.
2. In the form of a reduction of the ordinary state tax rate and a proportional reduction of taxable public transfers. Thus, it is a different design to the earned income tax credit to be implemented from 2004.

Source: Finansministeriet (2002c).

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broad interpretation of "compelling reasons" for revenue-neutral reforms could include measures to address "urgent needs" for increasing work incentives. The "urgent needs" might arise from a lower-than-anticipated impact on labour force participation of measures already implemented or from difficulty in putting in place additional measures to boost labour supply. Tax reforms addressing such problems could, for instance, involve reductions of taxes on earned income financed by base-broadening measures, such as cutting back the deductibility of interest outlays and/or jettisoning the nominal principle of the tax freeze.

Summing up

The current design of fiscal policies to ensure spending control and a lower tax burden could be improved. Changes in the composition of the population and increased competition from globalisation will require adjustments to the strategy to make it more efficient.

- By employing the strict version of the tax freeze, including the nominal principle for the property value tax and excise duties, the government has in reality prevented itself from using tax instruments to significantly increase the economic incentives to work, even though the reduction of excise duties and, to a lesser extent, the property tax also have some positive indirect impact on the labour supply decision. Better outcomes in terms of labour supply, and hence public finances, might be obtained if the nominal principle were replaced by a rule involving adjustment of property value tax and excise duties in parallel with price increases. This would provide more room to cut taxes on labour income.
- While the already announced tax cuts on earned income move in the right direction, any additional room for cuts could be used more efficiently, thereby putting the priority on measures that provide the greatest estimated boost to labour supply.
- It remains to be seen whether the system of sanctions on local governments that do not comply with the tax freeze will be sufficient to keep average tax rates at the local government level from increasing. If not, the government should stand ready to implement additional or alternative measures, which could involve sanctions applying only to tax-increasing local governments or looking more closely at the alternative of a tradable-permits scheme.
- It will be a challenge to achieve only a minor increase in public service standards while also maintaining solid surpluses on public finances. The current political consensus on the necessity of debt reduction or parliamentary support for the tax freeze could be at risk, when the effects of a more strict allocation of funds emerge.

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- The central government has announced that its main priorities for higher public service standards are health, child and elderly care and education, all of which primarily are administered at the local government level. However, the tax freeze and upward pressure on expenditure from population changes will make it difficult for local governments themselves to finance increased spending in these public service areas. The central government should ensure that appropriate funding is provided if it imposes standards on public services administered at lower levels of government.
- The restrictions on existing user charges and on the introduction of new user charges seem to close off the possibility of rearranging the financing of public services and risk excluding an important instrument in assessing the value of different services to the public. The decisions by the government's economic committee will be crucial in this respect. As advocated in the previous S*urvey*, some element of user charges could be usefully applied to a wide range of services provided by the public sector.

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V. Some environmental aspects of sustainable development

There is growing concern that long-run sustainable development may be compromised unless countries take measures to achieve balance between economic, environmental and social outcomes. This chapter looks at three issues of sustainable development that are of particular importance for Denmark: reducing greenhouse gas emissions, water pollution and household waste production. In each case, indicators are presented to measure performance and an assessment is made of government policies. The chapter also considers whether institutional arrangements are in place to integrate policymaking across the different areas of sustainable development (see Box 8).

Climate change

Main issues

The reduction of greenhouse gas (GHG) emissions has been a central feature of Danish energy policy since the early 1990s, with some initiatives taken even earlier. Within the EU burden-sharing agreement, Denmark agreed to achieve a reduction of emissions of 21 per cent from the base level of 1990 by the period 2008 to 2012, and the key emitting sectors have been subject to specific targets. In view of the demanding emission-reduction aims, the main issue in this area is to attain the objectives in a cost-efficient way.

Performance

Denmark's economy-wide greenhouse gas emission intensity is somewhat lower than in the OECD as a whole but above that in the European Union (Table 28), while the decoupling of emission trends from economic growth has been weaker than in the OECD area over the 1990s. Nonetheless, emission intensities have fallen comparatively strongly in electricity generation, industry and the household sector over this period (Figure 33). Emissions from agriculture, which represent one-fifth of the total, dropped 18 per cent in the 1990s as fertiliser use fell. These within-sector trends did not translate into steep cuts in economy-wide

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Box 8. The integration of environmental concerns into government policy*

The formal structure for integration of economic and environmental decision-making is well developed in Denmark. Legislative proposals are subject to a requirement to produce a Strategic Environmental Assessment (SEA), based on a checklist of possible effects on the environment and which is the responsibility of the ministry sponsoring the legislation. Under a prime-ministerial decree, introduced in 1993 and revised in 1995 and 1998, the impact of environmental and other legislation on the economy must be assessed when it is presented to parliament. Although these arrangements provide a good framework for analysis of policy, the process is sometimes treated as a mere formality except for budgetary impacts, with little independent checking of the accuracy of either the economic or environmental impact analysis. A report into the efficacy of environmental assessments in the legislative process concluded, at least in respect of the two case studies undertaken for the report, that the influence of environmental impact assessments on Parliament's decision to adopt legislation was "murky and vague" (Elling and Nielsen, 1997). The same report said that claims in the bills that they would bring beneficial environmental effects were not substantiated or explained.

Between 1997 and 2001, the Ministry of Finance was required to produce an environmental assessment of the finance bill. In practice, these reports focussed on the budgetary costs of environmental policy and provided a number of *ex post* cost-effectiveness studies. This requirement has been ended. However, a new Environmental Assessment Institute (IMV) has been established as an independent body and has the remit to produce broader studies that take into account benefits as well as costs. It has already reported on the balance between costs and benefits in several areas of policy.

Cost-benefit analysis of policies in a given area has sometimes been undertaken after the event. A recent study of 11 energy policies revealed that benefits had exceeded costs in only two cases (Det Økonomiske Råd, 2002b). The present value of the excess of costs over benefits of the nine policies (see below) amounted to over 2½ per cent of GDP in 1999, though, as with all studies, results are sensitive to assumptions. On the other hand, polices to address air pollution had been highly beneficial. The Council was unable to undertake any analysis of past water policies, as estimates of the benefits were scarce. Moreover when commitments were made to reduce water pollution from nitrogen, no model was available that linked nitrate runoffs to nitrate concentrations in the sea (Hägerhäll, 2001), though this situation has changed recently. As to waste management, the IMV examined packaging policy and found some aspects that had net costs for society (IMV, 2002a).

* The sections in this report dealing with climate change, water pollution and household waste are inputs into the Organisation's follow up on Sustainable Development as mandated by the Ministerial Council decision in May 2001.

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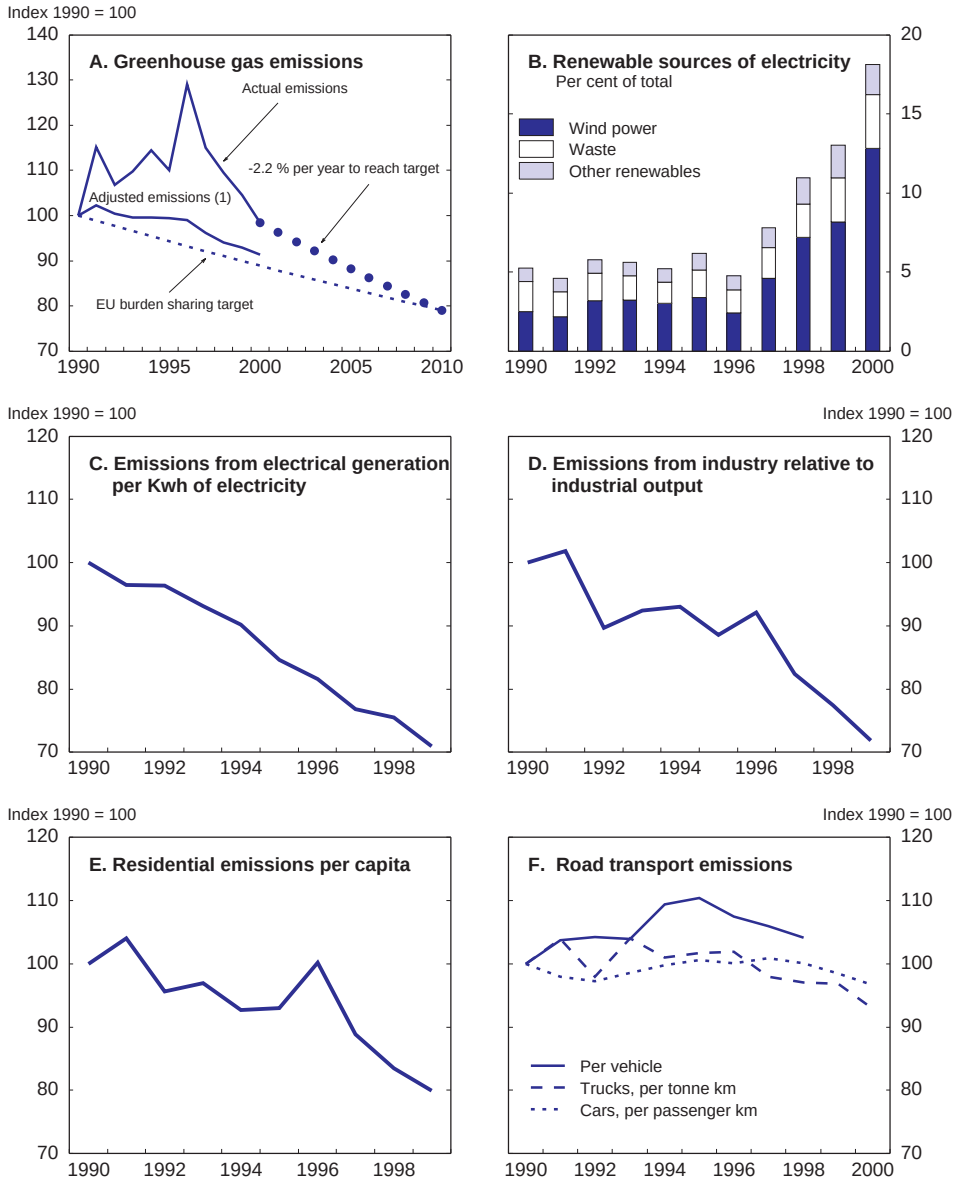
Table 28. **Main indicators: climate change**

Indicators of greenhouse gas (GHG) emission intensity, grams of CO_2 equivalent per $PPP of GDP, in 1995 prices

	Total GHG emissions	CO_2 emissions, electricity	CO_2 emissions, transport	Other GHG emissions	Total GHG emissions	CO_2 emissions, electricity	CO_2 emissions, transport	Other GHG emissions
	Level, 1999				Average annual percentage change 1990-99			
Australia	1 053	370	155	528	–2.07	–0.21	–1.93	–3.24
Austria	419	72	91	256	–1.87	–2.75	–0.52	–2.06
Belgium	617	97	101	419	–1.36	–2.12	0.16	–1.52
Canada	893	151	193	549	–0.98	–0.12	–0.36	–1.41
Czech Republic	1 058	457	88	513	–3.05	2.55	5.53	–6.93
Denmark	**549**	**194**	**94**	**261**	**–1.64**	**–1.43**	**–1.49**	**–1.85**
Finland	652	181	105	366	–1.88	–0.02	–1.29	–2.83
France	416	32	103	280	–1.69	–2.04	0.16	–2.26
Germany	536	169	96	271	–4.00	–3.86	–0.57	–5.05
Greece	813	275	130	408	–0.24	0.07	0.74	–0.73
Hungary	786	250	84	453	–2.33	1.44	0.38	–3.74
Iceland	395	4	88	303	–1.28	0.00	–2.31	0.81
Ireland	694	165	103	426	–4.27	–2.41	0.79	–5.75
Italy	439	105	92	242	–1.05	–0.82	0.37	–1.64
Japan	432	130	82	221	–0.30	–0.03	1.24	–0.99
Luxembourg	344	6	242	97	–11.46	–30.20	–0.45	–18.81
Netherlands	573	138	82	352	–2.38	–1.03	–0.94	–3.15
New Zealand	1 096	92	175	828	–2.28	4.58	0.65	–3.32
Norway	487	4	113	369	–2.54	1.31	–1.53	–2.87
Poland	1 195	481	90	624	–4.96	–6.63	0.50	–4.12
Portugal	540	149	106	285	0.41	2.58	3.37	–1.39
Slovakia	957	200	76	680	–4.47	–1.21	3.13	–5.78
Spain	537	127	130	280	0.41	1.12	1.28	–0.26
Sweden	358	41	112	204	–1.55	0.07	–0.65	–2.30
Switzerland	276	3	79	195	–0.62	–1.96	–0.28	–0.73
United Kingdom	526	132	108	287	–3.66	–5.30	–1.38	–3.61
United States	792	278	196	318	–1.89	–0.60	–1.18	–3.28
OECD total	**649**	**196**	**140**	**312**	**–1.80**	**–0.98**	**–0.38**	**–2.83**
EU	506	120	103	283	–2.36	–2.60	–0.16	–2.95

Source: National submissions to the UNFCCC and national publications; IEA; OECD.

Figure 33. **Greenhouse gas emissions**



1. Adjusted emissions smooth the year-to-year consequences of fluctuations in net electricity exports.
Source: UNFCCC (2001); IEA (2001); Energistyrelsen; Miljø-og Energiministeriet.

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emissions as electricity use and industrial output rose markedly in relation to GDP. Indeed, overall greenhouse gas emissions fell only 2 per cent over the 1990s. However, emission trends are obscured by year-to-year fluctuations in the export and import of electricity that depends on rainfall in Norway and Sweden. Correcting for fluctuations in net imports of electricity, the underlying fall in GHG emissions was 9 per cent. For the future, with existing measures that have been taken, there may be a slight increase in underlying emissions in the period until 2008-12 according to official projections (Fenham, 2003). With significant spare capacity in the fossil fuel sector of electricity generation (in part the result of growth in renewable production), the authorities expect a surge in electricity exports to the rest of the Nordic Pool electricity grid to boost GHG emissions by 18 per cent.[76] Overall, allowing for the low level of electricity production in the base year period,[77] the growth in electricity exports and the underlying increase in emissions, official projections suggest that emissions will be between 35 and 45 per cent (depending on assumptions about the base year) above the target set under the EU burden-sharing agreement for the period 2008-12.

Policies

Denmark has a form of carbon taxation with rates that are neither uniform across sectors (Table 29) nor across different types of energy. This rate structure would appear to be very inefficient from the point of view of cutting carbon emissions: the lowest rates are charged to the heaviest emitters, while small emitters are subject to high rates. Such discrimination has been introduced in order to avoid higher emitters moving their operations abroad or being out-performed by foreign competitors, potentially creating "carbon leakage". These distortions are increased by the rebates given under voluntary agreements targeted at energy-intensive enterprises, which grant considerable tax subsidies for the implementation of energy-saving measures agreed with the authorities. By 2001, over 300 enterprises had agreements with the Energy Authority, accounting for about 60 per cent of industrial electricity consumption. Even within

Table 29. **Dispersion of tax rates on carbon by sector**
Danish kroner per tonne of carbon dioxide

	1996	1997	1998	1999	2000
Heavy process no agreement	5	10	15	20	25
Heavy process with agreement	3	3	3	3	3
Light process no agreement	50	60	70	80	90
Light process with agreement	50	50	50	58	68
Space heating	100	100	100	100	100

Source: Energistyrelsen (2000).

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companies, different uses of fuel are subject to varying taxes: fuel consumption for space heating purposes is taxed at the same rate as that of households, which is a lower rate than applies to fuel used in manufacturing. Welfare-maximising tax rates would lie between these two values (Søbygaard, 2002).

Denmark has used an extensive mix of command-and-control and subsidy instruments to boost the production of renewable energy and increase energy efficiency. These policies have been criticised in previous *Surveys* for putting an undue burden on the economy, even if they have helped create a wind turbine industry whose productivity has increased through learning-by-doing. Analysis of the principal energy policies has shown that the value of the external benefits needed to justify the projects was extremely high;[78] ranging between EUR 150 and EUR 900 per tonne of carbon. Moreover present values for most projects were negative over a wide range of possible carbon prices and interest rates, though the calculations do depend on the nature of the baseline against which carbon savings are evaluated (Table 30).[79] Most estimates of the benefits of avoiding carbon emissions are below this range, suggesting that such programmes have lowered the level of sustainable consumption. Moreover, the existing high level of distortionary taxation adds to welfare costs of financing these projects. Outside the energy sector, a number of other policies appear to have high costs (Figure 34).

Denmark has now decided to rationalise its mix of policies to favour renewable energy to bring it onto market-based terms. From January 2003, electricity generated by new wind turbines (and gradually older ones too) receives a premium of 10 øre per kWh (1.35 euro cents per kWh) paid by all domestic consumers of electricity. This is equivalent to the highest CO_2 tax rate applied to electricity (including CO_2-free renewable-based). Such a premium is equivalent to EUR 58 per tonne of carbon used in a coal-fired power station, though if wind power reduces coal-fired production some benefits may flow from the reduction in local air pollution. However, if wholesale electricity prices rise above 26 øre per kWh, then the renewable energy subsidy will be reduced, one for one. The obligation on consumers to purchase fixed quantities of wind-generated electricity is being phased out, while the obligation to purchase electricity from small-scale CHP plants is being reconsidered.[80] Previous plans to establish a green certificate market for renewable electricity (see O'Brien and Høj, 2001 and OECD, 2002f) have been postponed indefinitely.

In the transportation sector where emission abatement costs are particularly high, targets for emission reductions have proven difficult to meet and the original targets set in 1990 have been abolished. The vehicle registration tax in Denmark is very high at 180 per cent of the vehicle's value. However, from 2000 the registration tax was reduced on new fuel-efficient cars,[81] although it remains high by comparison with the rest of the OECD. In addition, an annual "pollution tax",

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Table 30. **Present value of various policies introduced between 1992 and 1999**[1]
Billion DKK, 2002 prices

	Discount rate, 3 per cent		Discount rate, 6 per cent		
	Carbon price, € 23[2] per tonne	Carbon price, € 133 per tonne	Carbon price, € 23 per tonne	Carbon price, € 133 per tonne	Shadow price, EUR per tonne of carbon
Energy policies					
Subsidies to sustainable energy	–5.8	–5.0	–5.5	–4.8	913
Labelling policies	–9.2	–7.8	–8.4	–7.3	898
Subsidies for conversion to CHP	–4.2	–3.4	–4.8	–4.1	822
Subsidies to building CHP networks	–0.8	–0.4	–1.0	–0.7	356
Biomass power	–8.8	–4.6	–8.0	–4.6	280
Electrical heating subsidies	–1.7	–0.5	–2.0	–1.0	253
Local combined heat and power (CHP) plants	–12.6	–1.9	–14.7	–4.7	184
Non-utility owned land-based wind turbines	–10.4	0.1	–13.1	–4.0	182
Utility owned wind turbines	–3.3	1.3	–4.5	–0.7	153
Subsidies to business	2.8	9.1	–0.7	4.6	37
Carbon tax rebates	1.4	3.6	0.7	2.3	–22
Air pollution policy					
Sulphur control in electricity plants	99.7	99.7	92.7	92.7	n.a.

1. This table shows the present value of different government projects designed to reduce carbon emissions or raise energy efficiency. The projects have been evaluated including an external benefit from reduced carbon emissions that amounts to EUR 23 or EUR 133 per tonne of carbon avoided. The projects have been evaluated at real interest rates of 3 and 6 per cent, giving four possible results shown in the first four columns. The final column shows the external benefit (measured as price per tonne of carbon emissions avoided) that would have been necessary for the projects to have a zero present value when evaluated at a 6 per cent interest rate.
2. DKK 1 = EUR 0.1346.

Source: Søbygaard (2002).

based on fuel consumption of a new model, rises more than proportionally with fuel consumption. On balance, the very high taxes probably contributed to the low level pollution equipment on Danish cars at the beginning of the 1990s. By the end of the 1990s the proportion of Danish cars with control equipment was close to the European average, in part due to a scrapping subsidy in the mid-1990s designed to speed the renewal of the fleet. The combination of high taxes on use and ownership of vehicles has resulted in road transport in Denmark paying 70 per cent more than the estimated sum of user costs and externalities from pollution, accidents and noise but excluding traffic congestion (Nash *et al.*, 2002). Changing the price of transport taxation by introducing distance-based charging, cordon pricing[82] and reducing the reliance on private transport taxation would likely improve welfare (CEMT, 2003) but the role of taxes in this area is seen by the government as one of balancing costs and benefits.

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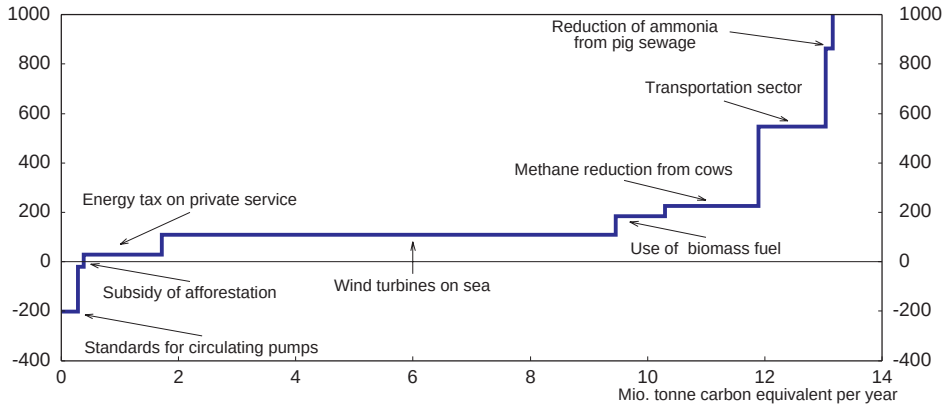
Figure 34. **Marginal abatement costs for greenhouse gas emissions**[1]

Cost per tonne of carbon equivalent



1. The diagram shows the cost of avoiding a tonne of carbon emissions for different projects, together with the output range for which this cost is valid. The policies are then grouped in ascending order of costs.

Source: Danish government submission.

In addition to implementing carbon taxation, Denmark has been a pioneer in domestic trading of carbon emissions permits. Its scheme began in 2001 and is limited to the electricity-generating sector. Companies received permits based on their emissions in the 1994-98 period and can bank or trade their excess permits in a given year, only paying a penalty of DKK 40 per tonne of CO_2 (EUR 19.8 per tonne of carbon) for emissions in excess of permits held. There has been a limited amount of trading, at prices well below those used in the evaluation of carbon abatement policies.[83]

The high costs of future domestic measures led the government to base its new strategy, adopted by Parliament in March 2003, on the use of flexible market-based instruments. In particular, the government has welcomed the freedom that the new EU quota system will give to some companies to achieve emission reductions in the most cost-effective way, either by abatement in Denmark, projects abroad or by purchasing permits once the EU trading system is working. In order to keep the strategy efficient the government is setting an overall cap of DKK 120 per tonne of carbon dioxide abated (EUR 59.3 per tonne of carbon) in deciding whether the costs of any project outside the EU trading sector outweigh benefits. Only projects whose benefits exceed costs will be implemented.

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The government has found that major extra abatement effort in those parts of the economy already taxed at rates above the likely benchmark price would be inefficient. Emissions in the transport, residential and commercial sectors are, thus, assumed to follow the recent unchanged policy projections of the government in the period to 2008-12 (Table 31). In order to meet its overall target, one strategy the government could follow would be to set the total quantity of allowances issued to eligible emitters as the difference between the Danish burden-sharing target and the projected emissions of the remaining sector in the compliance period. On this basis, eligible emitters would have to be allocated allowances equivalent to three-quarters of their 2001 emissions.

Significant scarcity rents will be generated by these allowances. Under the EU trading system, in which Denmark will participate, the bulk of allowances must be issued free according to a national allocation plan that has to be approved by the European Commission. Only 10 per cent can be auctioned in the period 2008-12. Given the government's assumption that the maximum price of the allowances is likely to be DKK 60 (in 2001 prices), the total value of the allowances is likely to amount to 0.17 per cent of GDP. Several studies suggest that only a fraction of this amount would be necessary to preserve the profits of the emitting industries covered by the trading scheme. The remainder would represent a windfall for the shareholders of the companies in this sector. Electricity prices would also rise significantly, resulting in improved profits for generators. One route that could be considered to recuperate some of this value would be to place a specific tax on the annual issue of such allowances that are not auctioned. This would permit part of this windfall to be reclaimed for society. The revenue could then be used to reduce other taxes, starting with the most distortionary ones.

Conclusions

After a long period when policies to reduce greenhouse gas emissions were far from optimal, government policy has begun to evolve significantly. Policies favouring wind power have been scaled down and made more rational. A similar approach should be applied to combined heat and power biomass-fuelled plants. At the same time, the government commitment to making the EU trading of carbon quotas the centre of its policy is a welcome step. The government should set the quota of allowances at a level that would ensure that Denmark would meet its EU burden-sharing agreement, on the assumption that emissions elsewhere are mainly determined on the base of current tax rates. There would then be no need for major additional greenhouse gas abatement policies, even though the government sees a role for them at costs of up to twice its expectation of the price for allowances in order to allow for uncertainties etc. More generally, these allowance prices could, as the market matures, be used as a benchmark for carbon taxes in the sectors of the economy not affected by trading in order to

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Table 31. **Projections of Danish greenhouse gas emissions and the value of allowances**

	2001	2008	2009	2010	2011	2012	Compliance period
	Gigagrams of Carbon dioxide equivalent						
Projections of emissions with no trading							
All greenhouse gases	69 792	79 558	80 017	80 134	80 236	80 360	80 061
Likely extent of CO_2 emissions covered by trading	55 894	67 117	67 714	67 992	68 2256	68 616	67 933
Energy	52 842	63 692	64 222	64 431	64 605	64 941	64 378
Manufacturing	3 052	3 425	3 492	3 561	3 620.5	3 675	3 555
Greenhouse gas emissions uncovered by trading	13 898	12 442	12 303	12 143	12 011	11 744	12 128
Calculation of allowances for the trading sector	–	–	–	–	–		
Target under burden sharing	–	–	–	–	–	–	55 000
BAU emissions for non-trading sector	–	–	–	–	–	–	12 128
Target less BAU emissions for non-trading sector	–	–	–	–	–	–	42 872
Emissions for non-trading sector in 2001	–	–	–	–	–	–	55 894
Initial allowance as a per cent of 2001 emissions	–	–		–		–	76.7
Price of allowances based on DKK 60 in 2001 prices	60	70	72	73	75	76	
Value DKK millions, current prices	–	3 014	3 074	3 136	3 198	3 262	3 137
Value as a per cent of GDP	–	0.17	0.17	0.17	0.17	0.16	0.17

Note: Projections of emissions with no trading taken from Fenham (2003).
1. Maximum emission price taken from Finansministeriet *et al.* (2003).
2. It has been assumed that all carbon dioxide emissions of the energy sector will included in the trading scheme and half of those from the manufacturing sector.
Source: Fenham (2003); Finansministeriet *et al.* (2003).

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equalise abatement costs. In particular, the premium given to renewable electricity may need to be re-evaluated. Finally, ways of capturing the value of the allowances for the community should be considered. A tax on newly-issued permits is one possibility. The resulting revenue could be used to cut distortionary taxes.

Reducing water pollution

Main issues

Since the late 1980s, the principal objective of policy in the area of water pollution has been to reduce the discharge of nutrients into groundwater, streams and lakes, thereby improving the quality of these resources. A further objective of policy has been to ensure that underground water (the almost exclusive source of the public water supply) does not contain nitrates above concentrations judged safe for drinking purposes. This policy was expected to limit the extent to which coastal waters, and in particular the Kattegat, would be subject either to very low oxygen levels or to an excessive growth of algae, both of which harm the environment. The key issue is whether the ambitious objectives in this area have been pursued in cost-efficient ways.

Performance

The past 15 years have been characterised by a marked fall in the discharge of nutrients to water systems. Nearly all households are now connected to wastewater systems with tertiary treatment (Figure 35), bringing Danish performance to a level considerably above the average OECD country. As a result, point-source emissions of nitrates fell by two-thirds between 1989 and 1999, and those of phosphates were reduced by over 80 per cent. Commercial fertiliser application has been falling though animal fertiliser use has been stable and now accounts for approximately 40 per cent of the total, and the excess application of nutrients has dropped three times as rapidly as elsewhere in the OECD area, but remains double the EU average. Concentrations of nitrates in most recently stored groundwater have peaked and even started to ease back. However, nitrogen concentrations in streams in agricultural areas fell only 7 per cent between 1989 and 1997. The reduction in phosphorus loading has also improved the quality of lakes, with less algae growing and water clearness increasing. The gains have been less marked in marine waters. Phosphate concentrations have fallen, both in lakes and in the sea, but this nutrient is not generally critical in marine water. Nitrate levels have fallen in the sea, but to a very limited extent, as air deposition represents well over half of total nitrogen deposition in the Baltic Sea.

Policies

In order to reduce the amount of nutrient discharge from point sources, a tax based on the nutrient content of effluents from wastewater was introduced

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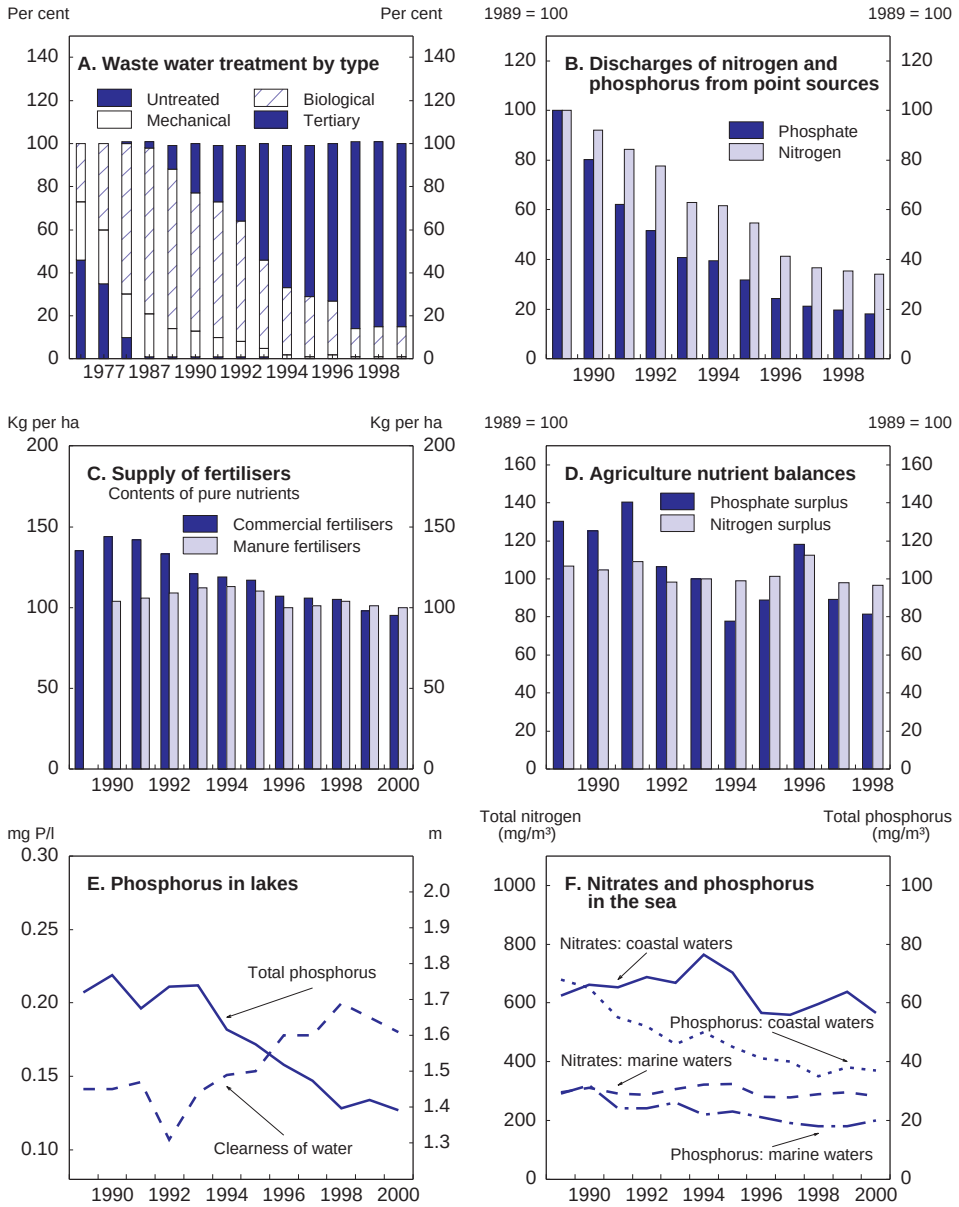
Figure 35. **Water pollution**



Source: Ministry of Environment.

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in 1997.[84] The tax was backed with new norms for wastewater treatment plants. By 2001, nearly all plants used a form of tertiary treatment to remove both phosphate and nitrogen from discharges. However, the major industrial polluters (associated with cellulose, sugar beet, organic pigments and fish processing) are allowed a reduction in the tax of between 70 and 97 per cent. These industries accounted for 88 per cent of the biological waste and between 50 and 60 per cent of nutrient waste discharged by all industries (Miljøstyrelsen, 1999).

Policies designed to reduce nutrient outflows have led to a sharp increase in the price of water for household users. The pre-tax cost of water rose 58 per cent in real terms between 1988 and 1999. Since 1993, water consumption has also been subject to a specific tax that accounted for 23 per cent of the retail price of water by 1998. Overall, real prices to households doubled between 1988 and 1999, with water consumption dropping by 28 per cent per capita in the same period. As most water is drawn from underground sources, reduced water use can result in increased outflow into streams so possibly reducing concentrations of pollutants.

The policy objective of halving run-offs from agriculture from their level in the mid-1980s (200 000 tonnes of nitrogen per year) is now close to being realised, but at a considerable cost. The first two plans to reduce water pollution from agriculture, introduced in 1987 and 1991 respectively, are estimated to have cut leaching by 60 000 tonnes per year. The second plan set fertiliser norms for each farm and taxed any surplus use. This policy continues in force (OECD, 2002f). In 1998, there was a further adjustment, with the introduction of the Action Plan for the Aquatic Environment II, which made use of a wide range of instruments aimed at changing land use as well as reducing fertiliser inputs (Table 32). The cost of these different measures varied considerably, and it is clear that the most cost-effective mix of policies was not introduced. The different programmes varied in cost by a factor of over 15 (Jacobsen, 2002). Overall, by 2001, the programme was estimated to be costing EUR 3.2 per kilo of avoided nitrogen leaching, in line with the wastewater tax, and over six times the cost of nitrogen fertilisers. Farmers paid slightly over 40 per cent of the EUR 77 million annual cost of the programme. The programme was expected to achieve a further annual reduction in leaching of about 40 000 tonnes of nitrogen by 2003. A third plan is expected to be introduced in 2004. In addition to these policies, Denmark has also been making progress towards implementing the EU Nitrates Directive (European Commission, 2002). Under this directive, *inter alia*, the application of manure is limited to 170 kg per hectare for cattle farms, although under a derogation granted in November 2002, this can be increased to 230 kg per hectare if the farmer respects stricter environmental regulations, and 140 kg per hectare for other farms.

The cost of the current programme might be lessened by using different economic instruments. If, as seems likely, the regulator cannot precisely estimate

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Table 32. **The cost of policy measures to reduce nitrogen leaching**

Policies	Area of land concerned	Reduced nitrogen leaching	Government costs	Farm costs	Abatement cost
	Hectares	Tonnes of nitrogen	Million euros/year		Euros per kilogramme of nitrogen
Persuade farmers to use feedstuff concentrates with more animo acids, so reducing nitrogen output from pigs[1]	–	3 100	–	–	0.0
Convert marginal land near rivers to year round wet land, thereby absorbing considerable nitrogen[2]	6 000	2 100	1.4	–	0.7
Require a 15 percentage point increase in spreading of cattle and pig manure on fields[3]	–	7 600	–	5.4	0.7
Require farmers to cover an extra 6 per cent of land in grass between August and October[4]	120 000	3 000	–	8.4	2.9
Provide subsidies for planting and operating a forest[5]	17 340	900	2.7	–	3.1
Lower nitrogen application norms to 10 per cent below the economic optimum, with fines for excess application[6]	–	5 100	–	17.0	3.4
Apply measures to environmentally sensitive areas near lakes and rivers, including conversion to pasture, a 60 per cent nitrogen norm and taking land out of production. All measures include compensation	29 195	900	9.3	–	10.4
Lower stocking density by obliging farmers to increase the amount of land available for spreading slurry from cows or pigs	–	100	–	1.5	15.0
Pay farmers a subsidy of € 204 to convert to organic farming[7]	155 700	1 600	31.7	–	19.9

1. Farmers adopted this measure as it implied no cost increase.
2. Only a fraction of the original target will be *met* as the payment to farmers was below the value of all but the poorest quality land.
3. To 65 per cent of nitrogen content for cattle farmers and 60 per cent for pig farmers.
4. The direct cost is low but farmers have to plant spring wheat instead of winter wheat so losing value.
5. Cost is overstated since the planting subsidy has not been amortised.
6. If application is less than 30 kg/ha above the norm a fine of DKK 10 is imposed per kg, above 30 kg/ha the fine is DKK 20.
7. Conversion targets are not being met as production of organic milk is twice demand.

Source: Jacobsen (2002).

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the optimal dosage of nutrient for each field, then the current norm-based systems of reducing fertiliser input can impose costs on farmers and, in addition, generate a significant administrative burden through the need for annual fertiliser plans (Hansen, 2001a). Simpler alternatives to a norm-based system exist and would impose compliance costs only on feed merchants, fertiliser companies and abattoirs. For example, a tax could be applied to all commercial fertiliser inputs and the nitrogen content of feedstuffs, with farmers receiving a rebate based on the nutrient content of their output.[85] Overall model simulations suggest that a tax/rebate system could deliver a 10 per cent reduction in nitrogen run-off for less than half of the simulated cost of a norm-based system (with a 10 per cent error by the regulator) (Hansen, 2001b), though it could create incentives to lower the nitrogen content of the soil. It would also generate major reductions in administrative costs. There might still be a need for a manure management regime. Current regulations add about 4½ per cent to the cost of pig meat. As yet the environmental gains can only be assessed with great uncertainty (OECD, 2002f), whereas the cost of the regulations is more apparent.

The impact of changes of agricultural policies on the use of nutrients may have been limited. The last reform of the CAP (Agenda 2000) reduced prices of a number of key agricultural products while lowering the amount of land that was subject to set aside. It is still too early to measure its impact on Danish farmers, but simulations of linked agricultural and environmental models suggest that the impact on fertiliser inputs may be quite limited, as they are already restricted by quotas, (Wier *et al.*, 2002). The overall the nitrogen uptake by the sea may be lowered by 1 per cent. Such a decrease in nutrients is not expected to result in any change in the oxygen content of marine waters.

Conclusions

The policies designed to reduce nitrogen run-off are now close to having achieved their stated aims, with a number of positive effects on the state of the aquatic environment. However, the impact on nitrogen concentration in coastal waters has so far been very limited. Further gains can be expected in future years as the impact of past policies is felt. But the targets for reduction in nitrogen loading have been established on the basis of only limited modelling of the linkage between nitrate run-offs, marine concentrations and environmental objectives. Moreover, to lessen the economic cost of reducing run-off from agriculture, it would be appropriate to make greater use of economic instruments, such as taxing commercial fertiliser inputs with rebates based on the nitrogen content of output. Greater effort needs to be made to quantify the benefits of different nutrient reduction and manure management policies, and in the meantime, a cautious and incremental approach to further policy initiatives would be advisable.

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Improving management of waste

Main issues

Public authorities in Denmark have been attempting both to reduce the growth of waste and to ensure that it is recycled. Policymakers were concerned about the growing pressure of landfills on available space and fears about dioxin emissions from incineration plants but have also wanted to limit the loss of resources from waste disposal. The main issue in this area is how to balance the environmental benefits against the total financial costs of the different disposal methods.

Performance

Household waste has risen by over 60 per cent between 1985 and 2000, although it fell 15 per cent relative to private consumption. This was similar to the decline experienced in other OECD countries in the same period (Table 33). In the past five years, the pace of the decline has increased somewhat. Domestic waste generation (including bulky and garden waste) was about 630 kg per capita in 2000, slightly higher than the average for municipal waste in other OECD countries. Since 1986, the provision of recycling services for paper and glass has been mandatory for local authorities, contributing to an increase in paper and cardboard recycling rates from 30 to 50 per cent between 1986 and 1997, with glass recycling rates rising from 32 to 70 per cent in the same period.

Policy

Denmark's waste policy embodies a waste hierarchy, employing both economic instruments and command-and-control measures to attain policy targets (Environmental Protection Agency, 2001). It is implemented by municipalities, who must charge fees to cover costs. The latest government plan sets targets for different streams of total waste disposal for 2004: 64 per cent of waste is to be recycled, 24 per cent incinerated and 12 per cent landfilled. In fact, these targets had already been met when the plan was introduced in 2000, suggesting that the targets were deliberately set to involve little further effort. The aggregate figures mask very different positions according to the source of waste.[86] There were difficulties in meeting targets for recycling domestic waste. The previous waste plan (1993-97) failed to achieve its target for this type of waste, with the recycling rate reaching only 29 per cent in 2000, against a target of 49 per cent. Moreover, most recycling of waste collected by local authorities takes the form of composting garden waste.

The Waste 21 plan envisages only an increase in domestic recycling to 30 per cent by 2004. The current plan includes the introduction of kerbside

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Table 33. **Performance indicators: waste**

	Municipal waste					
	Level	Relative to private consumption		Period	Recycling	Composting
	Kg per capita, latest available year[1]	Grams per US$, 1995 PPPs	Annual growth, per cent		Per cent of total, latest available year[1]	
Australia	. .	. .	. .	–	. .	. .
Austria	548	45.5	3.7	1990-1999	24	40
Belgium	534	47.0	3.0	1990-1999	37	15
Canada	. .	. .	. .	–	. .	. .
Czech Republic	334	54.2	. .	1990-2000	. .	. .
Denmark	**626**	**57.4**	**3.5**	**1994-1999**	**25**	**14**
Finland	464	49.0	–2.8	1990-1999	. .	. .
France	524	46.4	1.7	1993-1999	10	8
Germany	549	46.0	–1.5	1991-1998	41	. .
Greece	371	42.7	3.8	1990-1997	8	1
Hungary	444	85.7	–1.9	1990-2000	0	0
Iceland	718	52.7	2.8	1992-2000	9	2
Ireland	615	49.5	8.2	1995-2000	8	1
Italy	497	40.1	4.0	1990-1999	7	10
Japan	406	34.6	0.2	1990-1999	13	. .
Korea	364	50.3	–5.8	1990-2000	41	. .
Luxembourg	643	38.7	3.1	1991-1999	0	12
Mexico	315	63.5	4.3	1991-2000	2	0
Netherlands	613	53.3	2.9	1991-2000	23	24
New Zealand	. .	. .	. .	–	. .	. .
Norway	617	51.2	2.7	1992-2000	22	9
Poland	316	64.0	1.0	1990-2000	0	2
Portugal	437	50.7	4.2	1992-1999	4	5
Slovakia	316	58.2	. .	1992-2000	2	5
Spain	464	48.0	4.3	1990-1999	5	18
Sweden	452	47.5	2.8	1990-1998	25	8
Switzerland	655	41.8	1.4	1990-2000	32	14
Turkey	385	101.4	. .	1989-1998	0	1
United Kingdom	558	44.8	2.3	1990-1999	9	2
United States	764	35.7	1.3	1990-1999	22	6
OECD average	501	42.3	1.3	–	17	7
EU average	526	45.6	1.9	–	18	10

1. 2000 for Ireland.
Source: OECD, Eurostat. For Spain: national submissions.

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collection of paper and glass from settlements of more than 2 000 inhabitants but only if voluntary "bring" systems fail to achieve targets for the recycling of glass and paper. Brisson (1997) calculates that separate kerbside collection of different types of waste would increase the private costs of recycling considerably, even exceeding the reduction in the estimated external environmental costs from the disposal of waste. Even if voluntary "bring" systems result in good collection rates, there is evidence that it would be better to incinerate such paper than incur additional costs be re-utilising it (IMV, 2002b). In addition, electronic and electrical equipment will be collected separately to reduce the incineration and landfilling of copper and lead, so enabling higher recycling of incinerated waste. These measures may increase total annual costs by DKK 40 per household (Miljø- og Energi, 2000). However, the government has stopped short of mandating recycling of household plastic waste due to the high costs.

Attempts in some municipalities to reduce household waste have met with mixed results. While the municipalities that introduced fees based on the weight of household waste collected have seen refuse amounts fall, other municipalities with less complex systems have also experienced similar declines (Anderson *et al.*, 1997), casting doubt on the importance of weight-based charging. In addition, the Environmental Protection Agency concluded from the results of surveys that although collected household waste was reduced, there was a concurrent increase in waste deposited at recycling stations and illegal waste dumping (Environmental Protection Agency, 2002).

There are a number of taxes on waste and packaging that have been introduced to influence the treatment of refuse. Over time the tax rate on waste disposal has increased and become more differentiated (Table 34). The tax is set at a higher rate for landfill than incineration, thereby encouraging incineration over landfilling. However, a regulation from 1997 mandates municipalities to send burnable waste to incineration rather than to landfill (though the obligation is sometimes referred to as a ban). This increases costs substantially, since the operating expenses at landfills are only EUR 48 per tonne against incineration costs of EUR 92 per tonne (Miljøstyrelsen, 2002). Moreover landfill sites are

Table 34. **The waste tax**

DKK per tonne

	1987	1990	1993	1997	1999
Landfill	40	130	195	335	375
Incineration with electricity generation	40	130	160	210	280
Other incineration	40	130	160	260	310

Source: Environmental Protection Agency.

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operated inefficiently, offering significant scope for further cuts of up to 40 per cent, unavailable for incinerators. The waste tax also aims at increasing recycling and reuse. However, considering the high cost of recycling domestic waste, the tax on landfilling and incineration may still be too low to make recycling the least costly option for households and businesses. On the other hand, these taxes appear to exceed the levels of estimated externalities in both cases. This would suggest that the objectives concerning recycling do not appear to be driven by the social costs of incineration or landfill. As part of a greater drive to reduce waste, Denmark also introduced a tax on limited number of packaging products (bottles and jars) in 1998. A broader based tax was introduced the following year. The new tax covered a variety of packaging materials and containers. There were 26 different tax rates; some were based on the weight of the material used, while others were based on the volume of the containers. The tax rates for beverage containers were intended to encourage greater reuse rather than recycling, complementing the mandatory deposit scheme (in place since 1981) that has encouraged a recycling and reuse rate approaching 100 per cent, with the deposit scheme lowering waste by 20 per cent. The tax was revised again in 2001 on the basis of a study that estimated the externalities that occur in the life cycle of a packaging product. A tax rate was then chosen for glass and the tax on all other products was made proportional to the ratio of the life-cycle externalities of the respective product and those of glass. This methodology has produced relatively high taxes on aluminium. The tax rates will be revised again in 2004, following a further study of externalities.[87] Metal containers for beverages were banned until the beginning of 2002. Since then they have been made part of the mandatory deposit scheme, even though that scheme appears to place a net cost on society (IMV, 2002a).

Overall, the waste and various packaging taxes are significant both in relation to waste disposal costs and private consumption of goods. In 1998, for example, such taxes amounted to 63 per cent of the cost of household waste disposal (Table 35). With waste disposal externalities likely to be less than 10 per cent of the private costs of waste disposal (around EUR 15 per tonne according to a number of studies), this level of taxation would seem hard to justify on environmental grounds. It would suggest that there may also be welfare losses for the population from current waste policies.

Conclusions

The overall cost of waste collection, treatment and taxation in Denmark represents a significant cost to households. In part this is a result of the waste tax, though collection and disposal costs vary markedly across municipalities; even for the country as whole, collection and disposal costs are amongst the lowest in Europe. Current initiatives to increase separation at source are likely to push

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Table 35. **Household waste disposal charges and taxes**
1998

	Millions DKK	Per cent household consumption of goods
Disposable plate tax	56	0.02
Raw material tax	156	0.06
Retail packaging tax	809	0.30
Landfill and incineration tax	593	0.22
All taxes on waste	**1 614**	**0.60**
Waste disposal outlays (ex disposal taxes)	2 561	0.95
Total waste related outlays	**4 175**	**1.55**
Memorandum item:		
Private consumption of goods	269 278	–

Note: Landfill and incineration tax yield from domestic waste only.
Source: Statistical Yearbook of Denmark; Danish government submission.

waste collection costs up further. Although a wide range of economic instruments is used in Denmark, their effect is limited due to the regulations that also cover key areas. However, the level of taxation resulting from economic instruments has to be compared to the level of environmental externalities and the administrative costs of the tax. Overall, the current extent of taxation on packaging and waste would appear difficult to justify on grounds of external costs. Moreover use of regulations to control recycling implies an even higher valuation of the negative externalities of landfill and incineration and would not appear to be based on environmental concerns alone.

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Notes

1. Within the ERM2, the krone's rate against the euro shall be kept within a narrow band of +/–2.25 per cent around the central rate. If the krone reaches the upper or lower limit against the euro, both the ECB and the Danish National Bank must intervene to strengthen the weaker of the two currencies, so that the exchange rate is kept within the fluctuation band. In recent years, the krone has been kept close to its central rate.
2. It should be noted that the government does have one other instrument for adjusting demand, namely raising contribution rates to the Special Pension (which is a form of compulsory private savings). However, the difficulties of getting the timing just right to successfully reduce fluctuations in economic activity rather than making matters worse also apply to using this measure. Therefore it should be used only sparingly and when action is clearly warranted.
3. In Dang *et al.* (2001), working age is defined to be 20-64 years, youth 0-19 years and elderly 65 years or above. In the remainder of this chapter, working age is defined to be 15-64 years and youth 0-14 years.
4. Therefore, a great deal of attention has been devoted to ways of shielding public finances, in particular through reforms of pension systems.
5. The Danish Rational Economic Agents Model group is an independent, publicly funded institution administratively located within the Ministry of Finance. It maintains and develops a computable general equilibrium model of the Danish economy, which applies demographic projections as a basic input. While the population projection from DREAM has been widely used in recent years, demographic projections produced by Statistics Denmark still form the basis for planning by local governments, as they include projections for individual counties and municipalities. The projections from DREAM and Statistics Denmark differ slightly in assumptions and therefore outcomes.
6. The number of individuals in each generation is very small for descendants of immigrants from LDCs. Thus, this group is assumed to have the same age-specific participation rates as descendants from MDCs.
7. The general increase in female participation has also reduced the need for men to provide for the household and thereby made temporary or permanent withdrawal more feasible for men.
8. The last effect is likely to be more pronounced in the public sector where the bulk of female entry has been concentrated.
9. A reduction in working hours is also found in most other OECD countries (OECD, 2002a). Differences in sources make the data unsuitable for comparison in level terms, but they seem to indicate that the average annual number of working hours is relatively low in Denmark.

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10. Feyrer (2002) finds that countries with a large proportion of workers below 40 years of age have lower aggregate productivity levels than those with large cohorts in their 40s, whereas the proportion of the working force above 50 years is not seen to have a significant effect. The same relationship seems to hold when looking at productivity growth and changes in the composition of the workforce, *i.e.* countries with a change towards a larger part of the working force in the ages below 40 have lower aggregate productivity growth. The regressions of productivity on the population shares of different age-groups are based on panel data from a sample of all OECD countries covering five year intervals from 1960 to 1990.
11. Individual public consumption comprises expenditure that is – directly or indirectly attributable to and/or can be appropriated by individual users. This stands in contrast to public goods such as defence and security or the provision of street lighting.
12. More technically, the government should be able to satisfy its inter-temporal budget constraint, which means that current net debt cannot exceed the present value of future primary surpluses.
13. Real public service standards refer to real public consumption per user. Total real public consumption will change in parallel with the population if these standards are kept constant, and nominal public consumption per user will grow slower than wages because the price on public non-wage consumption increases less than wages. If nominal public service standards, *i.e.* nominal public consumption per user, follow wages, real public consumption per user would increase for the same reason.
14. Both the Economic Council and the Ministry of Finance incorporate "structural" increases in the labour force in their main scenarios that to some extent counteract the demographic developments. Also, the assumption of real growth (if any) in public service standards (either because nominal service standards follow wage increases, or explicit assumptions on real growth are made) can make for substantial differences in the outcome of sustainability indicators.
15. Changes in the discounting rate used to calculate net present values do not significantly affect the results.
16. This group is considered for cross-country analysis because the rules governing younger and older age groups are more difficult to compare across countries, not least because of differences in expectations about the importance or desirability of their being in work.
17. This includes unemployment benefits, social assistance cash benefits, educational support, sick leave, parental leave and early retirement (disability) benefits.
18. Once unemployment insurance entitlements have been exhausted, the long-term unemployed instead receive means-tested social assistance benefits.
19. These data include short-cycle higher education courses but not adult education.
20. In response to these issues, the Danish Government, along with the University and Counselling reform, has emphasised faster study completion.
21. Students in Denmark in 2002 were able to take out a state loan providing DKK 2 165 per month, compared with maximum monthly grants of DKK 2 103 for students living at home and DKK 4 231 for students living on their own. The interest rate is set by Parliament at a rate well below commercial levels, and loans must be repaid within 15 years of graduation or giving up study. In 1988 this scheme replaced an earlier system of student support provided as state guarantees for bank loans at more or less commercial interest rates. Most students at the time took out these loans

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and were heavily affected by the sharp increase in interest rates in the first half of the 1980s. The present scheme was explicitly intended to avoid the problems of severe debt build-up that had occurred for some students, with subsequent high default rates.

22. Pupils over 18 years of age in upper secondary education face no time limits on support, although they are required to attend classes and sit examinations.

23. These estimates are based on 49 353 students admitted to higher education in 2001 and 29 669 students completing higher education courses up to and including bachelor's degree (Ministry of Education, 2002).

24. Data are not available for the age group 60 to 66 years for that period.

25. For women the effect is illustrated by a slowly falling participation rate for 60 to 69 year-olds, in contrast to a marked increase in participation for 50 to 59 year-olds over the last 25 years.

26. This "lump of labour" fallacy was a relatively common motivation of policy around that time but is thoroughly discredited by economic analysis (see, for example, OECD 1994).

27. The termination of the transitional early retirement benefit scheme by 1 January 1996 was announced in the autumn of 1995, which led to a massive inflow to the scheme in the last months of 1995. This is reflected in the drop of participation rates for 50 to 59 year-old men and women from 1995 to 1996.

28. The calculation assumes that the trend decrease in age-specific participation rates for men and women during the period 1992-2001 is prolonged until 2010. The decline in participation rates in the late 1990s is to some extent due to the outflow from the transitional early retirement benefit scheme. This outflow will gradually fall and totally cease from 2007. Thus, prolonging the fall in participation rates in the last 10 years might exaggerate the underlying trend.

29. The Economic Council estimates that abolishing the VERB scheme for 60 to 61 year-olds would reduce the required increase in the ordinary state tax rate to ensure fiscal sustainability by 1.7 percentage points (Det Økonomiske Råd, 2002a).

30. The Swedish pension system has three separate components: the guaranteed pension for those on low or no income; the PAYG income pension where payments are determined by lifetime earnings (including transfer incomes) in a notional defined contribution scheme with returns based on the performance of the economy as a whole; and the premium pension, which is fully-funded, defined contribution and provided through market-based funds. The income pension component also has an automatic balance mechanism to ensure that the scheme remains financially sustainable. It would do this by spreading the burden of adjustment across society as a whole in a pre-defined manner.

31. No labour market attachment or earnings history is necessary, the only requirement is a residential one.

32. It can also be drawn by those over the age limit but still working, but only a few workers continue beyond that age.

33. This is intended to provide an economic incentive for municipalities to make greater efforts to get people on sickness benefit back to work before that point, and they must already shoulder half the costs out of their block grants after the first four weeks (see previous *Survey*).

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34. The gap is less than a day per year for state employees, more than 2 days for county employees, and around 3 days for municipality workers (Danish Employers Federation, 2002).

35. Short-term sick leave rates are marginally higher for those in the 25 to 35 years bracket than for older workers, but long-term sick leave rises significantly with age, while women tend to take more sick leave than men.

36. The measured improvement over time may also be biased upwards by those migrants from earlier cohorts who have left the country again (Edin *et al.*, 2000). More than 17 000 foreign nationals emigrated from Denmark in 2001: some 5 per cent of the stock of foreigners. Almost two-thirds were of working age, and although there were actually more foreigners from MDCs than from LDCs, those who left had a slightly lower employment rate than those who chose to stay. However, there is little further concrete information about the characteristics of those who departed or their reasons for doing so.

37. For example, the Integration Act 1999 imposed a requirement to attend a three-year introduction course (including a significant language component) on all immigrants, not just refugees, and an amendment to the Aliens Act in May 2000 tightened the rules on family reunification.

38. For example, Australia, Canada, New Zealand and the United States all operate quotas for total immigrant flows or for certain categories. In France and Japan, some immigrants are granted residence permits but are not permitted to work or face restrictions on the work they may do.

39. This assessment was in part based on a reappraisal of the extent to which the mere act of seeking asylum in another country put them in danger in Iraq, and of the consequences of refusing military service.

40. The United Kingdom has also operated a programme of voluntary assisted returns since early 1999. Most of those who have availed themselves of this facility were still working their way through the asylum process at the time. An evaluation by Deloitte and Touche found that this approach provided clear financial and human benefits, especially for those who would have been refused leave to stay at the end of the asylum process (see United Kingdom Home Office, *www.homeoffice.gov.uk/rds*).

41. Two ethnic groups stand out for such family reunification: Turks and Pakistanis, where some 80 to 90 per cent of young people marry someone from outside the country and seek to bring them to Denmark.

42. For example, the United States operates a significantly more generous policy towards family reunification, while Austria operates a quota for family reunification for those who arrived since the beginning of 1998 that is limited to the spouse and unmarried minors. Those who settled in Austria before 1998 may only bring in their spouse and children under the age of 14 years. The family member is not allowed access to the labour market until after four years residence, and even then, it is not automatically granted. In Canada, the family class includes grandparents as well as dependent children, but it does not include fiancés, who have recently been moved to the humanitarian and compassionate category to allow for more detailed scrutiny and avoid abuse of this provision.

43. This scheme provides for foreign, highly paid employees to pay 25 per cent gross income tax instead of normal (higher) rates of taxation on earned income, for a maximum of three years. The expert may choose to remain in the country for up to four more years, under normal tax rules.

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44. Contributions are required for unemployment insurance, although workers overseas can still be members of an unemployment fund. Access to cash benefits is now restricted to those who have been in the country for seven out of the previous eight years.
45. However, it should be noted that the government has also been working actively on addressing the elimination of all forms of discrimination on grounds of racial or ethnic origin. The government is working towards introducing legislation to implement the EU directive on equal treatment irrespective of race or ethnic origin, based on recommendations prepared by a committee set up in May 2001 by the then Minister of the Interior. In May 2002 the Danish Parliament adopted the Act that established the Danish Centre for International Studies and Human Rights, whose tasks include the promotion of equal treatment and which can publish independent reports. Awareness campaigns were also conducted within the context of the UN Year against Racism in 2001.
46. It not only provides financial independence, but also can provide valuable work skills and experience together with greater opportunities to mix with Danes and acquire language and customs (Husted *et al.*, 2001).
47. Although these workers are relatively poorly paid by host-country standards, for migrants the alternative is returning home to even less attractive economic prospects. A separate issue is the extent to which these immigrants have driven down wages for low-skilled native workers, although the evidence suggests this impact is over-stated. For example, Friedberg and Hunt (1995) find that a 10 per cent increase in the fraction of immigrants in the population reduces native wages by at most 1 per cent.
48. In the United States some 20 per cent of jobs pay less than two-thirds of the median hourly wage and 30 per cent pay less than three-quarters of the median wage (Economic Policy Institute website).
49. One study of black market activity in Denmark found that it was primarily being undertaken by young, skilled males, including students and involved on average less than 3 hours per week for each person engaged in such activity (Pedersen, 1998).
50. For example, 25 per cent of Pakistani and 8 per cent of Iranian and Turkish family-owned businesses are small-scale supermarkets, kiosks, etc, compared with less than 0.5 per cent for Danes (Bager and Rezaei, 2001).
51. This phenomenon can also be seen for Pakistani migrants to the United Kingdom (Clark and Drinkwater, 1999), where the self-employment rate is almost 30 per cent for those over 35 years old and almost 40 per cent for those aged 16 to 24 years.
52. Part of the difference can be explained by the average age difference, since amongst descendants there is a bulge at the younger end of the age group.
53. See, for example, a number of studies cited in H.S. Nielsen *et al.* (2002).
54. Jakobsen and Smith (2002) found that an additional five years of education for mothers (the equivalent of raising Turkish mothers' levels to those of ex-Yugoslavian mothers) would increase the probability of formal education by 10 percentage points for young women and 5 percentage points for young men. A very positive attitude to education among parents increases the probability by 18 percentage points for men and 31 percentage points for women.
55. For each year that marriage is postponed in the sample that Jakobsen and Smith (2002) examined, the probability of completing education rose by around 2 percentage points for men and 3 percentage points for women.

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56. In contrast, already in the early 1980s, some 25 per cent of Mexican American marriages in California were mixed, and this rate had risen to 50 per cent by 1997 (Duignan, 1998).
57. For example, English proficiency has been shown to have a significant impact on both employment probability and earnings in both the United Kingdom and the United States (Dustmann and Fabbri, 2000), while speaking Swedish well has been shown to increase chances of getting a job in Sweden (National Board of Health and Welfare, 1999). It is worth noting that for the United Kingdom it was found that verbal skills boosted employment prospects only slightly, while writing abilities make about three times as much difference. Language proficiency makes a difference to earnings as well, although the effect seems to be stronger in the United Kingdom than in the United States, Canada or Australia, while in Germany the difference is slight.
58. In contrast, a study for the United Kingdom, where language skills were also assessed by the interviewer, 78 per cent of immigrants arriving before the age of 10 years old, and 24 per cent of those arriving after the age of 24, were fluent. Among different minority groups, Pakistanis had the weakest language performance (Shields and Wheatley Price, 1999).
59. Bilingual education involves providing some proportion of the instruction in subjects other than Danish language in the student's home language.
60. In the United States, the question has become highly politicised and closely intertwined with the objective of preserving ethnic cultures as well as an issue for educational attainment (Duignan, 1998).
61. The rationale for home-language teaching has also varied between countries and over time, including: facilitating the return home of guest workers' children; recognition of separate cultural identity; building minority children's self-esteem and sense of security; and cognitive benefits of bilingualism.
62. The rest of this housing is allocated by the housing association itself.
63. This would be the result if a shift to market-based rents removed the incentive for people to hang on to good quality, larger flats, on which they pay cheap rents. This would boost turnover, and a greater supply of available accommodation would reduce market prices.
64. Average hourly wages for refugees in 1995 were 18 per cent lower than for Danes and for immigrants from Africa, Pakistan and Turkey, earnings were still 14 to 15 per cent lower than for Danes (Husted *et al.*, 2001).
65. For refugees, early and firm attachment to the labour market is a critical determinant of future wages; although their wage levels are much lower at the outset, they climb more steeply than Danish wages if firm attachment is obtained within 10 years of arrival. But because such attachment is difficult for refugees, average wages remain low. Indeed, the probability of a male refugee finding employment starts to fall after 10 years in Denmark and more sharply so after 20 years, although this may also be an age effect.
66. In addition to the tax freeze, a nominal ceiling on increases in land taxes (taxes levied on land, whereas property value tax is levied on owner-occupied houses) has been introduced. Previously, land tax payments to municipalities and counties would follow adjustments of the land value, given unchanged tax rates. By imposing a ceiling on the increase in the land value used when calculating the tax payment, landowners are protected against rises in land tax payments that substantially exceed their income growth. However, it also means that a fall in land values will not reduce land tax

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payments, if the ceiling has previously been binding. Therefore, the scheme provides a smoothing of land tax payments, but it does not erode the tax base, as is the case for the nominal ceiling on the property value tax.

67. In principle, the tax freeze is binding only for the current government, which – if re-elected in 2005 (at the latest) – will have to agree on a new programme. The tax freeze is implemented until 2010 in the government's medium-term survey and in the fiscal sustainability calculations. The tax freeze in itself lowers tax revenue by ¾ per cent of GDP (at the 2002 level) by 2010. It implies a permanent net revenue loss of 0.6 per cent of GDP when converted into a fixed annual amount (net present value) in 2003, compared with a scenario where excise taxes, etc., are indexed to prices.

68. The freeze does not imply that the overall tax burden will always decrease from one year to the next because increases in tax bases could exceed nominal GDP growth.

69. In theory, changes in consumption taxes (including excise duties) and labour income taxes affect the optimal labour supply decision for an average individual in a fairly similar manner. However, lower consumption taxes benefit everybody and will thus imply a certain deadweight loss. Also, income tax is progressive and the marginal excess burden rises sharply as the marginal tax rate increases. If the same reduction of tax revenue was applied to labour income, the labour supply effects would most definitely be larger. Also, it could be argued that individuals are more likely to react to clearly visible increases in their pay check than to lower increases in prices (due to the constant nominal excise taxes) on a number of goods, which to the broad public are not obviously identified. Joumard (2001) concludes that EU countries in general would gain by shifting from (relatively high) labour taxes to taxes on consumption and/or property.

70. Such manipulation can be constrained by strict application of Generally Accepted Accounting Practices (GAAP) for the public sector, along with a strong, independent auditor, such as operates in New Zealand.

71. See OECD (2002f) for a thorough description of the relations between central and local governments.

72. The bill giving the Minister for the Interior and Health legal authority to enforce sanctions on counties was passed by Parliament in February 2003. The government has announced that a similar bill concerning municipalities will be presented if a breach of the tax freeze occurs at the municipality level. The sanctions are therefore described as applying at both the county and the municipality level.

73. The sanctions scheme was not subject to formal game-theoretic testing (which could be very complicated) before its introduction.

74. See OECD (2002f) for a detailed description of the income tax system in Denmark.

75. The Economic Council has found that increases in the top state tax threshold provide most "value-for-money": their self-financing degree is somewhat higher than that found by the Ministry of Finance (Det Økonomiske Råd, 2001). The quantitative differences can be attributed to differences in methods and data, the most crucial difference probably being the financing rule and the applied income distribution (see Finansministeriet, 2002c).

76. By the period 2008 to 2012, the government projects emissions of 80 million tonnes of carbon dioxide equivalent, against a burden-sharing target of 55 million tonnes. Of the projected total, 10 million tonnes is the result of growth in electricity exports and 5 million tonnes represents a rebound from the abnormally low level of emissions in 1990.

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77. The availability of hydroelectricity in the Nordic countries was particularly high in 1990, boosting Danish imports of electricity and reducing domestic emissions in that year.
78. For the wind industry, previous subsidies have resulted in the industry being able to make significant technological progress that has lowered generating costs.
79. A cross-ministerial working group reached a different conclusion in 2001. The government has launched a new working group to reconsider the welfare benefits of investments in renewable energy and the benefits of other environmental instruments.
80. Together these requirements were known as the "Public Service Obligation" (see previous *Survey*).
81. High efficiency cars are those that consume less than 4 litres of fuel per 100 kilometres.
82. Cordon pricing requires payment for entry into certain zones that are, on average, highly congested. Congestion (or road) pricing takes into account where and when a car is being used and the extent of congestion at that point.
83. In 2001, there were 14 trades involving 0.5 million tonnes of CO_2. In 2002, permit trading was also weak, but one electricity company swapped Danish permits for UK permits, as the latter can be rolled forward to years beyond 2003. The market price for these permits was EUR 14.4 per tonne of carbon in 2001 and fell to EUR 10.4 in 2002 (Natsource, 2002).
84. The tax rates were DKK 11 per kg of biochemical oxygen demand equivalent (EUR 1.48), DKK 20 per kilogram of nitrogen (EUR 2.69), and DKK 110 per kilo of phosphorus (EUR 14.78).
85. Such an instrument places a tax on the nitrogen that does not leave agriculture in the food chain, staying either as excess nitrate in the soil or animal manure. The nitrogen content of each type of output is constant so that the appropriate rebate can easily be established.
86. Almost half of all waste is generated by power plants, sewage plants and construction. Such waste materials are reused on a commercial basis, though the waste tax and regulations introduced in 1995 have reduced the proportion of such waste going to landfill from 15 per cent in 1994 to 8 per cent in 2000.
87. Life-cycle analysis attempts to calculate the extent of pollution embodied in a given product, adding the pollution caused by the production process, the delivery process and other environmental impacts. One problem with this analysis is that a number of environmental externalities are already internalised in the market price of a product, either through regulations or taxation. With the advent of greenhouse gas trading in the EU, a further externality will be internalised, lessening the need for life-cycle analysis.

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Annex I
Macroeconomic situation and prospects

This annex briefly reviews the current macroeconomic situation and short-term prospects.[1] The pace of activity has slowed in the face of weak export growth, although domestic demand remains firm (Table A1). Growth prospects are expected to brighten as the international situation improves and firms regain sufficient confidence to increase investment and hiring. But unemployment is still lower than its structural rate, and labour shortages, accompanied by accelerating wages, could re-emerge as the expansion quickens.

Domestic activity remains solid, but exports have faltered

The Danish economy continued to deliver creditable GDP growth of just over 1½ per cent in 2002, despite the international slowdown. Danish exporters continued to gain market share, especially in the first half of 2002, although exports stalled in the second half. Private consumption continued its rebound, as household disposable incomes have recovered from the dampening effect of the Whitsun package tax increases from 1999 to 2001, and the savings ratio has returned to a comfortable 6 per cent. Car sales have been strong, but the pick-up in household spending has also been more broad-based (see Figure A1).

Business indicators suggest that production may remain sluggish in the first half of this year, as new orders have weakened again and confidence remains weak: this may lead to a postponement of some business investment plans until prospects brighten. Nevertheless, additions to capacity during 2000 and 2001 along with shrinking private sector employment over the course of 2002 have boosted labour productivity (Figure A2). Job creation has continued in the public sector, albeit at a reduced annual pace of around ½ per cent. This has attenuated the impact of cyclical weakness on the standardised unemployment rate, which stood at 4¾ per cent in the first quarter of this year, lower than the OECD estimate of its sustainable level. Compensation per employee has decelerated a little, but overall the labour market remains relatively tight, while output remains only slightly below estimated potential.

Monetary conditions will support growth

Monetary policy continues to mirror developments in the euro area, as the National Bank keeps the *kroner* within narrow bands around its central parity *vis-à-vis* the euro. Official interest rates have gradually been reduced from 5.4 per cent at the beginning of 2001 to 2.7 per cent in early March (Figure A3). However, the impact of this easing of monetary policy on economic activity has been attenuated by a rising effective exchange rate since around the middle of last year. In any case, monetary conditions in Denmark are expected to remain relatively easy over the next 18 months or so, since its interest rates are largely tied to the decisions of the European Central Bank, which will in turn be based on inflationary pressures and the amount of spare capacity in the euro area as a whole.

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Table A1. **Summary of projections**

	1999		2000	2001	2002	2003	2004
	DKK billion, current prices	Per cent of GDP	Percentage changes (1995 prices)				
Demand and output							
Private consumption	599.5	49.6	–1.9	0.4	2.1	1.9	2.2
Government consumption	312.1	25.8	1.1	2.1	0.9	1.0	0.9
Gross fixed investment	240.9	19.9	9.1	1.3	1.3	1.1	3.2
Final domestic demand	1 152.6	95.4	1.3	1.0	1.6	1.5	2.1
Stockbuilding[1]	–2.6	–0.2	0.6	–0.3	–0.4	0.2	0.0
Total domestic demand	1 149.9	95.2	2.0	0.8	1.2	1.7	2.1
Exports of goods and services	459.6	38.1	13.1	3.2	3.4	3.1	7.5
Imports of goods and services	401.8	33.3	11.9	1.8	2.5	3.5	7.1
Net exports[1]	57.8	4.8	0.9	0.7	0.5	0.0	0.6
GDP	–	–	2.8	1.4	1.6	1.6	2.6
GDP at current prices	1 207.7	100.0	5.9	3.5	2.7	4.1	5.3
Price, costs and employment							
GDP deflator	–	–	3.1	2.0	1.1	2.4	2.6
Consumer price index	–	–	2.9	2.4	2.4	2.4	2.3
Private compensation per employee	–	–	4.1	4.3	4.1	3.9	4.2
Total employment	–	–	0.3	0.2	0.1	0.0	0.5
Unemployment rate	–	–	4.4	4.3	4.5	4.7	4.4
Financial variables							
Short-term interest rate	–	–	4.9	4.6	3.5	2.5	2.5
Long-term interest rate	–	–	5.7	5.1	5.1	4.4	4.5
General government balance, per cent of GDP[2]	–	–	2.5	2.8	1.8	1.6	1.9
External transactions							
Export volumes (all goods)	–	–	10.8	2.2	6.1	3.8	7.9
Import volumes (all goods)	–	–	6.4	1.2	5.9	4.5	7.9
Trade balance ($ billion)	–	–	6.8	7.5	8.4	10.0	11.6
Current account balance ($ billion)	–	–	2.4	4.9	5.0	5.9	7.8
Memorandum items:							
Output gap[3]	–	–	1.2	0.2	–0.4	–0.8	–0.3
Household saving ratio	–	–	4.8	6.0	6.2	6.0	6.1

1. Contributions to changes in real GDP (percentage of real GDP in the previous period), actual amount in the first column.
2. Net lending as a per cent of nominal GDP.
3. Per cent of potential GDP.

Source: OECD.

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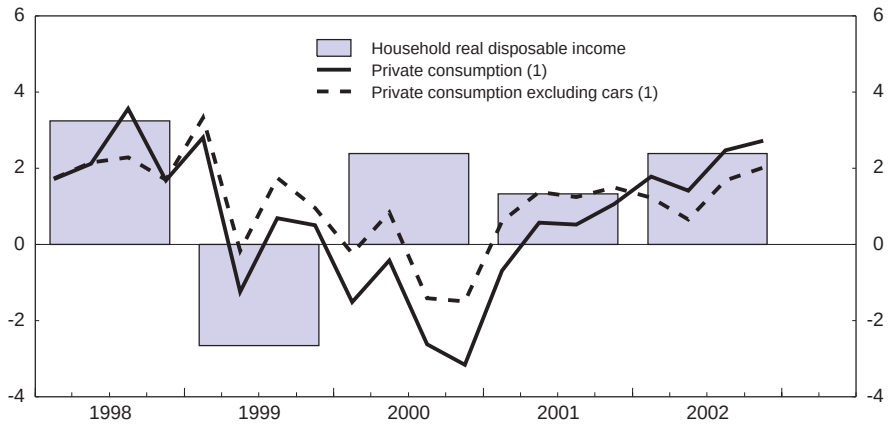
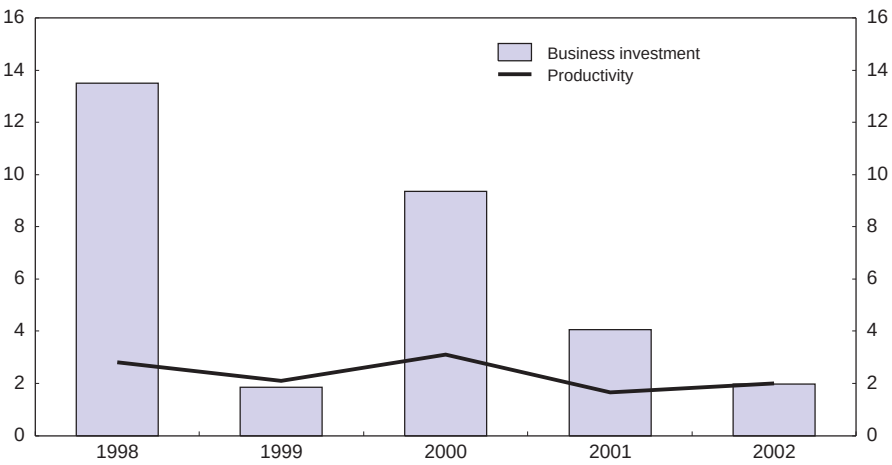
Figure A1. **Private consumption and household disposable income**
Annual per cent change



1. 4-quarter moving average.
Source: Statistics Denmark; OECD.

Figure A2. **Business sector investment and productivity**
Annual per cent change



Source: Statistics Denmark; OECD.

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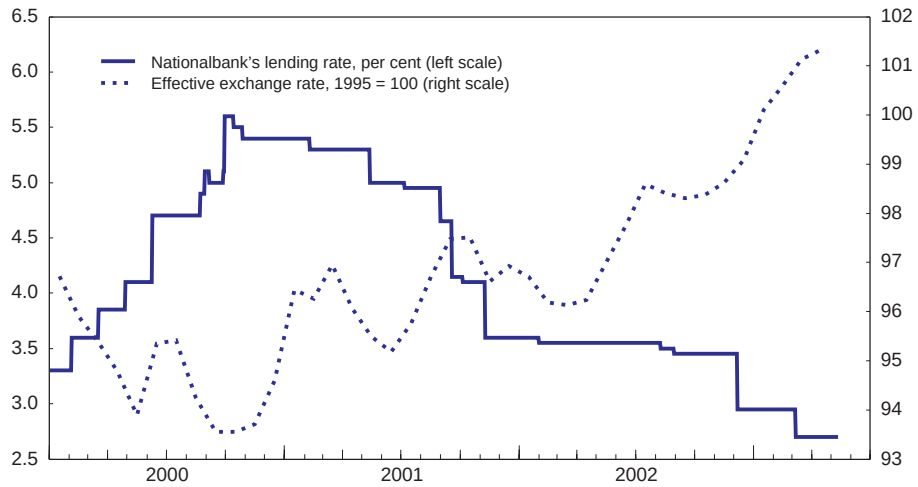
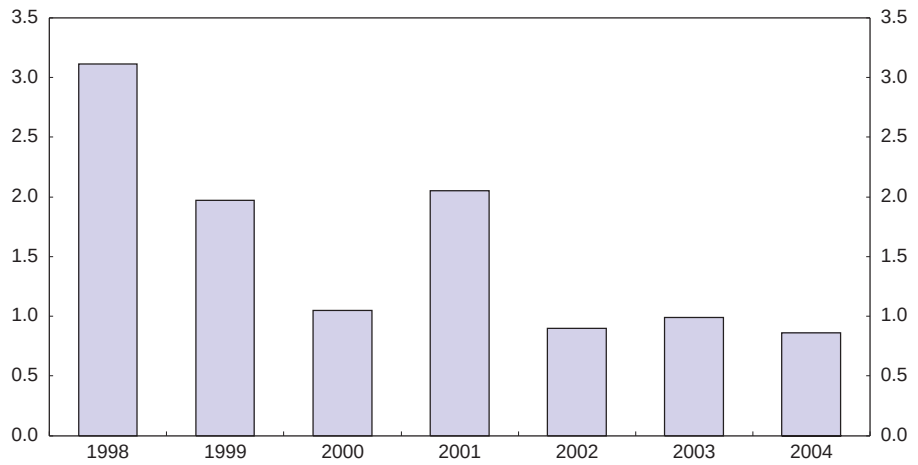
Figure A3. **Official interest rate and effective exchange rate**



Source: Denmark National Bank; OECD.

Figure A4. **Real growth in public consumption**
Annual per cent change



Source: Statistics Denmark; OECD.

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Tax cuts could add further stimulus

The overall general government budget surplus is expected to be only around 1½ per cent of GDP this year, a slight reduction compared with 2002 that is largely the result of weak economic activity.[2] Fiscal policy is expected to be broadly neutral this year, but next year it will add stimulus to activity unless the government succeeds in restraining public expenditure growth sufficiently to offset the announced tax cuts. Its stated objective is to constrain real public consumption growth to 0.7 per cent next year, which would require a significant break in trend (Figure A4). Tax cuts are planned out to 2007 and these could further erode the surplus over time, unless offsetting modifications are made to public spending or the "tax freeze" is adjusted to allow other taxes to be raised instead (see Chapter IV).

The pick up in activity could lead to supply pressures

The rate of expansion is expected to be modest in the first half of this year, as consumers and businesses postpone some spending, but then to quicken from the second half to reach around 2½ per cent in 2004 (Figure A5). Export growth is projected to rebound as global demand accelerates, while stimulus from monetary conditions and tax cuts should underpin private domestic demand. Employment is likely to expand again next year by around ½ per cent, with a corresponding fall in the unemployment rate. With real GDP increasing faster than its potential rate next year, the output gap is likely to almost close in 2004, and labour shortages could start to re-emerge, which might lead to some acceleration in wages and prices. However, the evolution of foreign demand will be a key determinant of Denmark's short-term prospects and this is the main area of uncertainty in the OECD's projections.

Notes

1. These projections correspond to those published in the OECD *Economic Outlook* 73 (OECD, 2003a).
2. It should be noted, however, that some technical revisions in the national accounts have reduced the government surplus in both 2002 and 2003 by around ¼ per cent of GDP.

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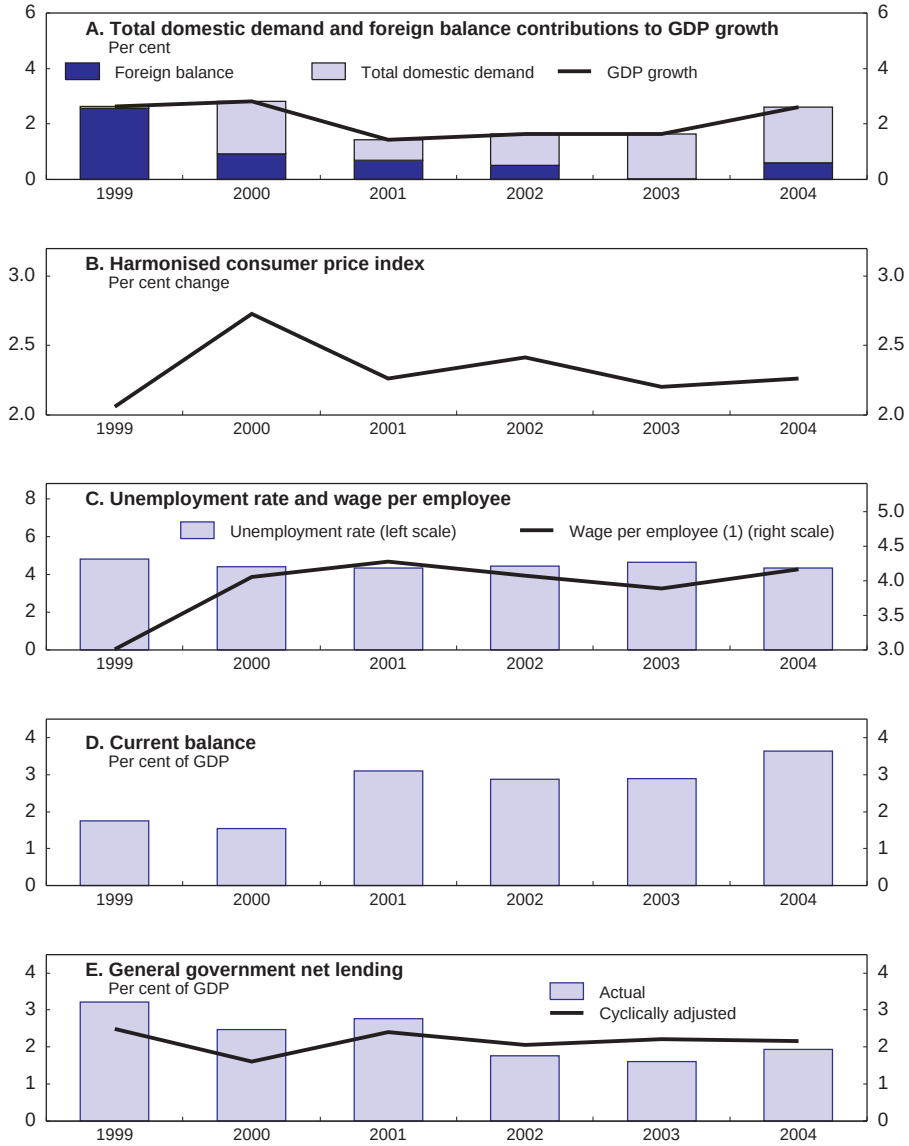
Figure A5. **Macroeconomic performance and prospects**



Note: 2003 and 2004 are OECD projections.
1. Annual per cent change.
Source: OECD.

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Annex II
Demographic projection assumptions

To a very large extent, demographic projections are determined by the age structure of those already alive and living in the country. Nevertheless, they are subject to significant uncertainties arising from assumptions about fertility rates, life expectancy and net migration. Although fertility patterns and life expectancy are only very indirectly under any sort of policy influence, immigration patterns are more amenable to deliberate choices. But immigration patterns can be to some extent unpredictable, and recent fluctuations suggest that some caution is warranted when evaluating the implications of the specific assumptions of the size and composition of net immigration. The assumed net migration of around 10 000 people per year in the DREAM population projection (DREAM, 2002) would ensure a total increase of population of almost 10 per cent between now and 2100. However, the shares of immigrants, descendants and the remaining population would change substantially: immigrants from less developed countries (LDCs) and their descendants, in particular, will increase both in numbers and as part of the total population. Thus, the proportion of immigrants from LDCs will almost quadruple, while the number of descendants relative to total population will be almost nine times higher, partly reflecting a relatively low starting point. Immigrants from more developed countries (MDCs) will also increase both absolutely and relatively, but not to the same extent.

The DREAM population projection distinguishes between nine groups depending on civil status:

- Immigrants from more/less developed countries with/without Danish citizenship.
- Descendants from more/less developed countries with/without Danish citizenship.
- Remaining population.

The *remaining population* consists of residents with at least one parent who is a Danish citizen and born in Denmark. Thus, it is irrelevant whether the person herself is a Danish citizen or born in Denmark. If a person does not belong to the *remaining population*, (s)he is either an *immigrant*, if (s)he is born abroad, or a *descendant*, if (s)he was born in Denmark.

More developed countries are: United States, Canada, Japan, Australia, New Zealand and all European countries, except Turkey, Cyprus, Azerbaijan, Uzbekistan, Kazakhstan, Turkmenistan, Kyrgyzstan, Tajikistan, Georgia and Armenia. *Less developed countries* comprise all countries that do not belong to the group of MDCs.

The main differences between groups are in fertility rates. Some convergence is assumed; while the fertility rate for immigrants from LDCs without Danish citizenship was more than double that of the remaining population in 2001, it is assumed to be only 40 per cent higher in 2050. The average fertility rate is assumed to stay around 1.7 until 2025, after which it gradually increases to just above 1.9 in 2065 and remains at that level throughout the projection period.

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The expected lifetime for men and women is assumed to increase by 5½ and 2 years, respectively, until 2100, which is less than in UN projections for other EU countries, in particular for women. It is a result of the applied forecasting model extending a relatively small increase in expected lifetime for women from 1981 to 2001.

Long-term net immigration is assumed to be around 10 000 people per year, which is in line with the population projection from the United Nations. The projection does not include any effects on immigration from the recent changes to the Aliens Act (family reunification, etc., see Chapter III), but immigration from LDCs is adjusted downwards compared with the historical average due to the drop in asylum seekers in 2002.

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Annex III

Basic mechanisms and assumptions in the projection of public finances

In addition to the effects from the change in the composition of the population, the projection of public finances presented in Table 3 is shaped by a number of mechanisms and assumptions:

- Public transfer rates and wages in the public sector are indexed to wages in the private sector in conformity with present legislation and wage agreements. The annual adjustment of public transfer rates and thresholds in the tax system ("satsreguleringen") is based on the increase in wages according to the Danish Employers Federation's structural wage statistics, which cover only members of that association. When calculating the adjustment, increases in pension contributions are excluded from the increase in salaries, thereby ensuring an adjustment that more closely resembles the increase in wage earners' disposable wage income. The wage agreements for public employees include adjustments of public-sector wage increases to the increases in the private sector ("reguleringsordningen"). The statistical basis for the adjustment is the quarterly summary wage statistics for the private sector published by Statistics Denmark. These adjustment mechanisms imply that public finances are broadly unaffected by the level of price inflation and productivity increases, as these are ultimately fully reflected in the wage increases in the private sector. Thus, while higher productivity growth would boost overall output and thereby living standards, it would not ease the long-term pressures on public finances.

- Collective public consumption (administration, defence, justice, etc.) is assumed to follow the evolution of the overall population.

- The government's medium-term objectives, as set out in its medium-term survey to 2010 (Finansministeriet, 2002a), are assumed to be achieved. These include: *i*) a modest increase in real public consumption; *ii*) a tax freeze until 2010 (see Chapter IV); *iii*) tax cuts implemented from 2004; and *iv*) a substantial increase in employment by 2010.

- After 2010, nominal public consumption per user follows the evolution of wages. That implies some growth in real public service standards because the price of public non-wage consumption increases less than wages.

- After 2010, tax revenue follows nominal GDP except for changes in tax bases relative to GDP. Maturing of labour market pension schemes will imply a significant increase in the base for taxation of pension payments. On the other hand, revenue from oil and gas extraction in the North Sea is assumed to decrease and finally cease in 2020 as the resources deplete and the costs from further operations become prohibitively large.

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Annex IV

Progress on structural reforms

This annex provides a compendium of recommendations carried forward from the previous *Economic Survey*, and records what action has been taken, along with any other major policy changes announced since the previous *Survey*.

Reducing structural unemployment

Previous recommendations

- Reduce benefit generosity: lower the highest statutory replacement rate and allow benefits to gradually decline with the length of the unemployment spell, for example, by implementing the so-called annual income model for benefit calculation.
- Extend the waiting period for benefits.
- Reduce UI benefit duration and abolish benefits for non-core purposes such as holidays.
- Improve the efficiency of active labour market policies:
  - Closely monitor that greater flexibility does not in practice weaken the testing of individuals' availability;
  - Extend the early activation requirement for the unemployed under 25 years to those aged 25 to 29 years;
  - Introduce competition to the public employment service for placement services and promote a more competitive environment for providing educational activation activities.

Action taken

- "More People in Work" package contained measures that improve the efficiency of ALMPs and that increase economic incentives to work for a sub-group of those receiving social assistance (see Annex V).

Increasing labour supply

Previous recommendations

- Reduce scope for early withdrawal from the labour force: continue to carefully monitor the inflow to the voluntary early retirement benefit scheme and the labour market status of those aged 60 to 64 years. Take further steps to strengthen incentives to postpone retirement if the effects of the reform fall short of expectations, or if labour market pressures intensify more than expected.

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- Promote a more inclusive labour market: strengthen efforts to remove barriers for establishing jobs on special terms for those with reduced working capacity, while ensuring that stringent eligibility assessments for the flexi-jobs scheme are maintained.
- Abolish the publicly financed educational leave scheme and make financing such leave a matter for the social partners.
- Introduce a waiting period for the sickness benefit scheme and enforce the current 12-month time limit.
- Reduce excessive time spent in the education system:
  - Consider introducing tuition fees for tertiary education.
  - Reduce the period for which students can receive educational grants, and set stricter time limits for completion of higher education.
  - Consider limiting the voluntary tenth grade of lower secondary education to those with obvious needs.
- Improve immigrant integration into the labour market.
  - Monitor closely the municipalities' implementation of integration policies.
  - Consider introducing a temporary in-work benefit targeted on immigrant groups facing the weakest economic incentives to seek work.

Action taken

- In May 2002, the social partners, municipalities and the government signed a new agreement, "The Integration Package for Immigrants". This focuses on rapid introduction to working life and acquisition of Danish language skills through the workplace itself. It is motivated by the belief that the workplace is the best environment for immigrants to learn the skills they need, including Danish language, for successful labour market participation. It also reflects a realisation that the workplace provides excellent opportunities for closer contact between immigrants and Danes. The necessary legislation was introduced to Parliament in March 2003 (see Chapter III).
- Income tax cuts are to be implemented gradually from 2004 to 2007. The cuts aim at increasing incentives both to join the labour force and increase hours worked (see "Action taken" under tax policies).
- "More People in Work" contained measures increasing incentives for labour force participation.
- Maternity leave has been extended from 28 weeks to 52 weeks of which 32 weeks can be freely divided between the parents. The parental paid leave scheme was abolished at the same time.

Distribution policies

Previous recommendations

- State equity objectives more clearly. Welfare gains from structural reforms should be allowed, more often, to take precedence over keeping static measures of inequality at current low levels. This is particularly relevant where reforms might provide most benefit to high-income groups but with little, or no, effect on low-income groups.

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Action taken

- The design of the announced cuts in income taxes indicates that equity concerns have prevented providing rate cuts to high-income groups where the gain in labour supply would probably be higher.

Tax policies

Previous recommendations

- Reduce marginal personal income tax rates. Increase the income threshold for paying the top tax rate. Consider the scope for further broadening of the tax base to allow rate cuts. For example, the tax-value of deductible interest outlays remains high by international standards.
- Remove asymmetry in capital income taxation. Align the tax treatment of positive net capital income with that of interest outlays. Further to this, reconsider the need for having a separate tax scheme for share income (dividends and capital gains) and in any case simplify the taxation of share income.
- Reduce the corporate tax rate. The corporate tax rate is well aligned with neighbouring countries. However, there may still be some leeway for reducing tax expenditures further, thereby making additional modest rate cuts possible.
- Safeguard effective taxation of homeownership. As a minimum, resist pressures for reducing or eroding property taxation as would be the result of, for example, nominally freezing tax assessment values. Introduce a property tax on rental housing.

Action taken

- The government has parliamentary support for a package of income tax cuts to be implemented between 2004 and 2007 that involves a rise in the threshold for intermediate tax, but do not shift the threshold for the top state tax. This means that all those earning incomes above the new threshold for intermediate taxes will receive higher after-tax incomes but face the same marginal tax rates as before. The package also provides an earned income tax credit. This will raise incentives for those at the bottom end of the income scale to find work, but as every worker gets this rebate, it has high dead-weight costs.
- The government's tax freeze erodes property taxation by nominally freezing the tax assessment values. Also, revenue from nominal excise duties will fall relative to GDP as they are not adjusted in parallel with inflation.
- The Special Pension scheme has been rearranged so that redistribution within the scheme no longer takes place and savings are purely individual. As a result, the contributions to the scheme are no longer treated as a tax in national accounts but as private savings (from 2002), and payments are likewise no longer a part of public sector expenditures.

Housing policies

Previous recommendations

- Extend property rights in social housing.

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- Remove the rent controls applying to private rental housing, while taxing away any induced economic rent to landlords resulting from higher property prices. Rent regulation in social housing should also be modified to eliminate unwarranted rent differentials within and across vintages.
- Clarify governance structures in the social housing sector. A strategy to deal with the expected wealth accumulation in the sector should be developed in order to avoid lock-in effects of capital and over-investment in housing. Increasing property rights might be one element.
- Improve incentives for cost reductions in urban renewal. Reconsider the current rules so as to avoid the perverse incentives in the present system. As a first step, implement policies that would increase cost awareness.
- Reconsider the housing benefit scheme. Benefits available to pensioners should be aligned with those for other groups.

Action taken

- A package of housing initiatives has been implemented. Pension funds and private investors can get a tax credit if investing in construction of new rental housing. The rent of new rooftop apartments can be freely set. At the same time, subsidies to urban renewal were reduced and the subsidies to private housing co-operatives will be phased out. Furthermore, funds accumulated in the social housing sector are made available for renovation and improvement of existing dwellings and they will now also cover some parts of the social housing sector's funding that was previously provided by the central government.
- The government is preparing an urban renewal reform and legislation allowing tenants in social housing to buy their apartments.

Education policies

Previous recommendations

- Secure basic skills, in particular at compulsory level. Consider making the introductory school year prior to primary school (børnehaveklasse) compulsory and strengthen its educational content.
- Ensure an increase in teachers' classroom hours towards the OECD average to reverse the trend increase in unit costs. In addition, make sure that resources are gradually shifted from the area of pre-school childcare to primary and lower secondary education in step with the entry into school age of the big cohorts born in the first half of the 1990s.
- Strengthen quality and cost-effectiveness in adult education and training. Introduce sizeable user charges on adult education and training for the employed, thereby shifting the main financing responsibility to the employers. Further narrow the range of courses receiving subsidies, in particular by eliminating those with a firm-specific content.

Action taken

- Adult education grants to vocational courses have been abolished.
- User charges on adult education have been harmonised by implementing a general floor and ceiling for charges.

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Competition policies

Previous recommendations

- The potential fines for violating the Competition Act remain modest, and an increase seems warranted.
- Strengthen competition in electricity generation. Remove the Public Service Obligation (PSO) and allow consumers to purchase electricity from the cheapest supplier. Convert the fixed tariff commitments given to wind-turbine owners into a diminishing fixed-term government subsidy.
- Further increase vertical separation in energy distribution sector. Further tighten the rules to reduce the extent to which municipalities can in practice continue to own production, network and distribution facilities and remove the current financial disincentives for divestment.
- Liberalise the market for all gas consumers as rapidly as possible, and expose DONG to further competition.
- Utilise competition to improve resource use in road and rail transport more consistently. In particular, long-distance bus services should be further deregulated. Develop a comprehensive strategy for increasing competition between different modes of transport that allows effective competition on all routes and involves converting current protection granted to rail into subsidies paid for the provision of social services, where these are deemed necessary.
- Verify whether competition in the telecommunications sector has strengthened further and access to the network become cheaper. Ensure that implementation of long run average incremental costs (LRAIC) model for interconnection tariffs and full number portability take place.

Action taken

- A competition package was introduced in January 2002 to simplify regulation and ease firms' administrative burdens. The package includes a pilot scheme on tax credits (within a total of DKK 25 million) for business R&D investments for the years 2002 and 2003. While positive externalities could provide a *raison d'être* for such measures, the costs and benefits should be scrutinised when considering a continuation and/or extension of the scheme as – in addition to administrative costs – it might involve dead-weight losses, reclassification of expenditure and small country leakage.
- The competition legislation has been strengthened, including a rise in the penalty level that brings it closer to the level in other EU countries.
- Direct subsidies to businesses have been reduced.
- Measures designed to achieve liberalisation of electricity and gas markets have been introduced. In a bill presented in February 2003, the framework for municipalities' use of proceeds from sales of their holdings in power utilities is laid down. It involves a smaller reduction of general government grants to municipalities if selling (fully or partly) their shares in power utilities. The incentive for municipalities to sell is thus increased.
- Measures to increase transparency for consumers on the financial markets have been introduced. In two bills presented in March 2003, regulations on these markets are simplified and modernised.

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- Granting of rights to operate long-distance bus routes is no longer based on evaluations of the need for establishing such routes, but requirements for maintaining routes for a minimum duration have been implemented.
- A number of railway lines have been contracted out from January 2003 (15 per cent of the passenger traffic).
- The LRAIC model for interconnection tariffs has been implemented.

Environment policies

Previous recommendations

- Use the environmental component of current water charges to equalise abatement costs, removing the major rebates given to industrial users. Shift to a tax on net nitrogen discharges from farms or establish a tradable permits scheme.
- Extend the cap-and-trade system to cover all carbon emissions with the aim of encouraging reductions to be made where abatement costs are lowest across all sectors.
- Re-examine the taxation of car transport in conjunction with road pricing.
- Abolish national targets, which are based on a rigid interpretation of the waste hierarchy, and instead allow economic incentives to achieve the optimal mix of recycling, incineration and landfill.

Action taken

- The mix of policies to favour renewably energy has been rationalised.
- A new climate strategy has been introduced that relies more heavily on the use of flexible market-based instruments for carbon emissions.

Budget processes

Previous recommendations

- An explicit target for total central government expenditure should be introduced. While determining its precise demarcation would warrant further analysis, to be effective it would need to be set in nominal terms and include grants to local governments. Achieving political agreement on the overall spending level prior to the budget negotiations with the local governments would strengthen top-down control.
- Improve the consistency with medium-term projections and use forecasting techniques to provide best possible projections of future expenditure.
- Evaluation of performance and productivity developments in individual spending areas should be linked closer to the budget process itself, possibly in tandem with a wider use of sunset clauses. In order to safeguard legitimacy, consideration should be given to concentrate more responsibility for evaluations with an independent body, for example by broadening the mandate of the National Audit Office.

Action taken

- In conjunction with the government's fiscal surplus target, the tax freeze will act as an indirect rule to control expenditures.

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– Accrual accounting has been introduced on a pilot basis in a number of institutions in 2003. The aim is to have implemented accrual accounting in all parts of the central government by 2005 and accrual budgeting in 2007.

Local governments

Previous recommendations

– Local governments' entitlement to set income-tax rates should be restricted, either through a centrally determined tax freeze or by imposing financial penalties on tax increases. To reconcile this with local-government autonomy, counties and municipalities could be allowed more discretion in setting property tax rates, the basic tax allowance and user charges.
– In policy areas where the central government wants to play a major role in setting service standards, it should take back the financing responsibility accordingly. On the other hand, where the central government does not want to reduce local governments' taxing powers, it should abstain from detailed regulation.
– Efforts to develop instruments encouraging inter-municipal co-operation, voluntary mergers and enhancing the role of market signals should be strengthened.

Action taken

– The tax freeze has been implemented in the agreements with local governments' associations.
– A sanction mechanism has been implemented. It comes into force if average tax rates are raised at either the county or the municipality level and involves a reduction of central government grants to local governments. It is designed to encourage co-operation within each local government level.
– In the central government budget for 2002 an amount of DKK 1½ billion was set aside for counties documenting that increased activity had reduced waiting lists for certain hospital treatments.
– Accrual accounting will be gradually introduced in local governments from 2004.

Strengthen competition and market signals for public services

Previous recommendations

– Encouraging competitive framework conditions is warranted for a wider range of services. Enforcement of existing provisions regarding central government agencies' contracting out should be strengthened, and local governments should have more freedom and stronger incentives to establish competitive production structures. More effort should be devoted to identifying and removing barriers to outsourcing.
– Develop a coherent strategy for user charges. User charges should in general be designed so as to optimise the demand response. While higher or more well-designed charges seem appropriate in areas such as home help, childcare and adult and tertiary education, reductions could be considered elsewhere. Local governments should be allowed more discretion in setting charges.
– Strengthen client choice to exploit synergies. Clients' choice may often in itself raise the perceived quality of services. In addition, synergy effects may arise, as allowing

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clients to determine the prevailing providers could also foster greater competition. However, for services with no imminent level of satiation, a necessary balance needs to be struck between clients' choice and expenditure control. More client choice would often need to be accompanied by user charges.

Action taken

- Client choice has been expanded in the health and care sector. Citizens are now entitled to treatment in a private and/or foreign clinic paid by the public sector if the public health care system is unable to provide treatment within two months. Legal barriers to the introduction and extended use of private providers in the public health and care sector have been removed.
- The tax freeze and the government's fee and user charge policy restricts the application of user charges, thus probably impeding a general rearrangement of the current structure.
- Legislation has been implemented granting private companies the right to offer provision of local and central government services ("Udfordringsret").

Administrative efficiency

Previous recommendations

- Broaden and deepen the use of performance-improving instruments, particularly at lower levels of government.
- Continue developing the strategy for e-government and ensure that savings are collected for centrally determined re-allocation.

Action taken

- All formal requirements in legislation preventing digital communication are to be removed by summer 2003.
- An "Infostructurebase" allowing for communication between government databases has been launched, and a joint IT structure for all government levels is under development. The central and local governments have agreed that all authorities have a right to communicate electronically across all levels of government from 1 September 2003 – the so-called eDay.
- The government has offered all Danes a free digital signature to be used for Internet services and secure communication from March 2003. An e-Portal for all government services to businesses is launched in May 2003. The portal provides a single entry point for information and services needed by businesses in order to perform their administrative tasks.
- Preparation of efficiency-improving strategies has been made mandatory in central and local governments. These strategies are to include policies for the use of outsourcing.

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Annex V

More People in Work

The government unveiled a package of measures in the autumn of 2002, which were passed by parliament in 2003. This annex provides details of the changes in different areas.

Unification of systems

Previous rules

– UI recipients were managed by the Public Employment Service (PES) while cash benefit recipients were under the municipalities. Referral differed between the PES and different municipalities. Different registration systems were used in municipalities and the PES.

More People in Work

– Harmonised rules for both UI and cash benefit recipients and all unemployed persons will be assessed according to the same principles (common method of referral). Common registration in digital database will provide better data access and avoid double registration; unemployed persons and firms have to submit information only once.

Faster and more direct process towards employment

Previous rules

– All unemployed UI recipients should have a plan of action set up before 1 year of unemployment. Youth (younger than 25 years) should have a plan of action set up before 6 months of unemployment. All cash benefit recipients with problems in addition to unemployment should have a written plan set up. Persons with unemployment as the only problem have the right to have a plan set up.

– Contact with UI recipients was to be scheduled after the first and the 6th month and when offered an activation programme. Contact with cash benefit recipients was to be scheduled for every 3 months at a minimum. Enrolment and registration of qualifications should take place by personal appearance at respectively the PES and the local government office.

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- All unemployed persons should have a plan of action set up no later than when receiving the first offer of activation. Personal contact (appearance) should be at least once every three months during the entire unemployment period for all unemployed persons. Contact can be made more frequently, and the content of the contact interview should be adjusted on an individual basis.
- All unemployed persons should submit a CV to the new job and CV database (jobnet.dk) no later than after one month of unemployment. Data on jobs/unemployed persons is free and easily accessible for private actors. Digital submission of CVs is possible.

Other actors

Previous rules

- Contracting out was done in each region, with other actors involved in carrying out activation programmes, primarily training courses and education. Payment of other actors was primarily based on the length of the programme and other actors were subject to many rules. Co-operation contracts were sometimes reached between PES, municipalities and UI funds.

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- Extended use will be made of other actors in the entire process, including job mediation and contact and in both PES and municipalities. Payment of other actors should be based on actual outcome as far as possible, while they should be free to use any method to achieve those outcomes. Contracting out arrangements can be made across regions and municipalities and the rules will be simplified. Co-operation contracts between parties should be extended in size and content.

Simple and active programmes

Previous rules

- For UI recipients activation could be offered only within a narrow economic budget, during the daily benefit period (first year of unemployment for those aged 25 and above and first six months of unemployment for those under 25 years). In the activation period (3 years following the daily benefit period) UI recipients should be in active programmes 75 per cent of the time. For cash benefit recipients, the local government could offer activation immediately. Persons younger than 30 years had a right and a duty to participate in active programmes after having received cash benefits for 13 weeks. Persons older than 30 years had a right and a duty to participate in active programmes after having received cash benefits for 12 consecutive months.
- Thirty-two different arrangements for activation and different tools used in municipalities and in the PES.
- Job training in a firm could be given to UI recipients for a maximum of 4 weeks. Cash allowance recipients could undertake individual job training with limits on duration. Persons in individual job training could only take on tasks that would otherwise not have been carried out.

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- Education could be taken by UI recipients, depending on the demand in the region. Cash allowance recipients had no access to courses of education that involved eligibility for education benefits for persons younger than 30 years. A "positive list" of courses set limits for the possibilities for both UI and cash benefit recipients.
- Eighteen wage subsidy arrangements were granted according to different principles. The costs of education related to job training in a firm and wage subsidies ("mentor") arrangements for UI-recipients in private sector job training were covered out of public funds. But municipalities did not cover costs for persons in rehabilitation who attended ordinary educational courses.
- There was no scope for responding with flexible initiatives for persons under notice of layoffs.

More People in Work

- A plan of action including the first offer of activation programme should be set up within the first year of unemployment for those aged 30 and above and within six months for those under 30 years. The activation rules will be harmonised for UI recipients and cash benefit recipients.
- All unemployed persons who are able to take a job are required to enter an activation programme every time they have been "out of touch with the labour market" for 6 months. All activation offers can be started flexibly at any time. Unemployed persons who are able to take a job cannot participate in educational activation lasting more than 6 weeks until after one year of unemployment. Danish language education can be started at any time.
- The distinction between daily benefit period and activation period is abolished. Three activation tools will be used for all unemployed: guidance and qualification, job training in a firm, and wage subsidies.
- UI recipients and cash benefit recipients who are able to take a job will be able to have 4 weeks job training in a firm. Besides that, it is possible to offer one month firm-based training up to three times as a part of a guidance/qualifying programme. Persons with other problems than unemployment (social, psychological, physical, abuse, etc.) can have job training in a firm for 13 weeks initially, with the possibility of extension to 26 weeks or even longer, if it is considered necessary. Unemployed persons should have the opportunity to perform meaningful work while the rules on duration and a targeted referral procedure should ensure against exploitation by firms hiring subsidised labour.
- Education can be taken for a maximum 6 weeks for the first year (with individuals free to choose the course of education). All persons younger than 30 years should, as a rule, receive educational benefits when taking a course of education. Grants of exception are possible for breadwinners aged 25-29 years without qualifying education, or with problems other than unemployment. Access to all courses of education that are targeted at labour market participation. The "positive list" is abolished.
- Wage subsidies shall be made under one system with 3-4 rates of subsidisation. The subsidy will be paid as a fixed amount of *kroner* per hour. It is possible for both PES and municipalities to grant financial support for education costs, including mentor arrangement (also in case of ordinary hiring, which is not possible today).
- In case of sizeable layoffs the PES can – within certain limits – grant an amount for re-training of employees during notice period.

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– Municipalities cover all costs involved in the purchase of education to people in rehabilitation. However, each year a list of "free" educational courses targeted at bottleneck areas is drawn up, which the municipalities can use free of charge.

Special efforts of young people

Previous rules

– The young people benefit rate was approximately DKK 1 000 higher per month than the education benefit rate.

– UI recipients younger than 25 years had a right and duty to education after 6 months while cash benefit recipients had a right and duty to activation (not necessarily education) after receiving cash benefits for 13 weeks. For 25-29 year-olds, "Labour market efforts" started no later than after 12 months of unemployment, with a right to an offer of a place in an activation program, but not necessarily education.

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– The cash benefit rate is reduced to the level of education benefits, when educational activation is offered after 6 months.

– Cash benefit recipients younger than 25 years without qualifying education, who are able to take a job, should have an offer of education no later than after 6 months of unemployment. Rules are harmonised to provide right and duty to education after 6 months of unemployment for all. Follow-up will be made every 3 months. For 25-29 year-olds, "Labour market efforts" are started after 6 months of unemployment for people receiving unemployment insurance. All persons who are able to take a job but have no qualifying education have a right to an offer of education after 6 months of unemployment.

Special efforts for people older than 50 years

Previous rules

– People aged 58-59 years could be exempted (by the local employment authorities) from the duty to participate in activation programmes.

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– People aged 58-59 years are also subject to the general rules about flexible and individual contact and activation offers.

Work should be economically attractive

Previous rules

– Labour income of DKK 11.50 per hour can be excluded from means testing when calculating the cash benefit rate.

– Breadwinners can receive a maximum amount of DKK 11 522 per month in cash benefits (corresponds to 80 per cent of maximum daily benefits).

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– Cash benefits, special benefits and housing benefits is granted without any ceiling on the total amount.

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– Labour income from ordinary employment of DKK 28.00 per hour can be excluded when calculating the cash benefit rate for married cash benefit recipients, who have received benefits for more than 6 months.
– The cash benefit rate for married breadwinners is reduced by DKK 500 to DKK 10 022 per month when cash benefits have been received for 6 months.
– The total amount received as cash benefits, special benefits and housing benefits can not exceed a certain amount, when cash benefits have been received for 6 months.

Simplification and reducing bureaucracy

Previous rules

– The unemployment period is divided into two: Daily benefit period of 1 year within the last 2 years; activation period of 3 years within the last 4 years. Unemployment in the daily benefit period is measured in hours, in the activation period it is measured in weeks.
– Access to daily benefits for graduates is achieved 4 weeks after the end of education, contingent on the person being enrolled in an UI fund no later than 2 weeks after the end of education.
– An individual benefit rate for graduates can be calculated after 3 month of employment.
– UI recipients: Duty to take a "suitable" job after the first 3 months of unemployment, and a "reasonable" job after 3 months. Requirement to take a job that involves up to 3 hours of transport per day.
– Different sanctions for daily and cash benefit recipients and for different offences (resignation, turning down offers of mediation and activation).
– The Labour Directorate handles complaints and performs random tests of UI funds' decisions on availability. The Social Appeal Agency makes some registration of the availability testing in municipalities.

More People in Work

– Daily benefit period of 4 years within the last 6 years. Unemployment is measured in weeks during the whole period.
– The right to daily benefits requires membership of an UI fund for 1 year including graduates. Admission to an UI fund during education is possible, with the introduction of a reduced subscription when membership during education.
– An individual benefit rate can be calculated 12 months after the right to graduates benefit rate is achieved.
– Job search and duty to take a "reasonable" job from the first day of unemployment. Job search during activation is required; exemption can be granted if the person participates in a special targeted education or re-qualification programme. Extended requirement of geographical mobility: up to 4 hours of transport per day, though it is possible to take into account special circumstances regarding breadwinner status, etc.

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Requirement of extended geographical mobility for medium or highly educated persons. Requirement of submitting a digital CV to the job and CV databases that is accessible to employers, PES and local governments.

- Intensified contact with PES/municipalities if doubts about availability. Testing of availability by introducing a duty to apply for open job offers if required so by PES, municipalities or the UI fund. Testing of availability by introducing special job projects that are not offered as ordinary jobs. Requirement of sickness reporting to PES/ municipalities/employers. Vacation should be reported with at least 14 days notice.
- Common first time sanction of 5 weeks for all offences: In the 5-week period the daily benefit recipient will receive no benefits, and the cash benefit recipient (if able to take a job) will receive only one-third. Common second time sanction in case of repeated offence within 12 months: Until the unemployed person has had 300 hours of ordinary work within 10 weeks, the daily cash recipient will receive no benefits, and the cash benefit recipient (if able to take a job) will receive only one-third. The reduction of benefits can, however, only last for 20 weeks maximum. In this sanction period, the cash benefit recipient can accept an offer, but he will continue to receive the lower benefit rate until the end of the sanction period. A sanction in one system will have effect in both systems: Daily benefit recipients under sanction can only receive cash benefits that he will be required to pay back.
- The Labour Directorate's control of UI funds' decisions is extended, so that the Directorate deals with every report from PES of UI funds refraining from sanctions. The central registration, surveillance and control of the administration in municipalities is extended to prepare for benchmarking and best-practice among municipalities. In the longer term, a control of the administration in municipalities, equivalent to the Labour Directorate's control of UI funds, will be introduced.

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Annex VI

Voluntary early retirement benefit reform

As a part of the agreement on the fiscal bill for 1999, a long-debated reform of the voluntary early retirement benefit scheme was eventually adopted by a broad majority in the Parliament (Table A2). Since raising the entitlement age (60 years) was not politically feasible, the design of the reform is rather complicated, but it aims to raise the average age of entry through a mix of carrot- and stick-type instruments, namely:

- The reductions of the benefit level for income from private pension schemes for entry before 62 years of age have been increased.
- A tax discount is given to wage earners who stay in work beyond 62 years of age. The tax discount increases gradually with the time worked beyond 62 years of age up to a maximum of DKK 100 000. It is given as a one-off payment at 65 years, when the person becomes entitled to an old-age pension.
- The annual employee-paid contributions to the system have been approximately doubled, while the qualifying period has been raised to 25 years within the last 30 years – up from 20 years within the last 25 years.

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Table A2. **Reform of the voluntary early retirement benefit system**[1]

	Old scheme	New scheme
Old age pension	Entry age 67. Means testing: wage income above threshold reduced by a 60 per cent rate in basic state pension.	Entry age 65. Means testing: wage income above threshold reduced by a 30 per cent rate in basic state pension.
Conditions for part-time work	Distinct transfer scheme for eligible wage earners working part time. 200 hours work annually allowed in the main scheme without benefit reduction.	Unified scheme. Continuous reduction in benefit for the number of hours worked.
Benefit level	100 per cent unemployment benefit of the first 2½ years. 82 per cent unemployment benefit of the last 4½ years. (Flat 100 per cent of unemployment benefit rate if withdrawal was postponed to 63 years).	Flat rate of 91 per cent of unemployment benefit. Flat rate of 100 per cent of unemployment benefit if withdrawal is postponed to 62 years.
Reduction of benefit for income from private pension schemes	Income from labour market pension schemes with continuous payments reduces benefits by a 60 per cent rate. No reduction if payments were voluntarily postponed to after age 67. No reductions for income from pension schemes with capital payments.	*Withdrawal before 62 years*: Equal treatment of all pension schemes. 60 per cent of actuarial pension income, calculated from pension wealth at 60 years, is subtracted from benefits. *Withdrawal after 62 years*: Old rules apply
Tax discount for postponement of withdrawal	No tax discount.	Tax discount if retirement is postponed 2 years or more. The discount equals 35 per cent of the early retirement benefit that could alternatively have been received. The maximum discount is a little above DKK 100 000.
Contributions and eligibility	Unemployment insurance contributions of 7¼ times maximum daily unemployment benefit rate – approximately DKK 4 000 per year before tax and DKK 2 400 after tax. Eligibility requirement: payment of unemployment insurance contributions in 20 out of last 25 years.	Unemployment insurance contributions are reduced to 6 times maximum daily unemployment benefit rate. New early retirement benefit contribution, at 7 times maximum daily unemployment benefit rate, is introduced. Eligibility requires payment of unemployment insurance and early retirement benefit contributions for 25 out of last 30 years.

1. Only the main principles of the schemes are specified in the table and transitional provisions are left out for ease of presentation. These include a three-year phase-in of the VERB contribution, a phased-out floor for the imputed actuarial pension income and easier terms for payment of VERB contributions for people who were above 35 years of age on 1 July 1999.

Source: OECD.

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Annex VII

Transforming disability into ability: an OECD-wide study

No single country in the OECD review of disability policies (OECD, 2003c) can be said to have a particularly successful policy for disabled people. Nevertheless, there are differences in outcomes that appear to be related to the policy choices that countries have made. From these observations, the following policy positions can be recommended.

Recognise the status of disability independent of the work and income situation

Societies need to change the way they think about disability and those affected by it. The term "disabled" should no longer be equated with "unable to work". Disability should be recognised as a condition, but it should be distinct from eligibility for, and receipt of, benefits, just as it should not automatically be treated as an obstacle to work. The disability status, *i.e.* the medical condition and the resulting work capacity, should be re-assessed at regular intervals. The recognised disability status should remain unaffected by the type and success of intervention unless a medical review certifies changes.

Introduce a culture of mutual obligations

Most societies readily accept their obligation to make efforts to support and (re)integrate disabled persons, but it is less common to expect disabled persons themselves and, if applicable, their employers to contribute to the process as well. This change of paradigm will require a fundamental rethinking and restructuring of the legal and institutional framework of disability policy in many countries. It will be effective only if it is accompanied by a change in the attitude of all those involved in disability issues. Note that the following three recommendations are to a varying degree related to this new culture of mutual obligations.

Design individual work/benefit packages

Merely looking after the financial needs of disabled people through cash benefits is insufficient; this would still leave many excluded from the labour market and sometimes even from society more generally. Therefore, each disabled person should be entitled to a "participation package" adapted to individual needs and capacities. This package could contain rehabilitation and vocational training, job search support, work elements from a wide range of forms of employment (regular, part-time, subsidised, sheltered) and benefits in cash or in kind. It could also in some circumstances contain activities that are not strictly considered as work, but contribute to the social integration of the disabled person.

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Introduce new obligations for disabled people

Benefit receipt should in principle be conditional on participation in employment, vocational rehabilitation and other integration measures. Active participation should be the counterpart to benefit receipt. Just as the assisting caseworker has a responsibility to help disabled persons find an occupation that corresponds to their capacity, the disabled person is expected to make an effort to participate in the labour market. Failure to do so should result in benefit sanctions. Any such sanctions would need to be administered with due regard to the basic needs of the disabled person and those of dependent family members. Furthermore, sanctions would not be justified in any case where an appropriate integration strategy had not been devised, or proves impossible to formulate, *e.g.* because of the severity or acuteness of the disability.

Involve employers in the process

Involving employers is crucial to the successful re-integration of disabled persons. Different approaches exist, ranging from moral suasion and anti-discrimination legislation to compulsory employment quotas. The effectiveness of the measures depends on the willingness of employers to help disabled persons stay in or enter work (which can be influenced through incentives aimed at raising labour demand), but also on the possibilities of circumventing legislation or paying the fines imposed for non-compliance.

Promote early intervention

Early intervention can in many cases be the most effective measure against long-term benefit dependence. As soon as a person becomes disabled, a process of tailored vocational intervention should be initiated, where appropriate including, *e.g.* job search, rehabilitation and/or further training. Where possible, such measures should be launched while the person is in an early stage of a disease or a chronic health problem. Preventive measures at the workplace could even be delinked from being temporarily out of work.

Make cash benefits a flexible policy element

The cash part of the work/benefit package needs to reflect the disabled person's capacity to work, but also needs to take into account whether the person has actually been able to find a job. Thus, cash benefits would have to be available with sufficient flexibility to take account both of different cases of remaining work capacity and of the evolution of an individual's disability status over time. In addition, benefit entitlements should be designed such that the disabled person is not penalised for taking up work.

Reform programme administration

A more individual approach will place a wide range of new demands on disability gatekeepers, *i.e.* the people who administer entitlement for, and arrays of, active and passive interventions offered to a disabled person. Caseworkers will need an extensive knowledge of the range of available benefits and services. More time will be required to assist individuals and follow each case. Implementation of a one-stop approach will help gatekeepers to manage the full menu of available intervention and promote equal access to all programmes for all people.

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Design disability programmes as active programmes

Often, disability benefit systems function as early retirement programmes, providing a route for quasi-permanent exit from the labour market. Emphasising activation and the mutual obligations of both society and the disabled person moves disability policy closer to the underlying logic of unemployment programmes, which expect an active contribution and effort from beneficiaries. Unreformed disability programmes are likely to attract applicants who may find it difficult to comply with the stricter obligations of unemployment schemes. There is a need for a consistent strategy in disability and unemployment policy that extends the culture of mutual obligations to all labour market programmes.

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Annex VIII

Refugees and asylum seekers in the Aliens Act 2002

The main changes to the Aliens Act concerning refugees involved the following:

- The "*de facto*" status is abolished and only asylum seekers who meet the criteria of the Geneva Convention will be granted refugee status. Those who do not meet these criteria but cannot be sent back to their home country due to reasons covered by the provisions of Article 3 of the European Convention on Human Rights or the Convention on Torture, will no longer be considered as refugee but can be granted a new "protection status".
- The number of "safe third countries" is increased. Furthermore, countries may be reclassified as safe after a period of unrest, if the political and social circumstances change.
- It is no longer possible to lodge a claim for asylum with a Danish diplomatic representation abroad and only those lodged on Danish territory are considered.
- Claims submitted by asylum seekers who abscond are no longer being processed.
- An increased number of claims are dealt with under the procedure for manifestly unfounded applications. It is also possible to process a case under this fast-track procedure within one day, for example in case of a large number of arrivals from a country whose nationals are normally not granted asylum. The cases vetoed by the Danish Refugee Council are referred to the Chairman of the Board, who will have the possibility to rule alone on the case.
- The Refugee Appeals Board is reduced to three members – *i.e.* one judge, acting as Chairperson, and one representative of each the Ministry of Interior and the Bar Association. The Danish Refugee Council and the Danish Ministry of Foreign Affairs are no longer part of the decision-making process at the appeal stage.
- The existing 15-day period for a rejected asylum seeker to leave the country voluntarily is abolished, those whose cases are rejected are obliged to leave the country as soon as the final decision on their case is made.

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Annex IX

Sanctions in 2003

In 2003, the sanctions imply that the local governments in Copenhagen and Bornholm will have their state grant (their part of the block grant) increased by DKK 12.8 million, respectively, 0.8 million as a compensation for their loss of revenue from lowering tax rates (Table A3). Meanwhile, the local governments of Frederiksberg and West Zealand County, which raised taxes, will have their grants reduced by DKK 12.3 and 60.5 million, respectively. This leaves DKK 59.1 million of the breach to be offset by a reduction in the block grant. As an exception, the block grant in 2003 is not being reduced, whereas the reduction in 2004 will cover both 2003 and 2004. From 2005 the adjustment of individual grants ceases and the full adjustment will fall on the block grant, which is reduced by DKK 118.3 million (twice the DKK 59.1 million).[1]

It should be noted that the local governments of Copenhagen and Frederiksberg have separate negotiations with the central government, as they are members of neither the

Table A3. **Effect of sanctions on counties**

Million DKK in 2003 prices

	Copenhagen[1]	Frederiksberg[1]	West Zealand	Bornholm[1]	Total
2003 and 2004					
Change in tax revenue	–25.7	24.6	133.2	–1.7	130.3
Change from sloping tax ceiling	–	–	–12.2	0.1	–12.1
Change in revenue from adjustment of tax rates	–25.7	24.6	121.0	–1.6	118.3
Reduction of state subsidy	–	–12.3	–60.5	–	–72.8
Increase in state subsidy	12.8	–	–	0.8	13.7
Net revenue after adjustment of individual state subsidies	–12.8	12.3	60.5	–0.8	59.1
Reduction in block grant distributed to all counties	–	–	–	–	59.1
2005 and afterwards					
Reduction in block grant distributed to all counties	–	–	–	–	118.3

1. The figures relate to the county part of the local governments of Copenhagen, Frederiksberg and Bornholm.

Source: Folketinget (2002).

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municipality nor the county association.[2] Since the total revenue effect from tax cuts in Copenhagen and rises in Frederiksberg approximated zero, these changes would *ceteris paribus* be in compliance with the tax freeze. However, when West Zealand County adopted a budget, which – despite a reduction of tax rates by the local government of Bornholm – breached the agreement with the County Association and thereby implied an infringement of the tax freeze, the bill of sanctions was put forward by the government. The bill affects all counties, irrespective of the status of their negotiations with the central government. Therefore, both the local government of Frederiksberg and West Zealand County are punished for raising taxes. In special cases, the Minister for the Interior and Health can exempt a county from sanctions for raising taxes if it faces severe economic difficulties. The county would also not be included in the calculation of the average county tax rates forming the basis of an identification of breaches to the tax freeze.

Notes

1. The penalty amount is stated in 2003 prices, but it will be adjusted in parallel with general wage and price increases in subsequent years.
2. These two local governments both have municipal and county status, and 40 per cent of their tax rates is associated with their county status. The same applies for Bornholm, as the municipalities on the island were merged from 1 January 2003 following an advisory referendum in 2001.

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